Option-Only Portfolio Optimization with Greek-Induced Covariance and Conditional Risk Premia

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Abstract

How should an allocator choose among listed calls and puts when the instruments are funded, expiring, state-contingent cashflows rather than ordinary asset return columns? This paper builds the option-only analogue of Markowitz for that problem. Each option is represented by premium-funded NAV exposure, payoff and settlement convention, conditional option-risk-premium forecast, Greek exposure, implementation screen, and residual risk. The covariance matrix is induced by the Greek map,

$$r_O = \mu_{O,t} + B_t f_{t+1} + \varepsilon_{t+1}, \qquad \Sigma_{O,t} = B_t \Omega_t B_t^\top + \Sigma_{\varepsilon,t},$$

and the optimizer applies the familiar efficient-frontier algebra to premium weights scaled by portfolio NAV.

In this sample and under these conventions, the empirical evidence is intentionally mixed. The framework is coherent, point-in-time, and auditable. Gross option-only portfolios can look attractive, but gross Sharpe is a research diagnostic, not a trading result. VIX options can improve the design as volatility instruments only when VX-forward modeling and exact VRO/SOQ settlement support the relevant rows. Post-cost survival is scenario-dependent: the leading sleeves remain profitable under executable-cost scenarios but fail under a conservative stress-cost layer, and family-wise multiple-testing corrections cannot certify skill on sixty months of data. A follow-on breadth and capacity diagnostic sharpens the limitation: at \$1 million NAV, the 57-group VIX-enabled book moves from -1.837 net Sharpe in the uncapped paper configuration to $+1.499$ net Sharpe (gross 1.915) only after the historical-mean estimator is regularized away and contract-level liquidity caps bind net positions. Above about \$2 million, representative month-end option liquidity rather than optimizer algebra becomes the binding constraint. The locked E1 breadth candidate is best read as a small-account research candidate rather than a live trading strategy. `larger+VIX` and `orig+VIX` pass the main robustness screens; `larger` without VIX is mixed; and `orig` without VIX is capacity-infeasible under the hard cap budget. The `larger+VIX` result survives 12-group CPCV/PBO diagnostics, resampled and refit Monte Carlo, tail-path simulations, and true rolling 36-month refits with positive net Sharpe, while the synthetic repricing stress is explicitly labeled as a historical-cost overlay rather than synthetic NBBO. The portfolio-construction contribution is therefore not a live-trading claim. It is a disciplined way to decide whether convexity, volatility exposure, or short-premium risk deserves capital after premium accounting, costs, settlement, beta audits, Greek budgets, liquidity, and out-of-sample validation.

Key Findings

- Option-only Markowitz is feasible when options are treated as premium-funded cash-flows with Greek-induced covariance, not as generic return columns.
- Gross option portfolios can appear attractive, but post-cost survival is scenario-dependent: the leading sleeves survive executable-cost scenarios, fail the conservative stress-cost layer, and do not clear family-wise multiple-testing hurdles.
- VIX options are useful as conditional volatility instruments only when VX-forward modeling and exact VRO/SOQ settlement support the evidence.
- The locked E1 result is a robust small-account research candidate, strongest with VIX, not a live-tradability proof. At $\$1$ million, diagonal residual covariance, N-scaled shrinkage, structural-only means, and net liquidity caps lift the 57-group book from -1.837 to $+1.499$ net Sharpe. The locked 57-group VIX candidate remains positive under resampled histories (net Sharpe p05 0.989), refit stability (p05 1.262), and rolling 36-month OOS refits (net Sharpe 1.217). Against the two simplest baselines, ordinary underlying Markowitz and capped naive option allocation, the VIX-enabled E1 books beat both; `larger` without VIX is mixed because it essentially ties capped equal-risk naive (0.551 versus 0.550 net Sharpe); `orig` without VIX is capacity-infeasible; and all deployable evidence remains limited to small-account scale in the representative contracts studied.

Impact Statement

Option research often begins with a named trade: covered calls, crash puts, variance spreads, or volatility hedges. This paper begins one layer earlier. It gives a reproducible template for turning arbitrary listed calls and puts into NAV-normalized cash-flows, Greek exposures, conditional premia, covariance estimates, and constraints before asking whether they deserve capital. The conclusion is useful precisely because it is restrained: option-only Markowitz is mathematically clean, but empirical claims survive only if premium accounting, settlement, costs, inference, and equity-beta audits support them. The breadth experiment adds the same lesson for liquidity: more names help only when the estimator and the pre-trade capacity model are disciplined enough to keep the optimizer from spending scarce option depth on noise.

1. Introduction

An investor restricted to listed options has a simple question with a difficult answer: which calls and puts actually deserve premium capital? The usual asset-allocation template starts from return columns. Options do not fit that template cleanly. They are funded, expiring, state-contingent cashflows whose economics depend on premium paid or received, strike, expiry, exercise style, settlement, volatility, convexity, and the state of the underlying.

This paper builds a premium-weighted option-only Markowitz framework for that choice. The universe can contain arbitrary listed calls and puts across equities, strikes, expiries, and moneyness buckets. Positions can be long or short. Cash is not optimized as a separate risky asset; it is the NAV numeraire and collateral account used to scale option premium exposures. The premium is still paid or received. A long option that expires worthless loses its premium weight, and a short option that finishes in the money pays the payoff through the same NAV accounting. The allocation problem is therefore not "which stock return column has the best mean," but whether a funded option cashflow deserves capital after settlement, costs, beta exposure, Greek budgets, and validation gates.

1.1. Research Question and Contributions

The research question is: how should an allocator choose among listed calls and puts when every instrument is a funded, expiring, state-contingent cashflow rather than a standard asset return column? The proposed answer is a premium-weighted option-only Markowitz framework that maps listed options into NAV-normalized cashflows, payoff and settlement conventions, Greeks, residual risk, conditional option-risk-premium forecasts, implementation screens, and portfolio constraints. On the sixty-month held-out sample, the validated empirical claim is that the framework is coherent, point-in-time, auditable, and useful for diagnosing whether option convexity, VIX exposure, or short-premium risk deserves capital. The non-claim is equally important: the article does not claim live alpha, production tradability, broker-executed performance, live margin parity, or deployable option trading performance.

The organizing thesis is that option-only portfolio construction should be written as conditional option portfolio optimization. A listed option is not a stock with a different ticker. It is a state-contingent claim with delta, gamma, vega, theta, exercise, settlement, liquidity, and marking conventions. Two options can share the same underlying and still contribute very different risks. A put and a call can leave the book with little net delta and large volatility or convexity exposure. The covariance matrix must therefore come from the instrument map, not only from historical option-price correlations.

The theoretical contribution is the option-only analogue of Markowitz. Classical portfolio theory starts with expected returns and an asset covariance matrix (Markowitz, 1952). This paper starts with option premium weights, a local Greek return map, conditional option-risk-premium forecasts, and a residual covariance term. Once those objects are built, the mean–variance algebra is familiar: the unconstrained maximum-Sharpe direction is proportional to $\widehat{\Sigma}_{O,t}^{-1}\widehat{\mu}_t$. The difference is the construction of $\widehat{\Sigma}_{O,t}$ and $\widehat{\mu}_t$, not the final quadratic geometry.

What is new is the object rather than a single trade. The paper is not a defense of long calls, short puts, covered calls, protective puts, or a named volatility spread. It asks how to allocate across an arbitrary option universe once each contract has a mark, a payoff convention, a Greek exposure vector, a conditional expected return, and an implementation ledger. The expected-return side is also explicit: covariance optimization alone cannot create a high Sharpe ratio. A useful option allocator needs forecastable premia such as carry, variance risk premia, volatility risk premia, skew compensation, tail-insurance pricing, relative value, and regime-dependent convexity.

1.2. Related Work and Positioning

The starting point is mean–variance allocation. Markowitz (1952) showed how expected returns and a covariance matrix define an efficient frontier, while Sharpe (1966) made reward-to-variability central to performance evaluation. Listed options require a different first step because the columns are not ordinary asset returns. Black–Scholes–Merton option valuation (Black and Scholes, 1973; Merton, 1973) and the Black–76 forward option model (Black, 1976) provide local pricing and Greek maps, not a complete physical-measure portfolio allocation rule. The present framework uses those maps to translate options into delta, gamma, vega, theta, VIX-forward exposure, and residual risk before applying Markowitz geometry.

The closest antecedents optimize option portfolios directly. Faias and Santa-Clara (2017) construct optimal option portfolios by simulating option returns to expiry and maximizing expected utility over the simulated distribution, and Driessen and Maenhout (2007) evaluate option-pricing anomalies from an empirical portfolio perspective, asking how much of an investor's wealth optimally flows into index puts and straddles. This paper differs on both inputs: risk comes from a Greek-induced covariance matrix rather than simulated return distributions, and positions are premium-funded NAV weights rather than expiry-payoff shares, so the same machinery prices long and short option capital symmetrically. The premium-weight convention matters economically because option premia embed leverage that varies enormously across strikes and tenors (Frazzini and Pedersen, 2022); a premium weight is the natural unit in which that embedded leverage is charged against the portfolio budget.

Expected option returns have their own economic sources. Empirical option-return research links compensation to carry, volatility risk premia, skew, crash insurance, and tail-risk states (Coval and Shumway, 2001; Bakshi and Kapadia, 2003; Carr and Wu, 2009; Bollerslev et al., 2009; Broadie et al., 2009). This article uses that spine to motivate conditional option-risk-premium forecasts rather than treating the optimizer as a source of alpha. Robust covariance estimation also matters because inverse-covariance portfolio rules amplify noisy inputs; shrinkage follows Ledoit and Wolf (2004). Evaluation uses both reward-to-volatility and downside-aware measures, including Sortino (Sortino and Price, 1994), Omega (Keating and Shadwick, 2002), Calmar and drawdown (Magdon-Ismail and Atiya, 2004), and active risk (Grinold and Kahn, 2000).

Implementation frictions and validation discipline determine the strength of any empirical claim. Option overlays are sensitive to spread, settlement, borrow, assignment, margin, and capacity assumptions. Backtest overfitting and multiple testing can turn optimizer variants into false discoveries, so the analysis uses bootstrap inference and reality-check ideas consistent with Bailey et al. (2017) and López de Prado (2018). The distinction from strategy-specific option papers is therefore deliberate. The object is not a covered-call, put-write, crash-put, or volatility-spread test. The object is an instrument-level cashflow accounting and allocation framework for arbitrary listed options before asking whether any option sleeve deserves capital.

The empirical result is intentionally mixed. The gross point-in-time backtest tests whether the option-only Markowitz object behaves coherently on the local OPRA-derived panel. Post-cost evidence is reported at two intensities: executable-cost scenarios that charge entry frictions only, and a conservative stress-cost layer that additionally charges round-trip spreads, slippage, borrow, capacity, simulated margin funding, and assignment-risk penalties. VIX options are included as volatility derivatives with VX-forward Black–76 Greeks, but VIX option evidence supports listed-settlement claims only when exact VRO/SOQ settlement covers the relevant rows. If the coverage table reports proxy rows, the same evidence is reported only as diagnostic hedge-sleeve evidence rather than a listed-settlement claim.

The final breadth/capacity diagnostic makes the bridge to practical portfolio management more explicit and narrows the claim. The original 56-name expansion fails because the optimizer is fed too many noisy historical-mean directions and no pre-trade contract-cap budget. With VIX included, using diagonal residual covariance, N-scaled covariance shrinkage, structural-only means, and net liquidity caps moves the 57-group \$1 million book from -1.837 to $+1.499$ net Sharpe, with gross Sharpe 1.915. Without VIX, the same regularization is not enough to establish optimizer edge: the 56-name book reaches 0.551 net Sharpe against 0.550 for capped equal-risk naive. The spread ledger uses exact historical panel CBBO for the eight baseline equity names and a point-in-time inferred CBBO proxy for missing broad-name and VIX spreads, so the result is a calibrated execution-sensitivity simulation, not broker-executed tradability evidence. A locked-candidate robustness run then validates the four E1 capped universes with 12 blocked groups, 66 CPCV splits, one-month purge/embargo, Monte Carlo resampling/refitting/repricing, path simulations, drawdown breach tests, reality-check inference, and rolling 36-month monthly refits. The `larger+VIX` and `orig+VIX` books pass as small-account research candidates; the `larger` no-VIX book remains mixed; and the `orig` no-VIX row is diagnostic because full deployment is not supported by its hard-cap budget. The conclusion reports this as a visual final scoreboard against two deliberately simple baselines: ordinary underlying stock Markowitz and capped naive option allocation.

The article proceeds in the same order as the research workflow. Section 2 defines the data and empirical design. Section 3 builds the option cashflow and Greek-risk framework. Section 4 describes the conditional portfolio construction model. Section 5 sets out the metrics, point-in-time validation, inference, and claim gates. Section 6 reports the empirical results, breadth/capacity diagnostic, and negative evidence. Section 7 probes distributional robustness through blocked and combinatorial purged cross-validation and resampled and repriced Monte Carlo universes. Section 8 concludes with the limits of the current research implementation. Appendix A contains supplemental figures, option/VIX diagnostics, robustness tables, instrument conventions, proof sketches, and reproducibility notes.

2. Data

The empirical design is a monthly, point-in-time allocation study over listed equity options and VIX options. The investable equity-option universe is restricted to AAPL, AMZN, GOOGL, JPM, META, MSFT, NVDA, and TSLA, using OPRA/Databento-derived option panels after basic mark, expiry, Greek, and liquidity filters. The VIX-option universe is built from raw VIX chain shards and kept separate because VIX options are volatility derivatives, not options on an equity spot price. Table 2 records the sample size, observation count, train/test split, settlement status, and post-cost layer used in the baseline run.

The breadth extension reported in Section 6 is a diagnostic rather than a replacement for this baseline universe. It reuses the same local OPRA-derived equity-option panel and adds 48 available single-name underlyings to the eight baseline names, producing 56 equity underlyings and, when the VIX sleeve is included, 57 option-underlying groups. The eight baseline equity names are not part of the spread-fallback problem: `p3_spread_source_coverage.csv` records 5,777 historical panel-CBBO equity-option cost rows across 49 asset IDs and all eight baseline underlyings, with median relative spread about 1.89%. Thus the no-VIX baseline `orig` net cells use historical CBBO for their equity-option spread inputs. The VIX-enabled baseline `orig+VIX` uses the same exact equity-option CBBO rows, while its VIX option spread rows use the same inferred liquid-option spread proxy described below because the local VIX chain cache contains OHLCV chain records, not VIX option bid/ask CBBO.

For the additional 48 breadth names, measured historical CBBO is used when present; when it is absent, the breadth code now uses a point-in-time inferred CBBO proxy rather than a blanket class-default fill or stale current-chain quotes. For each decision row, the proxy uses only historical CBBO spread-surface observations available no later than that decision date and matches by moneyness bucket and tenor where possible. The calibration surface contains the local full-day CBBO names AAPL, AMZN, GOOGL, IWM, JPM, META, MSFT, NVDA, QQQ, SPY, and TSLA. In the regenerated P3 ledger, `p3_spread_source_coverage.csv` reports 23,740 added-name equity-option cost rows across 213 asset IDs and 47 underlyings with source `inferred_cbbo_proxy`; their median relative spread is about 1.97%. VIX proxy rows have median relative spread about 2.53%. The stale `current_option_spread_assumptions.csv` file is retained only as an audit/rebuild artifact; the cost loader rejects weekend, holiday, and after-hours Cboe snapshots and the regenerated breadth tables do not consume those off-hours rows. The breadth robustness artifacts reuse this same policy: the full validation ledger has no `current_cboe_liquid_quote` rows and no `default` spread rows, and repriced synthetic net paths subtract a realized full-cost overlay rather than inventing synthetic NBBO/CBBO quotes.

The spread ladder is therefore part of the claim boundary. OPRA is the consolidated U.S. listed-options quote and last-sale dissemination plan[1]; a historical broad-universe execution proof would need market-hours NBBO/CBBO matched to each backtest decision row. Cboe DataShop's option summary product documents 15:45 and end-of-day NBBO snapshots with displayed size, and describes the 15:45 snapshot as the more representative liquidity cut for analytics.[2] That paid history is the right replacement for the inferred broad-name proxy. Until such broad matched history is purchased or otherwise supplied, the proxy rows in Table 1 are calibrated execution-sensitivity inputs, not final execution truth.

Scope	Spread source	Rows	Underlyings	Median rel. spread	Claim use
Baseline 8 equity names	historical panel CBBO	5,777	8	1.89%	exact historical equity-option spread input
`orig+VIX` VIX rows	inferred CBBO proxy	536	1	2.53%	calibrated execution sensitivity
Broad added equity rows	inferred CBBO proxy	23,740	47	1.97%	calibrated execution sensitivity
`larger+VIX` VIX rows	inferred CBBO proxy	536	1	2.53%	calibrated execution sensitivity

Table 1: Breadth spread-source methodology. Counts and medians are taken from `p3_spread_source_coverage.csv` and the locked robustness spread ledger. The baseline eight equity names use historical panel CBBO. VIX and missing broad-name rows use a point-in-time inferred CBBO proxy calibrated from the liquid historical CBBO surface; those rows support sensitivity analysis, not live tradability.

[1] Options Price Reporting Authority overview: `https://www.opraplan.com/`.
[2] Cboe DataShop Option EOD Summary: `https://datashop.cboe.com/option-eod-summary`.

Item	Value
Raw OPRA-derived equity option rows	160,315
Raw VIX option rows after filters	103,650
Monthly snapshot dates	129
Training dates	69
Test dates	60
Primary equity underlyings	AAPL, AMZN, GOOGL, JPM, META, MSFT, NVDA, TSLA
VIX option treatment	VX-forward Black-76 Greeks; VIX/VVIX state only
VIX settlement source	exact VRO/SOQ (536 rows)
VIX headline status	exact VRO/SOQ settlement (excluded expiries disclosed in Section 2)
Post-cost layers	executable-cost scenarios (mid, half, full spread) and conservative stress-cost layer
Broker execution status	not broker-executed live evidence
Rolling option bucket assets	54
Train/test split	through 2020-12-31 / after 2020-12-31

Table 2: Empirical design summary. The table identifies the option universe, monthly snapshots, train/test split, VIX settlement status, and post-cost conventions used in the baseline run.

The train/test split is frozen at December 31, 2020. Training observations through that date estimate expected option returns, covariance objects, residual risk, factor exposures, and implementation screens. Test observations after that date are reserved for out-of-sample portfolio evaluation. The out-of-sample window runs from January 29, 2021 to April 30, 2026: 60 monthly snapshots spanning 64 calendar months. Four calendar months (May 2021, March 2022, March 2024, and March 2025) have no ledger row because no representative contract passed the eligibility screens at those decision dates; they are absent from the return series rather than imputed. Months in which the book holds only cash collateral are recorded at the zero financing rate, that is, as exact zero returns under the zero-financing numeraire of Remark 3.2; the counterfactual hurdle/no-trade ledger (`artifacts/no_trade_periods.csv`) additionally records, for each snapshot, the expected-return hurdles at which the optimizer would have stood aside entirely. Rebalancing is monthly. At each decision date, the research implementation sees only the option panel, state variables, underlying prices, VX curve, forecast inputs, and cost inputs observable at or before that date. The realized option return is then measured at the next monthly realization date or at listed expiry, according to the holding ledger.

Equity-option rolling buckets convert expiring contracts into stable empirical objects without pretending that options are perpetual assets. For each underlying, right, moneyness, tenor, and liquidity bucket, the decision date selects a representative listed contract using contemporaneous marks and filters. The selected option can be a call or a put, and the optimizer can hold it long or short subject to gross NAV, concentration, Greek, beta, short-premium, liquidity, stress, and assignment-risk constraints. The return includes the premium paid or received: a long option that expires worthless loses its premium weight, while a short option that finishes in the money pays the realized payoff through NAV accounting.

The VIX-option construction follows a different convention. Raw VIX chain shards are stacked, OSI symbols are parsed, duplicate (trade date, symbol) rows are coalesced, and each option is aligned to the relevant VX futures forward. Black–76 Greeks are computed with respect to the VX forward rather than spot VIX. VIX, VVIX, VIX9D, and VIX3M enter only as state variables. Exact VRO/SOQ settlement is a claim gate: a VIX option becomes headline-grade only when the expiry row uses `vro_soq_exact`. Proxy rows, if any, remain appendix diagnostics and cannot support listed-settlement claims.

The settlement gate has a selection cost that should be stated plainly. Of the 630 representative VIX option rows selected by the monthly screens, 536 (about 85%) carry a parsed exact VRO/SOQ settlement and enter the holding ledger; the remaining 94 rows (about 15%) lack a parsed exact settlement and are *excluded* from settlement-gated evidence rather than proxied with the VIX close. The exclusions are concentrated in 2015–2018—including 20 required 2017 expiries whose SOQ files could not be parsed exactly (Table 25)—so the retained VIX rows under-represent an eventful era for the VIX complex, and the training-window VIX evidence carries a potential coverage bias in the direction of calmer settlement prints. The expiry-level exact-versus-proxy comparison is written to the machine-readable audit `artifacts/vix_settlement_audit.csv`.

Settlement source	Rows	Headline eligible
vro_soq_exact	536	yes

Table 3: VIX settlement coverage. VIX option evidence supports listed-settlement claims only when every VIX expiry row used in the main tables has `vro_soq_exact` settlement.

Marks, payoffs, settlement, and costs are treated as first-order design choices. Option weights are premium exposures per dollar of NAV. The mark source, payoff date, settlement source, Greek model, state snapshot, and train-end date are recorded before performance is reported. Post-cost evidence is produced at two deliberately different intensities. The *executable-cost scenarios* charge entry frictions only: midpoint, half-spread, and full-spread quoted-spread crossing plus explicit fees, with no-fill rules and capacity limits. The *conservative stress-cost layer* additionally charges round-trip spread crossing, slippage, borrow proxies, capacity penalties, simulated margin/stress-capital funding, and assignment-risk penalties; it is a stress bound rather than an execution estimate. Section 6 reconciles the two layers, and Table 4 summarizes the accounting contract each instrument must satisfy. These conventions are adequate for a research paper on allocation mechanics because they test whether the option-only Markowitz object can be constructed, audited, stress-screened, and evaluated out of sample. They are not adequate for production trading claims because they do not contain broker-routed orders, live quote depth, fill probabilities calibrated from orders, live margin previews, or broker reconciliation.

Object	Accounting rule	Main failure if omitted
Equity option	Premium weight times mark or payoff return	Treats the option as free convexity
Short option	Premium received plus future liability	Hides tail and assignment exposure
VIX option	Black–76 exposure to matched VX forward	Treats spot VIX as tradable
Cash/collateral	NAV numeraire, not optimized asset	Converts the model into a stock-cash allocation
Executable-cost scenarios	Mid/half/full quoted-spread entry frictions plus fees	Overstates net returns if spreads are crossed in full
Stress-cost layer	Round-trip spreads, fees, slippage, borrow, capacity, margin funding, assignment flags	Promotes gross research returns to false trading evidence

Table 4: Option accounting contract. Each object is represented by the cashflow convention that must be specified before the optimizer can compare it with another option. The two post-cost rows are distinct layers: the executable-cost scenarios are the realistic research estimate, the stress-cost layer is a conservative bound (Section 6).

3. Option Cashflow and Risk Framework

The central modeling step is to express every option as a NAV-normalized cashflow. Let W_t be beginning portfolio NAV, $q_{i,t}$ the signed premium weight assigned to option $i \in \{1, \dots, n\}$, and $C_{i,t} > 0$ the decision-date mark. The contract count is

$$N_{i,t} = \frac{q_{i,t} W_t}{C_{i,t}}.$$

Ignoring transaction costs and slippage for the gross theory object, but not ignoring the option purchase price, the next-period NAV contribution is

$$\frac{\Delta W_{i,t+1}}{W_t} = q_{i,t} \frac{C_{i,t+1} - C_{i,t}}{C_{i,t}} = q_{i,t}\, r_{i,t+1}, \qquad r_{i,t+1} := \frac{C_{i,t+1} - C_{i,t}}{C_{i,t}},$$

which defines the one-period option mark return $r_{i,t+1}$. These are premium weights. A long option whose mark falls to zero loses exactly the paid premium weight. A short option is scaled by the premium it sold and then pays the future mark or payoff liability. Summing over contracts and the residual cash account turns the contribution into an exact NAV recursion.

Proposition 3.1 (NAV accounting identity)**.** *Let $q_t \in \mathbb{R}^n$ be the premium-weight vector, r_{t+1} the option mark-return vector, and $r_{f,t}$ the one-period financing rate earned on the residual cash $W_t(1 - \mathbf{1}^\top q_t)$. Then*

$$\frac{W_{t+1}}{W_t} - 1 = q_t^\top r_{t+1} + \big(1 - \mathbf{1}^\top q_t\big) r_{f,t}, \qquad \frac{W_{t+1}}{W_t} - 1 - r_{f,t} = q_t^\top \big(r_{t+1} - r_{f,t}\mathbf{1}\big).$$

Proof. At $t+1$ the option book is worth $\sum_i N_{i,t} C_{i,t+1} = W_t \sum_i q_{i,t}(1 + r_{i,t+1})$ and the cash account is worth $W_t(1 - \mathbf{1}^\top q_t)(1 + r_{f,t})$. Adding the two, dividing by W_t, and subtracting 1 (and then $r_{f,t}$) gives both displays. $\square$

Remark 3.2 (Zero-financing numeraire). The baseline empirical run sets $r_{f,t} = 0$: residual cash and intra-month expiry proceeds earn nothing. Every Sharpe object reported in the paper is therefore a raw-return Sharpe ratio measured against a zero threshold, not an excess-of-cash Sharpe ratio. With $r_{f,t} \neq 0$, Proposition 3.1 shows the correct objective is the excess return $q^\top(r_{t+1} - r_{f,t}\mathbf{1})$, and the unconstrained maximum-Sharpe direction (Proposition A.6) becomes $\Sigma_{O,t}^{-1}(\mu_{O,t} - r_{f,t}\mathbf{1})$. Over the 2021–2026 sample, short rates were materially positive for much of the window, so the zero-financing convention forgoes collateral yield in the numerator while also applying a zero rather than a positive hurdle in the excess return; the two effects act in opposite directions. All strategies and benchmarks are reported under the same convention, so comparisons remain internally consistent, and readers can restate any row against a cash hurdle using the identity above.

Two notational conventions apply throughout. The operator Δ applied to a state variable always denotes a one-rebalance-period difference, for example $\Delta S_u = S_{u,t+1} - S_{u,t}$ and $\Delta\sigma_i = \sigma_{i,t+1} - \sigma_{i,t}$; the Black–Scholes and Black–76 hedge ratios carry a superscript, Δ_i^{BS} for equity options and Δ_j^{VX} for VIX options, so the two uses never collide. The underlying simple return is $R_u = \Delta S_u / S_u$.

For a call or put on equity underlying u, the mark change is decomposed by the same Ito–Taylor accounting that underlies Black–Scholes–Merton option valuation (Black and Scholes, 1973; Merton, 1973; Shreve, 2004):

$$\Delta C_i = \Delta_i^{\mathrm{BS}}\,\Delta S_u + \frac{1}{2}\Gamma_i(\Delta S_u)^2 + \nu_i\,\Delta\sigma_i + \Theta_i\,\Delta t + \eta_i,$$

where η_i is *defined* as the exact remainder, so the display holds with equality rather than approximately. The paper does not require Black–Scholes to be a globally correct pricing model for American single-name options. It uses the weaker local statement: an option mark can be expanded in spot, volatility, and time, with vanna, volga, higher-order terms, early-exercise premium movement, and marking error left in η_i.

Dividing by the decision-date mark $C_{i,t}$ gives the return exposure map

$$r_{i,t+1} = \underbrace{\frac{\Delta_i^{\mathrm{BS}} S_u}{C_{i,t}}}_{b_{i,t}^{\Delta}} R_u + \underbrace{\frac{\Gamma_i S_u^2}{2C_{i,t}}}_{b_{i,t}^{\Gamma}} R_u^2 + \underbrace{\frac{\nu_i}{C_{i,t}}}_{b_{i,t}^{\nu}} \Delta\sigma_i + \underbrace{\frac{\Theta_i \Delta t}{C_{i,t}}}_{\theta_{i,t}^{*}} + \epsilon_{i,t+1}, \qquad \epsilon_{i,t+1} = \frac{\eta_i}{C_{i,t}},$$

again with equality because the scaled remainder is retained. The contract-level implied-vol change is mapped to its underlying's at-the-money factor through $\Delta\sigma_i = \beta_{i,t}^{\sigma}\,\Delta\sigma_{u(i)}^{\mathrm{ATM}} +$ (smile remainder), with the smile remainder absorbed into $\epsilon_{i,t+1}$. The coefficients are dimensionless option-return exposures. Deep out-of-the-money options can have large return exposure because the premium denominator is small. That is not a numerical nuisance. It is the reason option-only portfolios need explicit gross, per-contract, Greek, beta, and stress budgets.

Remark 3.3 (Unit conventions). Marks $C_{i,t}$ and Greeks are quoted per share; contract-level quantities carry the 100-share multiplier, which cancels in every premium-weight ratio above. Volatilities are in decimal units (one vol point $= 0.01$), Θ_i is annualized, and Δt is the rebalance horizon in year fractions. Contract counts $N_{i,t} = q_{i,t} W_t / C_{i,t}$ are fractional in the theory object; rounding to integer contracts perturbs each premium weight by at most $C_{i,t}/W_t$ per contract, which the implementation treats as part of the cost and capacity layer rather than of the risk model.

Stacking contracts gives the conditional factor form. The factor vector collects underlying returns, centered squared returns, at-the-money implied-volatility shocks, skew and tail-slope shocks, and the matched VX-forward returns with their centered squares,

$$f_{t+1} = \Big(R_u,\ R_u^2 - \mathbb{E}_t\big[R_u^2\big],\ \Delta\sigma_u^{\mathrm{ATM}},\ \Delta\mathrm{skew}_u,\ \Delta\mathrm{tail}_u,\\ R_T^{VX},\ \big(R_T^{VX}\big)^2 - \mathbb{E}_t\big[(R_T^{VX})^2\big]\Big)_{u\in\mathcal{U},\, T\in\mathcal{T}},$$

where $\mathcal{U}$ is the set of equity underlyings and $\mathcal{T}$ the set of VX-forward maturities. The matching rows of B_t are $\big(b_{i,t}^{\Delta},\, b_{i,t}^{\Gamma},\, b_{i,t}^{\nu},\, b_{i,t}^{skew},\, b_{i,t}^{tail},\, b_{i,t}^{\Delta,VX},\, b_{i,t}^{\Gamma,VX}\big)$. Writing $\Omega_t = \mathrm{Var}_t(f_{t+1})$ for the conditional factor covariance,

$$r_{O,t+1} = \mu_{O,t} + B_t f_{t+1} + \varepsilon_{t+1}, \qquad \Sigma_{O,t} = B_t \Omega_t B_t^{\top} + \Sigma_{\varepsilon,t},$$

where $\Sigma_{\varepsilon,t} = \mathrm{Var}_t(\varepsilon_{t+1})$ is the residual option-return covariance. The residual term is part of the model. Omitting it would assume that Greeks span all marking risk. The covariance identity requires one formal statement about what ε_{t+1} is.

Assumption 1 (Projection-residual definition of the residual)**.** $\Omega_t = \mathrm{Var}_t(f_{t+1}) \succ 0$, and B_t and ε_{t+1} are defined by the population linear projection of the centered option return on the factors,

$$B_t := \mathrm{Cov}_t\big(r_{O,t+1}, f_{t+1}\big)\,\Omega_t^{-1}, \qquad \varepsilon_{t+1} := r_{O,t+1} - \mu_{O,t} - B_t f_{t+1}.$$

Consequently $\mathrm{Cov}_t(f_{t+1}, \varepsilon_{t+1}) = 0$ holds by construction, and $\Sigma_{O,t} = B_t\Omega_t B_t^{\top} + \Sigma_{\varepsilon,t}$ is an identity (Proposition A.5), not an approximation.

Remark 3.4 (The Greek map is an estimator of B_t, not its definition). The analytic loadings in the return exposure map are the structural estimator of the projection coefficient B_t. The wedge between the two—vanna and volga terms, the higher-order Taylor remainder, and early-exercise premium movement in American contracts—is measured empirically in the approximation-diagnostics table (Table 53). If B_t were instead *fixed* at the analytic Greek loadings, the residual would in general correlate with the factors and the covariance would acquire the omitted cross term $B_t\Sigma_{f\varepsilon,t} + \Sigma_{f\varepsilon,t}^{\top} B_t^{\top}$. The paper therefore treats the Greek map as an estimator whose adequacy is reported, rather than assuming the cross term away.

Remark 3.5 (Rebalance-horizon approximation). The Taylor decomposition is applied at the monthly rebalance horizon with decision-date Greeks held fixed; it is a working risk model whose adequacy is tested empirically in the approximation diagnostics, not an asymptotic small-Δt statement.

VIX options require a separate convention. They are not options on an equity spot. The traded forward-risk anchor is the matched VX futures curve, and Black's futures-option logic is the natural local model (Black, 1976). For VIX option j with matched forward $F_{T_j}^{VX}$, the local decomposition is

$$\Delta C_j^{VIX} = \Delta_j^{VX}\,\Delta F_{T_j}^{VX} + \frac{1}{2}\Gamma_j^{VX}\big(\Delta F_{T_j}^{VX}\big)^2 + \nu_j^{VIX}\,\Delta\sigma_j^{VIX} + \Theta_j^{VIX}\,\Delta t + \eta_j^{VIX},$$

with η_j^{VIX} the exact remainder. Dividing by the mark $C_{j,t}$ gives the premium-normalized form that enters the factor model,

$$r_{j,t+1} = b_{j,t}^{\Delta,VX} R_{T_j}^{VX} + b_{j,t}^{\Gamma,VX} \big(R_{T_j}^{VX}\big)^2 + b_{j,t}^{\nu,VX}\, \Delta\sigma_j^{VIX} + \theta_{j,t}^{*} + \epsilon_{j,t+1},$$

$$b_{j,t}^{\Delta,VX} = \frac{\Delta_j^{VX} F_{T_j}^{VX}}{C_{j,t}}, \qquad b_{j,t}^{\Gamma,VX} = \frac{\Gamma_j^{VX}\big(F_{T_j}^{VX}\big)^2}{2C_{j,t}}, \qquad b_{j,t}^{\nu,VX} = \frac{\nu_j^{VIX}}{C_{j,t}},$$

where $R_{T_j}^{VX} = \Delta F_{T_j}^{VX} / F_{T_j}^{VX}$ is the matched VX-forward return entering f_{t+1} on equal footing with the equity underlyings, and $\epsilon_{j,t+1} = \eta_j^{VIX}/C_{j,t}$. VIX, VVIX, and term-structure indices are useful state variables; VX futures and VIX options are the traded volatility instruments. Exact VRO/SOQ settlement is required before VIX option expiry P&L can be promoted from settlement proxy to headline listed-settlement evidence.

4. Portfolio Construction Model

The optimizer is chosen only after the cashflow and risk map are built. Covariance is not enough. A high Sharpe ratio cannot come from $\widehat{\Sigma}_{O,t}$ alone; it must come from a conditional expected-return vector. Using the same loadings that generate the covariance in Section 3, the paper writes the option mean as

$$\widehat{\mu}_{i,t} = b_{i,t}^{\Delta}\widehat{\lambda}_t^{S} + b_{i,t}^{\Gamma}\widehat{\lambda}_t^{var} + b_{i,t}^{\nu}\widehat{\lambda}_t^{vol} + b_{i,t}^{skew}\widehat{\lambda}_t^{skew} + b_{i,t}^{tail}\widehat{\lambda}_t^{tail} + \theta_{i,t}^{*},$$

where $b_{i,t}^{\Delta}$, $b_{i,t}^{\Gamma}$, and $b_{i,t}^{\nu}$ are the delta-return, gamma-return, and vega-return rows of B_t, and $b_{i,t}^{skew}$ and $b_{i,t}^{tail}$ are contract i's loadings on the skew-shock factor Δskew_u and the tail-slope factor Δtail_u (put-wing and VIX-call-wing repricing). Delta loads on equity compensation. Gamma and vega load on variance and volatility risk premia. Put skew and VIX call wings load on crash-insurance and tail premia. Theta is carry under unchanged state variables. The option-return literature documents why these components can carry expected returns rather than only hedge noise (Coval and Shumway, 2001; Bakshi and Kapadia, 2003; Carr and Wu, 2009; Bollerslev et al., 2009; Broadie et al., 2009). In implementation the combined signal is estimated only from training-window or decision-date information and is shrunk toward zero.

The mean equation is the conditional expectation of the return exposure map, so its bookkeeping must be consistent with the centering used in Ω_t:

$$\widehat{\lambda}_t^{var} = \mathbb{E}_t\big[R_{u,t+1}^2\big], \qquad \widehat{\lambda}_t^{vol} = \mathbb{E}_t\big[\Delta\sigma_{u,t+1}^{\text{ATM}}\big], \qquad \mathbb{E}_t\big[R_{u,t+1}^2 - \mathbb{E}_t[R_{u,t+1}^2]\big] = 0.$$

Here $\widehat{\lambda}_t^{var}$ is the physical second moment of the underlying return; it is not itself a premium, and the centered squared-return factor entering Ω_t has conditional mean zero by construction. The carry term $\theta_{i,t}^{*} \approx -b_{i,t}^{\Gamma}\sigma_{IV,i,t}^2\,\Delta t$ is risk-neutral gamma rent, so

$$b_{i,t}^{\Gamma}\widehat{\lambda}_t^{var} + \theta_{i,t}^{*} \approx b_{i,t}^{\Gamma}\big(\sigma_{P,u,t}^2 - \sigma_{IV,i,t}^2\big)\Delta t$$

recovers the variance risk premium exactly once: the premium lives in the gap between the physical second moment and implied variance, not in either term separately, and there is no double counting between carry and the gamma channel.

Remark 4.1 (Implemented mean estimator). The displayed mean equation is the framework object; the baseline implementation is deliberately simpler, and the paper reports the estimator actually run. Three channels are active: carry $\theta^*_{i,t} = \Theta_i \Delta t / C_{i,t}$; the delta channel $b^{\Delta}_{i,t} \widehat{\lambda}^S_t$, with $\widehat{\lambda}^S_t$ the training-window mean underlying return damped by a fixed scale (0.35); and a *vega-routed* variance/volatility premium, $-\kappa_v\, b^{\nu}_{i,t}\big(\sigma_{IV,i,t} - \widehat{\sigma}_{RV,u,t}\big) + \kappa_v\, b^{\nu}_{i,t} \widehat{\lambda}^{vol}_t$ with $\kappa_v = 0.10$, which prices the implied-minus-realized gap through vega rather than through $b^{\Gamma}_{i,t} \widehat{\lambda}^{var}_t$; the separate gamma-times-λ^{var} term is set to zero in the baseline mean. The skew and tail premia enter as *characteristics*, not as covariance factor loadings: near- and wing-moneyness puts receive $-\kappa_s |b^{\nu}_{i,t}|$ $(\kappa_s = 0.02)$ and near/wing VIX calls receive $-\kappa_{\mathcal{T}} |b^{\nu}_{i,t}|$ $(\kappa_{\mathcal{T}} = 0.015)$, so long positions pay the insurance premium and short positions receive it through negative q; a small relative-value tilt on the per-underlying implied-volatility z-score completes the structural sum. The combined signal is a fixed blend of the training-window historical mean (weight 0.25) and the structural sum (weight 0.75), shrunk 60% toward zero and clipped at ± 0.35 monthly. One consequence deserves a name: because the skew/tail terms are priced outside the covariance factor span—Δskew_u and Δtail_u do not appear in the implemented $\widehat{\Omega}_t$—the optimizer treats any premium attached to them as near-diversifiable and can over-allocate to crash-sensitive wings. This over-allocation bias is a stated limitation of the baseline implementation, bounded in practice only by the gross, per-contract, and stress budgets. The breadth diagnostic in Section 6 varies this estimator explicitly; its best-performing large-universe configurations set the historical-mean weight to zero and raise shrinkage-to-zero to 0.75, showing that the training mean is signal at eight names but becomes optimizer noise when the contract dimension rises sharply.

Given $\widehat{\mu}_t$ and $\widehat{\Sigma}_{O,t}$, the gross option-only Sharpe problem is

$$\max_{q \in \mathcal{K}_t \setminus \{0\}} \mathrm{SR}(q) = \frac{q^\top \widehat{\mu}_t}{\sqrt{q^\top \widehat{\Sigma}_{O,t}\, q}}, \qquad \widehat{\Sigma}_{O,t} = \widehat{B}_t \widehat{\Omega}_t \widehat{B}_t^\top + \widehat{\Sigma}_{\varepsilon,t}.$$

The feasible set has one mathematical description. Collecting the gross-NAV, short-premium, per-contract, per-underlying-concentration, Greek/beta, and scenario stress budgets,

$$\begin{aligned}\mathcal{K}_t = \Big\{ q \in \mathbb{R}^n : \ & \|q\|_1 \le g_t, \quad \textstyle\sum_{i:\, q_i<0} |q_i| \le s_t, \quad \|q\|_\infty \le c_t, \quad |q_i| \le \bar{q}_{i,t}\ \ \forall i, \\ & \textstyle\sum_{i:\, u(i)=u} |q_i| \le g_{u,t}\ \ \forall u, \\ & \big|a_{k,t}^\top q\big| \le b_{k,t}\ \ \forall k, \quad \ell_{j,t}^\top q \ge -m_t\ \ \forall j \Big\},\end{aligned}$$

where the $a_{k,t}$ rows hold NAV-normalized delta, SPY-beta, eight-underlying-beta, gamma, equity-vega, and VIX-vega exposures, and the $\ell_{j,t}$ rows hold scenario stress P&L. Every constraint functional is positively homogeneous of degree one, so $\mathcal{K}_t$ is a compact convex set that caps scale along any direction; only sign restrictions (for example a long-only sleeve, $q \ge 0$) restrict direction. The optimizer can therefore choose calls, puts, long positions, and short positions, but it cannot hide large exposure behind a small option premium.

In the baseline paper run, $\bar{q}_{i,t} = c_t$ is just the scalar per-contract cap. The breadth/capacity diagnostic replaces it with a pre-trade liquidity bound

$$\bar{q}_{i,t} = \min\left\{0.18,\ \frac{X \cdot \widehat{V}_i^{\text{train}} \cdot M_{i,t} \cdot 100}{\text{NAV}}\right\},$$

where X is the participation parameter, $\widehat{V}_i^{\text{train}}$ is the training-window median option volume for the rolling bucket, $M_{i,t}$ is the option mark, and 100 is the equity-option multiplier. If no train-window volume is available, the diagnostic falls back to the scalar 0.18 bound rather than importing future liquidity. The cap must bind the net option position q_i. Applying the same bound to the separated long and short legs of the homogenized conic formulation allows offsetting long/short pairs to consume scarce contract capacity as pseudo-cash; the shipped diagnostic therefore imposes $-\bar{q}_{i,t} \leq q_i \leq \bar{q}_{i,t}$ after netting the split variables.

The production-candidate problem is the same geometry with costs and operational constraints moved into the objective and feasible set:

$$\begin{aligned}
\max_{w \in \mathcal{C}_t} \quad & \widehat{\mu}_t^\top w - \frac{\gamma}{2} w^\top \widehat{\Sigma}_{O,t} w - c_t^\top |w|, \\
\mathcal{C}_t = \{w : \quad & \text{premium, Greek, beta/VIX, stress, and margin/collateral budgets hold,} \\
& |w_i| \leq \bar{q}_{i,t} \text{ for every contract } i\}.
\end{aligned}$$

Here c_t is the contract-level execution-cost vector, including spread, fee, slippage, borrow, assignment, and margin-drag components under the chosen accounting layer. The implemented E1 breadth candidate can be read as a maximum-Sharpe proxy for this robust conic allocation: it regularizes $\widehat{\mu}_t$, diagonalizes residual covariance, scales shrinkage with N, and binds net liquidity caps before any performance number is reported. Appendix A.8 records the propositions that justify the claim boundary: higher costs and tighter constraints cannot improve the objective, net caps are representation-invariant while split-leg caps are not, and historical sample means become dangerous when n/T is large.

If the feasible set is unconstrained and $\widehat{\Sigma}_{O,t}$ is positive definite, the maximum-Sharpe direction is proportional to $\widehat{\Sigma}_{O,t}^{-1}\widehat{\mu}_t$ (Proposition A.6). This is exactly the Markowitz tangency formula; the option work is in building the conditional option return objects the formula consumes. Because premium weights carry no unit-sum budget—cash is the numeraire, so $\mathbf{1}^\top q$ is unrestricted—the option-only efficient frontier itself degenerates from the familiar hyperbola to a ray.

Proposition 4.2 (The option-only frontier is a ray)**.** *Write Σ for $\widehat{\Sigma}_{O,t} \succ 0$ and $\mu \neq 0$ for $\widehat{\mu}_t$. For a target mean $m > 0$, the minimum-variance premium-weight vector subject to $q^\top \mu = m$ is*

$$q^*(m) = \frac{m\,\Sigma^{-1}\mu}{\mu^\top \Sigma^{-1}\mu}, \qquad \sigma(m) = \sqrt{q^*(m)^\top \Sigma\, q^*(m)} = \frac{m}{\sqrt{\mu^\top \Sigma^{-1}\mu}}.$$

The unconstrained frontier in (σ, m) space is the ray through the origin with slope $\text{SR}^* = \sqrt{\mu^\top \Sigma^{-1}\mu}$*, attained along the tangency direction $\Sigma^{-1}\mu$; the budget constraints in $\mathcal{K}_t$ select the admissible segment of the ray.*

Proof. For $\min_q q^\top \Sigma q$ subject to $\mu^\top q = m$, the first-order condition $2\Sigma q = \gamma\mu$ gives $q = \frac{\gamma}{2}\Sigma^{-1}\mu$, and the constraint pins $\frac{\gamma}{2} = m/(\mu^\top \Sigma^{-1}\mu)$; the objective is strictly convex, so this point is the unique minimizer, and substitution gives $\sigma(m)^2 = m^2/(\mu^\top \Sigma^{-1} \mu)$. Without a $\mathbf{1}^\top q = 1$ budget there is no second linear constraint to bend $m \mapsto \sigma(m)$ into a hyperbola: the map is linear through the origin, and every frontier portfolio has Sharpe ratio SR^*. □

The constrained problem inherits this geometry. The Sharpe objective is scale-free, so the implemented solver either maximizes the ratio directly over $\mathcal{K}_t$ under a gross-NAV normalization or solves the equivalent convex program of Proposition A.7 and rescales the solution to the gross budget.

Strategy	Predicted monthly vol	Train realized vol	Test realized vol	Test / predicted
Equity-option Greek Markowitz	0.334	0.267	0.307	0.918
Greek Markowitz + VIX	0.363	0.261	0.262	0.722
Beta/delta-neutral + VIX	0.354	0.272	0.239	0.677
Cost-aware Sortino + VIX	2.720	1.361	1.429	0.525
Equal premium	1.509	0.625	0.542	0.359
Equal risk	0.668	0.394	0.368	0.551
VIX hedge sleeve	4.648	4.092	0.417	0.090

Table 5: Greek-induced covariance calibration. Predicted monthly volatility is $\sqrt{q^\top \widehat{\Sigma}_{O,t}\, q}$ using the training-window Greek risk model; realized volatility is measured on train and test option returns.

The estimator follows the standard risk-model route, with two implementation simplifications stated exactly. First, the implemented $\widehat{\Omega}_t$ is block-diagonal across the underlying-return, centered squared-return, and volatility-shock blocks: cross-block covariances—including the spot–vol leverage covariance—are set to zero, and the squared-return block defaults to the joint-normal approximation $\mathrm{Cov}(R_u^2, R_v^2) = 2\, \mathrm{Cov}(R_u, R_v)^2$ when a direct estimate is not supplied. Whatever cross-block covariation this omits is left to $\widehat{\Sigma}_{\varepsilon,t}$ and the shrinkage target rather than modeled. Second, positive semidefiniteness can break at one specific estimation step: $\widehat{\Sigma}_{\varepsilon,t}$ is estimated from training residuals on an unbalanced contract panel using pairwise-complete moments, and pairwise-complete moment matrices need not be positive semidefinite. The estimate is therefore projected onto the PSD cone (eigenvalue clipping at a small nonnegative floor) before it enters $\widehat{\Sigma}_{O,t}$, and the combined matrix is shrunk toward a diagonal target; the factor term $\widehat{B}_t \widehat{\Omega}_t \widehat{B}_t^\top$ is PSD by construction and never needs repair (Proposition A.5). This follows the same concern that motivates covariance shrinkage in ordinary portfolio problems: noisy matrices become dangerous when the optimizer inverts them (Ledoit and Wolf, 2004).

5. Metrics and Validation

The validation design asks several different questions. The realized path asks what happened in the held-out monthly window. The point-in-time audit asks whether decisions used only information available at the rebalance date. The regression asks whether the option book is mostly equity drift or volatility exposure. The attribution asks whether P&L comes from delta, gamma, vega, carry, VIX-forward exposure, skew/tail terms, or residual marking error. Cost stress asks whether the gross result survives realistic research frictions.

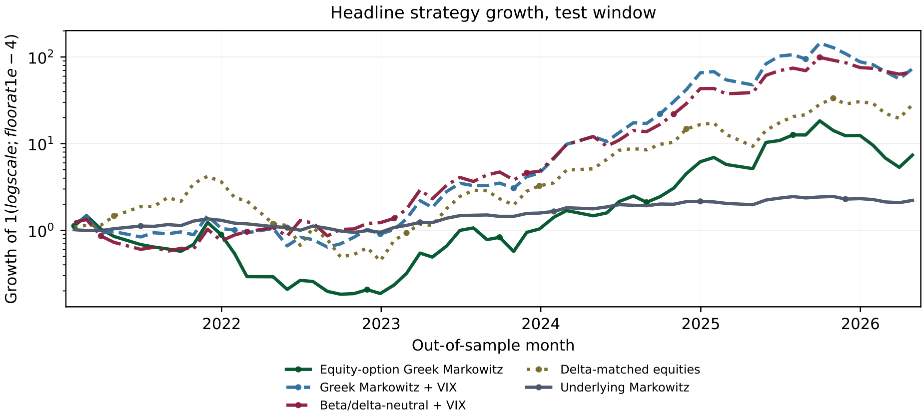

Figure 1: Headline out-of-sample growth paths. The figure focuses on the central option allocators and equity benchmarks; the all-strategy growth diagnostic appears in Appendix A. The log-scale floor prevents failed strategies from disappearing visually.

Performance is reported with complementary measures. Sharpe follows the reward-to-variability tradition (Sharpe, 1966). Sortino replaces total volatility with downside deviation (Sortino and Price, 1994). Calmar scales return by maximum drawdown (Magdon-Ismail and Atiya, 2004). Omega compares gains above a threshold with shortfalls below it (Keating and Shadwick, 2002). Information ratio measures active return against SPY (Grinold and Kahn, 2000). The zero monthly threshold is used for Sortino and Omega because cash is a numeraire in the model, not an alpha sleeve.

The inference layer is intentionally conservative. Sharpe, Sortino, Calmar, Omega, regime statistics, and leave-one-out differences are reported with block-bootstrap intervals where meaningful; the circular block bootstrap preserves short-run serial dependence in the spirit of the stationary bootstrap of Politis and Romano (1994). Factor alpha uses HAC/Newey–West standard errors (Newey and West, 1987). These intervals do not prove economic significance, but they prevent the paper from reading point estimates as certainties. The pre-production extension also reports probabilistic Sharpe ratios, deflated Sharpe ratios, and a block-bootstrap max-Sharpe screen across the tested gross and executable-cost variants. The headline estimator is stated formally so its units and conventions are not left implicit.

Definition 5.1 (Sharpe estimator and annualization)**.** Let $x_1, \ldots, x_n$ denote the monthly ledger-grid strategy returns: marks and expiry proceeds are struck on the monthly rebalance grid, and intra-month expiry proceeds are held at the financing rate $r_{f,t}$ until month end, so each observation spans exactly one month (the baseline sets $r_{f,t} = 0$; Remark 3.2). With sample mean $\bar{x} = n^{-1} \sum_t x_t$ and standard deviation $s_x = \left[(n-1)^{-1} \sum_t (x_t - \bar{x})^2\right]^{1/2}$, the reported annualized Sharpe ratio is

$$\widehat{\mathrm{SR}} = \sqrt{12}\, \frac{\bar{x}}{s_x}.$$

Three reporting conventions follow from Definition 5.1 and prevent misreadings of the more extreme table entries. First, annualized returns are arithmetic: the monthly mean is scaled by twelve, not compounded, so annualized returns below -100% per year are possible and several appear in the conservative stress-cost layer of Table 6. Second, compounded wealth paths are floored at zero, an absorbing default state: once a path loses its NAV it stays at zero rather than going negative or rebounding. The always-long VIX hedge sleeve hits this floor in the realized sample, which is why its maximum drawdown and Calmar entries are undefined ("–") in the gross diagnostics (Table 37). Third, the tail-path simulation tables report the share of simulated paths absorbed at zero as an explicit "defaulted path share" column (Table 43), so divergent sleeves appear as default frequencies rather than as astronomical terminal-wealth statistics. The predicted-versus-realized volatility calibration of the Greek risk model itself is reported in Table 5.

The $\sqrt{12}$ scaling is exact only when monthly returns are serially independent; under serial dependence, square-root-of-time annualization can bias the ratio in either direction (Lo, 2002). Probabilistic and deflated Sharpe ratios follow Bailey and López de Prado (2014) and are computed in per-period (monthly) units. The deflation trial set for each variant is the strategy family evaluated at the *same* cost basis (gross, midpoint, half-spread, or full-spread), and the trial count is reported per row: pooling near-perfectly correlated cost replicas of the same strategy would mechanically inflate the trial variance and hence the deflation hurdle. The multiple-testing screen is a centered max-statistic block bootstrap in the spirit of the reality check of White (2000) and the superior-predictive-ability test of Hansen (2005): each variant's monthly returns are demeaned to impose the null of no skill, the maximum Sharpe statistic across the variant set is bootstrapped on circular blocks, and the headline variant's statistic is compared against that null distribution.

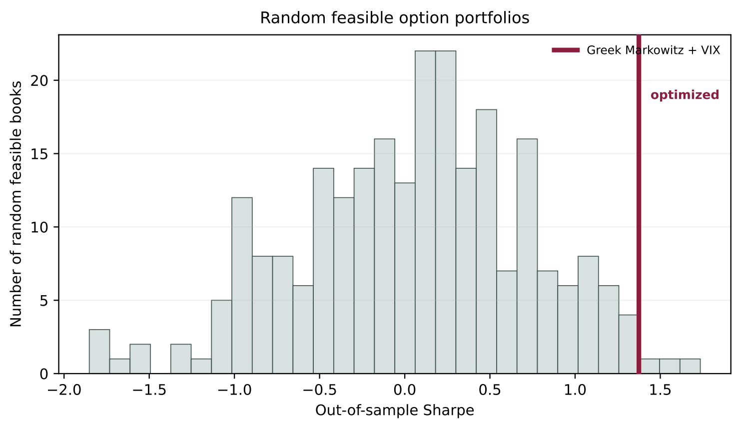

Figure 2: Random feasible comparison. The histogram shows out-of-sample Sharpe ratios for random gross-NAV-feasible option books; the vertical marker shows the optimized Greek-Markowitz-plus-VIX sleeve. The exhibit asks whether optimization adds information beyond arbitrary feasible option exposure.

The supplemental "rolling 36-month" diagnostic in the appendix (Table 51) is a genuine walk-forward exercise: at each evaluation date the representative contracts, conditional premia, and covariances are re-estimated on the trailing 36-month window ending at that date, and weights come from the same constrained maximum-Sharpe solver as the headline strategies.

An earlier draft of this table silently truncated every training window at the global 2020-12-31 split; the pipeline now threads explicit window bounds and asserts that no training observation postdates the window end. The breadth robustness runner applies the same rule to the production-candidate E1 books: CV, refit, and rolling rows rebuild representative contracts, mean and covariance estimates, residual covariance, and liquidity caps from each fold's training window before scoring the test months. Capacity-infeasible rows are kept as diagnostics and are not promoted to deployable headline books.

The point-in-time contract is strict. Decision dates precede return dates. State snapshots, Greeks, forecasts, covariances, residuals, betas, borrow proxies, and cost inputs are drawn from the training window or the decision date. Expiry spot, future IV, future VIX/VVIX, future VX curves, and future option returns are not allowed to enter portfolio construction. Cost-scenario ledgers, required-capital denominators, hurdle/no-trade records, liquidity-tier diagnostics, assignment-risk filters, and reality-check inputs obey the same rule. The audit layer reads the machine-readable outputs independently and fails if core lineage, settlement, cost, inference, claim, or manuscript checks fail.

6. Results and Discussion

The strongest result is conceptual rather than promotional: the option-only Markowitz object is coherent, auditable, and measurable on the held-out panel. The empirical investment result is more conditional. Gross option portfolios can look attractive, but gross Sharpe is a research diagnostic rather than a trading result. VIX options can improve the design as volatility instruments and hedge candidates only when exact VRO/SOQ settlement coverage supports the reported rows.

The body keeps the central evidence in view: out-of-sample performance in Table 6, post-cost survival in Table 9, factor exposure in Table 10, the realized path in Figure 1, the random feasible comparison in Figure 2, the VIX settlement gate in Table 3, claim strength in Table 11, and the claim audit in Table 12. Detailed exposure, liquidity, attribution, regime, forecast-ablation, simulation, and multiple-testing diagnostics appear in Appendix A.

6.1. Headline evidence

Return basis	Strategy	Ann. return	Ann. vol	Sharpe	Sortino	Calmar	Omega	Info. ratio
Gross before costs	Equity-option Greek Markowitz	0.894	1.062	0.842	1.770	1.020	1.888	0.598
Gross before costs	Greek Markowitz + VIX	1.246	0.907	1.374	3.385	2.034	2.784	1.200
Gross before costs	Beta/delta-neutral + VIX	1.172	0.829	1.414	3.410	2.057	2.948	1.147
Gross before costs	Cost-aware Sortino + VIX	5.020	4.949	1.014	3.349	4.993	2.471	0.971
Gross before costs	Equal premium	-0.056	1.878	-0.030	-0.055	-0.056	0.977	-0.120
Gross before costs	Equal risk	0.104	1.275	0.082	0.146	0.107	1.064	-0.058
Gross before costs	VIX hedge sleeve	-9.285	1.444	-6.429	-3.089		0.040	-6.132
Gross before costs	Delta-matched equities	1.095	0.931	1.176	2.317	1.225	2.367	1.041
Gross before costs	Underlying Markowitz	0.179	0.196	0.909	1.856	0.590	1.929	0.072
Post-cost research	Equity-option Greek Markowitz	-7.900	5.476	-1.443	-1.347	-6.357	0.098	-1.545
Post-cost research	Greek Markowitz + VIX	-6.079	4.827	-1.259	-1.198	-5.136	0.116	-1.345
Post-cost research	Beta/delta-neutral + VIX	-3.896	2.841	-1.371	-1.300	-3.895	0.151	-1.508
Post-cost research	Cost-aware Sortino + VIX	-157.425	148.976	-1.057	-1.019	-1.985	0.016	-1.003
Post-cost research	Equal premium	-3.165	2.378	-1.331	-1.502	-3.051	0.359	-1.372
Post-cost research	Equal risk	-1.276	1.367	-0.934	-1.199	-1.276	0.505	-1.082
Post-cost research	VIX hedge sleeve	-14.364	4.209	-3.413	-2.445	-12.391	0.016	-3.357
Post-cost research	Delta-matched equities	1.095	0.931	1.176	2.317	1.225	2.367	1.041
Post-cost research	Underlying Markowitz	0.179	0.196	0.909	1.856	0.590	1.929	0.072

Table 6: Gross and stress-cost out-of-sample performance. Gross rows test the option-only Markowitz object before frictions. The "Post-cost research" rows apply the *conservative stress-cost layer*: full spread crossing on every leg, explicit fees, slippage, borrow, capacity penalties, simulated margin funding, and assignment-risk penalties, compounded monthly. This layer is a stress bound, not an execution estimate; the milder *executable-cost scenarios* appear in Table 9. Annualized returns are arithmetic monthly means scaled by twelve, so stressed rows can fall below -100% per year (Section 5 records the conventions).

Table 6 separates two questions. The gross rows ask whether the mathematical allocation object works on held-out option returns. The stress-cost rows ask whether the same object would still look attractive if every implementation friction were charged at its most conservative research setting simultaneously. Between those poles, Table 9 asks whether the result survives midpoint, half-spread, and full-spread executable-cost scenarios with required-capital denominators, no-fill diagnostics, and margin/stress-capital screens. Gross Sharpe is not treated as a trading result under any of these readings.

On the gross ledger, the three optimizer-based option sleeves earn annualized returns of 0.894, 1.246, and 1.172 (per unit of NAV) at annualized volatilities near 0.8–1.1: out-of-sample Sharpe ratios of 0.842 for the equity-option Greek Markowitz sleeve, 1.374 with the VIX overlay, and 1.414 for the beta/delta-neutral variant. The circular-block-bootstrap 90% confidence intervals are wide—$[0.083, 1.679]$, $[0.602, 2.222]$, and $[0.677, 2.220]$ respectively (Table 37 and the full ledger in Table 39)—so the point estimates should be read as one draw from a sixty-month window, not as stable population quantities. The naive allocations confirm that construction, not option exposure per se, drives the result: equal-premium and equal-risk books earn gross Sharpe ratios of -0.030 and 0.082, and the always-long VIX hedge sleeve loses its premium almost every month (Sharpe -6.429, maximum drawdown undefined because the compounded path absorbs at zero wealth).

The benchmark comparison is closer than the option-sleeve numbers alone suggest. The delta-matched equity benchmark, which holds the same underlying names at the sleeves' decision-date delta exposure without any options, earns a gross Sharpe of 1.176; the plain underlying Markowitz portfolio earns 0.909. The option sleeves' advantage over delta-matched equities is therefore roughly 0.2 Sharpe units at the point estimate, well inside the bootstrap intervals. The framework's contribution is the accounting and the risk decomposition, not a claim that these particular sleeves dominate their equity shadows.

6.2. Two cost layers

The paper reports two deliberately different post-cost accountings, and they must not be conflated. The *executable-cost scenarios* (Tables 9 and 27) charge entry frictions only—quoted spread at midpoint, half, or full crossing plus explicit fees—at an average monthly cost, under full-spread crossing, of roughly 0.7% to 3.2% of NAV for the option sleeves (about 8% to 38% annualized). These scenarios fail closed: a position rejected for missing quotes, capacity, or assignment risk forfeits its gross return contribution instead of keeping it, so a rejection can move net performance in either direction. The *conservative stress-cost layer* (the "Post-cost research" rows of Table 6, with components in Table 26) additionally charges round-trip spread on every rebalance leg, slippage, borrow, capacity penalties, simulated margin funding on gross margin of roughly 1 to 4.6 times NAV, and assignment-risk penalties. Its annualized cost drag is 5.1 to 8.8 NAV units per year for the optimizer sleeves—one to two orders of magnitude larger than the executable scenarios—which is why the same strategies show net Sharpe ratios of $+0.91$, $+0.29$, and $+0.68$ in Table 9 but -1.44, -1.26, and -1.37 (with annualized returns as extreme as -790% under the arithmetic-mean convention) in Table 6.

The reconciliation is a statement about claims, not about arithmetic. The executable-cost scenarios are the realistic post-cost evidence for a small, patient allocator paying quoted spreads once per month; under them the leading sleeves survive with full-spread Sharpe ratios of 0.913, 0.294, and 0.678, while the naive books and the hedge sleeve fail. Two features of these numbers deserve comment. The equity-option sleeve's net Sharpe exceeds its gross because the trades the fill screens reject were, on net, losers for that sleeve; and the VIX overlay pays the largest toll because its wing positions cross the widest spreads and hit capacity screens most often. The stress-cost layer is a bound: it shows that if margin funding, capacity, and round-trip frictions are all charged at conservative research settings, no option sleeve in this study survives. The realistic post-cost outcome likely lies between the two layers and is scenario-dependent; the claim audit (Table 12) therefore records post-cost viability as a conditional research finding rather than a robust result, and no wording in this paper should be read as post-cost survival under all cost assumptions.

6.3. Breadth and capacity diagnostic

The breadth experiment asks whether the weak large-universe result is a real economic failure or an implementation failure. The answer is mixed but useful. At \$1 million NAV, breadth can pay with VIX after two fixes are applied together: the historical-mean component of the conditional return estimator is removed, and a pre-trade liquidity cap binds net contract exposure. With VIX included, the source-audited 57-group diagnostic improves from the paper configuration's net Sharpe of -1.837 to $+1.499$ when the combined regularized estimator uses diagonal residual covariance, N-scaled covariance shrinkage, structural-only forecasts, and shrinkage-to-zero 0.75. The corresponding gross Sharpe is 1.915, and the optimizer beats the best capped-naive book by 1.232 net Sharpe. Without VIX the 56-name book also turns net-positive, but capped equal-risk naive reaches 0.550 net Sharpe against the optimizer's 0.551, so the no-VIX optimizer edge is economically negligible. The net cells are calibrated execution-sensitivity estimates: they use point-in-time inferred CBBO proxy rows wherever historical bid/ask CBBO is absent, not the stale off-hours Cboe delayed-chain file or the old 10%/15% class defaults.

This caveat does not apply to the no-VIX eight-name equity baseline. The `orig` rows use measured historical panel CBBO for all equity-option spread inputs: 5,777 cost rows across 49 asset IDs and the eight baseline underlyings, with median relative spread about 1.89%. The `orig+VIX` rows use the same exact equity-option spread inputs for AAPL, AMZN, GOOGL, JPM, META, MSFT, NVDA, and TSLA; only the VIX option spread component remains proxied from the liquid-option CBBO distribution.

Config	Paper net	Capped-naive net	Capped-reg. net	Capped-reg. gross	Note
orig+VIX	-1.196*	-0.853*	1.287*	1.613	diagonal-residual/N-scaled row
larger+VIX	-1.837*	0.266*	1.499*	1.915	optimizer edge 1.232
orig	-1.443	0.350	0.720	0.961	cap budget below one NAV
larger	-1.991*	0.550*	0.551*	0.810; 0.924 gross-max	capped naive tie

Table 7: Breadth/capacity decision table at \$1 million NAV with impact costs on. "Capped-reg." denotes the regularized Greek-Markowitz optimizer with $X = 0.05$ pre-trade liquidity caps. "Capped-naive" is the best equal-premium/equal-risk book after applying the same cap vector at $X = 0.05$. The `orig` row is exact historical panel CBBO for equity-option spreads. Asterisks mark configs that use `inferred_cbbo_proxy` for VIX rows or for added-name equity-option rows without historical panel CBBO. The proxy is point-in-time: each row is calibrated from historical liquid equity/ETF CBBO observations available no later than the decision date. For the larger no-VIX configuration, the net-max regularized row has gross Sharpe 0.810, while the gross-max regularized row has gross Sharpe 0.924 and net Sharpe 0.530.

The P1 regularization sweep identifies the dominant problem. For the 57-group larger+VIX universe, dropping the historical mean and using structural-only forecasts raises gross Sharpe from 0.765 to 1.427. Residual-covariance regularization is the second lever: diagonal or Ledoit–Wolf residual covariance reaches roughly 1.2–1.24, whereas plain covariance shrinkage alone plateaus near 1.0. The pre-specified combination—diagonal residual covariance, N-scaled covariance shrinkage, historical-weight zero, and shrinkage-to-zero 0.75—reaches 1.501, above the 9-group VIX gross bar of 1.374. The same mechanism is not a free lunch: without VIX, the 56-name best gross Sharpe is 0.834 against the 8-name bar of 0.842, and at eight names removing the historical mean lowers gross Sharpe from 0.842 to 0.725. The training mean is signal when T/n is healthy and noise when breadth makes the inverse problem thin.

The P2 liquidity sweep shows that the post-hoc capacity penalty was a symptom rather than the disease. Turning it into a hard pre-trade cap of the form $\min\{0.18, X \cdot \widehat{V}_i^{\text{train}} \cdot M_i \cdot 100/\text{NAV}\}$ raises the original VIX-enabled paper configuration from net Sharpe -1.196 to $+1.080$ at $X = 0.05$, while cutting maximum capacity ratio from 187.8 to 32.6. For the larger+VIX default-estimator book, the same cap already reaches $+1.109$ net Sharpe at \$1 million under the inferred spread proxy; the estimator fix in P3 raises the best capped net result further to $+1.499$ and supplies the cleaner breadth mechanism. Increasing participation to $X = 0.10$ deploys more notional but earns less in this sample, so the combined experiment uses $X = 0.05$. The formulation detail matters: the cap is on the net position, not on the separate long and short legs used internally by the solver. A split-leg cap can burn offsetting exposure as pseudo-cash and consume good contracts' capacity; binding $|q_i|$ removes that degeneracy and leaves non-binding cases unchanged.

Scale remains the negative result. At $5 million and above, the sum of train-volume liquidity caps falls below one NAV for every configuration at the tested participation levels, so a fully deployed option-only book is no longer feasible in these representative month-end contracts. Relaxed positive-Sharpe rows beyond that scale deploy only a few basis points of NAV and are economically vacuous. Breadth raises the cap budget by roughly 40%, not by the 6x implied by the underlying count, because many of the added names' eligible options are thin. The breadth result is therefore a small-account diagnostic: regularization plus net liquidity caps can turn the $1 million larger+VIX book from negative to strongly positive net Sharpe under the inferred liquid-option spread proxy, but the result remains a research simulation rather than broker-executed evidence. Market depth rather than optimizer form still blocks scaling.

Three caveats travel with Table 7. First, large-universe net Sharpe still uses inferred spread rows wherever historical CBBO is missing. The proxy is calibrated from liquid equity/ETF CBBO buckets and is much more realistic than the old 10%/15% defaults or stale off-hours Cboe snapshots, but it is not matched historical NBBO/CBBO for every added name and VIX option. Gross Sharpe and capacity arithmetic are cleaner because they do not depend on this spread proxy. Second, the best knob is selected in-sample from a 22-point diagnostic grid, although the selected mechanism is consistent with the pre-registered failure modes. Third, the P1–P3 decision table itself uses the same single static train/test split and static-weight protocol as the headline paper; the follow-on robustness runner below is what tests the locked E1 candidate across alternate folds, resamples, synthetic worlds, and rolling refits.

config	verdict	deployable	net_sharpe	net_sortino	cpcv_net_p05	cpcv_net_p50	mc_resampled_net_p05	mc_refit_net_p05	rolling_net_sharpe
orig	diagnostic_capacity_infeasible	False	0.720	1.629	-0.171	-0.134	0.140	0.540	0.584
orig+VIX	pass	True	1.287	3.256	-0.147	-0.128	0.768	1.118	1.007
larger	mixed	True	0.551	1.196	-0.480	-0.349	-0.038	0.491	0.466
larger+VIX	pass	True	1.499	4.004	-0.541	-0.420	0.989	1.262	1.217

Table 8: Locked E1 breadth robustness summary. Every row uses the corrected full-cost stack, $X = 0.05$ liquidity caps, $1 million NAV, current-Cboe spread inputs disabled, and inferred CBBO proxy rows enabled where historical panel CBBO is missing. Deployable status is false when the full static hard-cap book cannot deploy. CPCV columns are diagnostic complete-path Sharpes, not tradable OOS returns.

Table 8 locks the E1 candidate rather than choosing knobs in the test folds. The full run uses 12 chronological groups, 66 CPCV splits, 78 total CV/PBO splits per configuration, purge/embargo of one month, 1,000 resampled paths, 200 refit paths, 1,000 repriced paths, tail-path simulations, drawdown breach rates, reality-check inference, and rolling 36-month monthly refits. The spread audit passes: zero rows use `current_cboe_liquid_quote` and zero rows use `default` spreads. The strongest result is still `larger+VIX`: net Sharpe 1.499, net Sortino 4.004, MC resampled p05 net Sharpe 0.989, MC refit p05 net Sharpe 1.262, and rolling net Sharpe 1.217. The `orig+VIX` book also passes (1.287 net Sharpe and 1.007 rolling net Sharpe). The no-VIX `larger` row is mixed because the realized and refit Sharpes are positive but weaker and the optimizer does not beat capped equal-risk by an economically meaningful margin, while the `orig` no-VIX row is diagnostic because the static cap budget is below one NAV. The CPCV complete-path net Sharpes are negative for all E1 rows even when gross CPCV medians are positive, which is a short-window cost-timing fragility rather than a claim that the fixed $1 million book is tradable. Repriced synthetic net rows are especially conservative and use a circular-block sample of realized full-cost drag; they are not synthetic NBBO/CBBO quotes.

6.4. Inference and multiple testing

The inference layer keeps the gross numbers honest. In per-month units, the probabilistic Sharpe ratios of the leading gross sleeves are high—0.979, 0.999, and 1.000 for the equity-option, VIX-overlay, and beta/delta-neutral variants—so each sleeve's Sharpe is individually distinguishable from zero at conventional levels (Table 40). Two multiple-testing corrections then remove most of that comfort. First, the centered reality-check family p-value across all evaluated variants is 0.078: the observed maximum studentized statistic of 3.161 sits below the bootstrap 95th percentile of 3.424, so the family-wise test fails the 5% hurdle. Second, the deflated Sharpe ratio is 0.000 for every gross variant and at most 0.51 across the executable-cost variants. After deflating by the dispersion of the evaluated variant set—which includes the -6.4-Sharpe hedge sleeve and therefore a very wide cross-sectional spread—no variant's Sharpe ratio is distinguishable from the expected maximum under the null of no skill. These corrections support, rather than undermine, the paper's restrained thesis: sixty months of data cannot certify skill for any variant in this family, and the paper does not claim otherwise.

Strategy	Gross Sharpe	Net Sharpe	Net Calmar	Avg spread cost	Capacity used	Survives?
Equity-option Greek Markowitz	0.8421194565895301	0.9134753475315867	0.9694779980961207	0.004454581057950615	0.9901002345178692	yes
Greek Markowitz + VIX	1.374388539995489	0.29431840025235295	0.2563821686403965	0.02189086570174581	0.99884496167564	yes
Beta/delta-neutral + VIX	1.4135317239912606	0.6779192503574191	0.8572440159658693	0.02646870683234331	0.9997072467259026	yes
Cost-aware Sortino + VIX	1.0144053186224946	0.046563442621254766	0.06800016356402226	0.011989850550872589	0.9984987277951501	yes
Equal premium	-0.02955676336526383	-0.15639381474734906	-0.1855093533148556	0.020340974748224574	0.9993803841618195	no
Equal risk	0.08183086586465663	-0.10651899845248258	-0.09449665379223078	0.01598309160474632	0.9990134907217686	no
VIX hedge sleeve	-6.429184989935395	-1.7767478772756302	-1.450654430103997	0.022500000000000003	0.9885078059495911	no

Table 9: Post-cost survival under the full-spread executable-cost scenario. This layer charges entry frictions only (quoted spread crossed in full plus fees); it is the milder of the two cost layers and is distinct from the conservative stress-cost layer of Table 6. A strategy survives only if net performance remains positive after these frictions together with capacity screens, settlement gates, and risk-capital checks.

6.5. *Factor exposure and benchmarks*

Strategy	Ann. alpha	Alpha HAC t	R^2	Residual ann. vol	Beta SPY	Beta VX front	Beta dVIX	Beta dVVIX
Equity-option Greek Markowitz	0.405	1.005	0.618	0.634	0.005	-1.368	0.046	0.003
Greek Markowitz + VIX	0.675	1.814	0.513	0.616	1.908	-0.767	0.056	-0.001
Beta/delta-neutral + VIX	0.845	2.425	0.429	0.601	0.715	-0.522	0.038	-0.001
Cost-aware Sortino + VIX	4.091	2.422	0.451	3.650	13.774	-4.302	0.195	0.008
Equal premium	-0.416	-0.499	0.253	1.647	2.522	-0.749	0.032	-0.001
Equal risk	-0.183	-0.294	0.253	1.115	2.041	-0.379	0.020	-0.003
VIX hedge sleeve	-8.807	-12.880	0.252	1.264	3.562	-1.180	0.093	-0.013
Delta-matched equities	0.000	1.990	1.000	0.000	-0.000	-0.000	-0.000	0.000
Underlying Markowitz	0.000	2.378	1.000	0.000	-0.000	-0.000	-0.000	0.000

Table 10: Headline factor regressions with HAC (Newey–West) standard errors. The table asks whether option returns are explained by equity drift or volatility state controls before any alpha language is considered; the full eight-underlying regression appears in Appendix A (Table 41). The two equity benchmark rows are exact linear combinations of the regressors ($R^2 = 1.000$, zero residual variance), so their alpha t-statistics are numerical artifacts and should be ignored.

The regression is an audit, not a causal proof. Against SPY, the eight underlying equities, VX front returns, and VIX/VVIX changes, the annualized alphas of the three optimizer sleeves are 0.405 (HAC $t = 1.005$), 0.675 ($t = 1.814$), and 0.845 ($t = 2.425$), with R^2 of 0.618, 0.513, and 0.429 and residual annualized volatility near 0.6—the factor block explains roughly half of the option-book variance and leaves an economically large unexplained remainder (Table 41). The t-statistics carry the small-sample $n/(n - k)$ HAC correction and are referred to a $t(n - k)$ distribution. Only the beta/delta-neutral variant clears a conventional two-sided 5% threshold, and none of these point estimates survives the family-wise corrections above, so the paper treats them as exposure diagnostics rather than alpha claims. The delta-matched-equities and underlying-Markowitz rows are spanned by the regressors by construction; their reported 0.000 alphas carry numerical-artifact t-statistics from dividing by near-zero standard errors and are excluded from any inference.

6.6. *Robustness reading*

Two robustness cuts summarize the appendix evidence. Leaving out any single underlying moves the VIX-enabled Greek Markowitz gross Sharpe within a narrow 1.311–1.476 band around the baseline 1.374, while excluding META, NVDA, and TSLA jointly lowers it to 1.115 (Table 50); the result is not a one-name artifact, but it does lean on the volatile mega-cap complex. Across SPY-return terciles, the VIX overlay earns its keep in down markets (Sharpe 0.888 down, 0.747 flat, 2.421 up, versus -0.973 down for the equity-option sleeve without the overlay), and the delta-matched equity benchmark loses in the same states (-2.753 down) (Table 46, Figure 10). The walk-forward rolling 36-month diagnostic re-estimates the whole stack each quarter and still finds a positive out-of-sample Sharpe (Table 51), though on only twenty evaluation months. The regime split is descriptive—terciles are formed on the realized sample—but it is the pattern a volatility-instrument overlay should produce.

6.7. Claim audit

Strength	Claim	Boundary
Strong	Option cashflow accounting, Greek-induced covariance, premium-weighted Markowitz formulation, and validation discipline are coherent.	Mathematical construction and reproducibility checks.
Moderate	The point-in-time research implementation produces useful gross and post-cost diagnostics in this sample.	Monthly OPRA/Databento-derived panels, exact-settlement rows, and research cost assumptions.
Weak/conditional	Convexity, VIX exposure, and short-premium sleeves help only in selected states after premium, skew, hedge cost, settlement, and risk-budget checks.	State- and settlement-dependent; exact VRO/SOQ rows present.
No claim	Live alpha, production tradability, executable capacity, broker-realistic fills, or live margin parity.	No broker-routed orders, live fills, live margin preview, or broker reconciliation.

Table 11: Claim-strength summary. The table distinguishes durable framework claims from sample-dependent diagnostics and from live-trading claims that are not made.

Claim	Type	Evidence	Status
Option covariance equals $B\Omega B^\top + \Sigma_\varepsilon$	Theorem	Displayed proof and standard covariance algebra	Proved
Unconstrained option tangency is $\Sigma_O^{-1}\mu_O$	Theorem	Displayed Lagrange multiplier proof	Proved
Greek Markowitz Sharpe exceeds random feasible p95	Generated empirical	performance table and random feasible Sharpe artifact	Not supported after exact-expiry PIT audit
Result is not only long-call equity drift	Generated diagnostic	exposure, regression, regimes, equity benchmarks, leave-one-out	Not supported as an alpha-independence claim
Option premium cost is included in P&L and Sharpe	Accounting identity	NAV weights, mark-return construction, exposure premium columns	Implemented by construction
Empirical returns use point-in-time option choices	Generated diagnostic	timing diagnostics, trading-data audit, and holding-return detail artifacts	Checked against train/test dates
Stock splits do not create option windfalls	Generated diagnostic	raw-close split factors and exact-expiry payoff construction	Checked for detected split/unit jumps
Greek risk model fully explains option P&L	Rejected overclaim	approximation diagnostics and P&L attribution artifacts	Not claimed
Post-cost research returns include implementation frictions	Generated empirical	Generated mid, half-spread, full-spread, fee, capacity, required-capital, and assignment-risk ledgers	Implemented as conservative research simulation
Pre-production results are broker-executed live evidence	Rejected overclaim	No live fills, order routing, broker margin preview, or broker reconciliation	Not claimed
VIX option results are headline-grade listed-settlement evidence	Conditional empirical	All VIX headline rows use exact VRO/SOQ source flags	Supported
Strategy is production tradable after costs	Production claim	No live fills, live broker margin preview, order routing, or broker reconciliation	Not claimed
VIX options are modeled as volatility derivatives, not equity underlyings	Modeling convention	VIX option panel uses VX-forward Black-76 Greeks and VIX/VVIX state variables	Implemented
VIX option expiry P&L is exact listed settlement P&L	Generated empirical	All VIX expiry rows use exact Cboe VRO/SOQ settlement	Supported

Table 12: Claim-audit summary. The table separates mathematical claims, empirical diagnostics, settlement-gated VIX claims, post-cost research claims, and production claims that are not claimed.

The claim audit is part of the result. The article can claim a mathematically explicit option-only efficient frontier and a point-in-time empirical implementation. It can report gross, exact-settlement VIX, and pre-production post-cost research diagnostics when the audit layer supports those outputs. It cannot claim production tradability, live fills, live broker margin parity, or executable capacity because those controls are not broker-run in this article.

7. Distributional robustness and cross-validation

7.1. Motivation and claim boundary

The headline design fixes a train/test split and asks whether the option-only optimizer, cost layers, and settlement gates survive a single frozen out-of-sample path. This section asks a different question: how fragile are those conclusions to alternative month orderings, refit samples, and synthetic repricing worlds? The answer is reported through blocked cross-validation, combinatorial purged cross-validation (CPCV), resampled universes, and repriced Monte Carlo universes. The tables carry the numerical evidence; the text here records the design and the claim boundary.

Honesty box

CPCV is deliberately non-PIT. By construction it trains some folds on observations after the fold being tested, so it is a distributional-robustness and overfitting diagnostic (path distribution and PBO), not a tradable out-of-sample claim. The headline claim boundary from the results and conclusion sections is unchanged.

The contract-universe eligibility filter is also fixed at the headline anchor: contracts must have at least 36 training observations through the global `TRAIN_END`. That same eligible universe is reused in every fold. This is benign for the diagnostic because the fold exercise is not a tradable signal, but it is disclosed because the fold membership is not re-estimated from each fold's own training set.

Per-fold CV refits cover the six option strategies that the option-contract evaluator can score: Equity-option Greek Markowitz, Greek Markowitz + VIX, Beta/delta-neutral + VIX, Equal premium, Equal risk, and the VIX hedge sleeve. Cost-aware Sortino + VIX is excluded from per-fold CV because its per-fold cost-aware solver is expensive; it is included in the fixed-weight Monte Carlo layer. Delta-matched equities and Underlying Markowitz remain headline-only in CV because their weight vectors are underlying weights, not option contract weights.

7.2. Blocked folds and CPCV design

The realized monthly return grid is split into 12 contiguous groups. The blocked k-fold diagnostic tests one group at a time. The CPCV diagnostic tests $k = 2$ groups at a time, giving $\binom{12}{2} = 66$ splits. Complete CPCV paths are then assembled so that each path includes each return month exactly once; with this design the complete-path count is $\binom{11}{1} = 11$, while incomplete paths are kept in the ledger but excluded from the path distribution.

The purge rule is intentionally simple and auditable. Option labels' payoffs realize at most 44 calendar days after the decision date. Each fold uses purge $= 1$ month and embargo $= 1$ month, which creates at least a two-snapshot train/test gap. In the shipped monthly grid that is at least 57 calendar days, and $57 > 44$, so the payoff window of a test label cannot overlap a training label kept by the fold. The verifier recomputes this schedule invariant from the fold ledger rather than trusting this paragraph.

Within each fold the option strategies are refit on the fold's training months and scored on the fold's test months. Strategy metrics use limited-liability absorption at zero, so a path that loses all wealth remains defaulted rather than sign-flipping. Table 13 reports the blocked k-fold refits, and Table 14 reports the CPCV path distribution against the headline realized path.

Fold	Equity-option Greek Markowitz	Greek Markowitz + VIX	Beta/delta-neutral + VIX	Equal premium	Equal risk	VIX hedge sleeve
kfold_00	-0.45 / -1.69	-0.78 / -2.21	-0.47 / -2.29	-0.13 / -0.91	0.23 / -0.99	-6.64 / -2.38
kfold_01	0.11 / -0.51	-2.53 / -1.02	-2.49 / -1.19	0.47 / 0.75	0.34 / 0.82	-0.60 / -1.56
kfold_02	3.37 / 2.04	2.42 / 1.89	2.07 / 1.44	-0.91 / 0.07	0.12 / 0.68	-2.27 / 0.00
kfold_03	-0.67 / -0.73	0.06 / -2.36	-1.10 / -2.72	0.35 / -0.06	0.62 / 0.12	0.04 / 0.36
kfold_04	0.90 / 0.69	-0.17 / -0.09	0.58 / 0.41	0.20 / -0.49	0.42 / -0.41	-1.03 / 0.23
kfold_05	2.66 / 2.16	1.42 / 0.13	0.91 / -0.26	2.15 / 1.93	2.69 / 2.21	0.72 / 0.96
kfold_06	0.69 / 0.56	1.24 / 0.61	1.06 / 0.16	-1.27 / -3.01	-1.49 / -2.58	-5.82 / -1.38
kfold_07	-0.73 / -1.49	-0.05 / -0.40	1.13 / 0.75	-0.16 / -0.85	-0.36 / -0.91	-4.70 / -1.63
kfold_08	1.53 / 2.37	1.25 / 1.70	1.60 / 2.70	0.96 / 0.22	1.19 / -0.05	-20.28 / -2.50
kfold_09	2.49 / 2.59	3.78 / 1.42	2.86 / 1.42	0.65 / 1.21	1.30 / 1.34	-19.89 / -1.42
kfold_10	1.53 / 1.29	2.00 / 1.07	2.29 / 1.24	-0.12 / -0.04	0.03 / -0.64	-4.36 / -2.71
kfold_11	-0.19 / -0.19	0.34 / 0.13	0.67 / -0.05	-0.56 / -0.02	-0.62 / -0.22	-5.40 / -1.85

Table 13: Blocked k-fold refit performance. Rows are diagnostics for refitted option strategies and do not alter the headline out-of-sample claim.

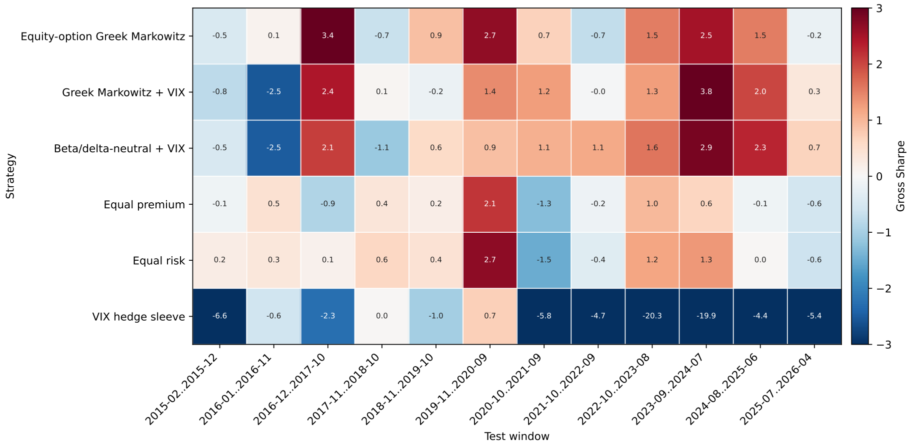

Figure 3: Blocked k-fold gross Sharpe by strategy and chronological test window. The color scale is clipped at ±3 so cross-fold variation in the equity-option strategies stays visible; cell annotations report unclipped values (the hedge sleeve reaches -20 in the 2022–2024 folds).

Strategy	Realized Gross Sharpe	CPCV Gross P05	CPCV Gross P50	CPCV Gross P95	Realized Full Spread Sharpe	CPCV Full Spread P05	CPCV Full Spread P50	CPCV Full Spread P95	PBO Gross	PBO Full Spread
Beta/delta-neutral + VIX	1.414	0.587	0.710	0.887	0.678	-0.032	0.046	0.165	0.192	0.205
Cost-aware Sortino + VIX	1.014				0.047				0.192	0.205
Delta-matched equities	1.176				1.176				0.192	0.205
Equal premium	-0.030	0.236	0.236	0.236	-0.156	0.156	0.156	0.156	0.192	0.205
Equal risk	0.082	0.410	0.465	0.499	-0.107	0.156	0.195	0.274	0.192	0.205
Equity-option Greek Markowitz	0.842	0.788	0.936	1.009	0.913	0.545	0.703	0.782	0.192	0.205
Greek Markowitz + VIX	1.374	0.602	0.766	0.954	0.294	-0.052	0.074	0.164	0.192	0.205
Underlying Markowitz	0.909				0.909				0.192	0.205
VIX hedge sleeve	-6.429	-0.404	-0.404	-0.404	-1.777	0.069	0.069	0.069	0.192	0.205

Table 14: CPCV complete-path distribution. The table compares realized-path metrics with the complete CPCV path distribution under gross and full-spread post-cost bases.

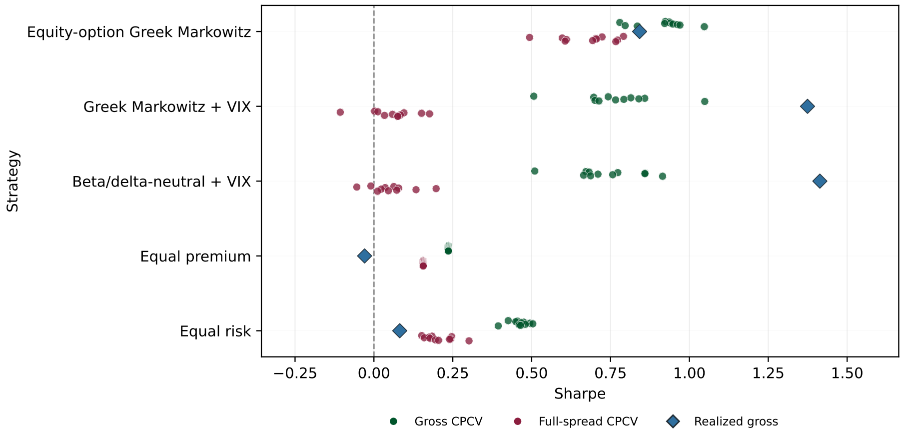

Figure 4: Complete-path CPCV Sharpe distributions with the realized static-split gross Sharpe marked for each CV strategy. The standalone VIX hedge sleeve (realized gross Sharpe -6.4) is omitted for scale; its fold-level values appear in Table 13 and Figure 3.

7.3. Probability of backtest overfitting

The combinatorially symmetric cross-validation (CSCV) probability of backtest overfitting (PBO) partitions the strategy set into in-sample and out-of-sample halves, selects the best in-sample strategy, and records the logit rank it attains out of sample. PBO is the fraction of splits with below-median out-of-sample rank. In this paper the gross variant selects on gross in-sample Sharpe and evaluates gross out-of-sample Sharpe. The full-spread variant selects on gross in-sample Sharpe and evaluates full-spread post-cost out-of-sample Sharpe, so it asks whether pre-cost selection survives a conservative executable-cost spread layer.

7.4. Regime-tagged performance

CPCV path months are also tagged by VIX terciles and by three named event windows: Volmageddon in February–March 2018, the COVID crash in February–April 2020, and the 2022 tightening bear market. The regime table is descriptive; the labels summarize path outcomes and do not enter expected returns, constraints, or strategy selection.

Regime family	Strategy	Group	Metric	Estimate	CI lo	CI hi	N	Seed
VIX tercile	Beta/delta-neutral + VIX	High VIX	Sharpe	0.704	0.207	1.653	43	20260625
VIX tercile	Beta/delta-neutral + VIX	Low VIX	Sharpe	0.625	-0.706	1.781	43	20260625
VIX tercile	Beta/delta-neutral + VIX	Mid VIX	Sharpe	0.986	0.169	1.795	43	20260625
VIX tercile	Equal premium	High VIX	Sharpe	0.529	-0.905	1.399	43	20260625
VIX tercile	Equal premium	Low VIX	Sharpe	0.365	-0.263	0.898	43	20260625
VIX tercile	Equal premium	Mid VIX	Sharpe	-0.304	-1.046	0.178	43	20260625
VIX tercile	Equal risk	High VIX	Sharpe	0.471	-0.962	1.368	43	20260625
VIX tercile	Equal risk	Low VIX	Sharpe	0.843	0.319	1.398	43	20260625
VIX tercile	Equal risk	Mid VIX	Sharpe	0.066	-0.491	0.494	43	20260625
VIX tercile	Equity-option Greek Markowitz	High VIX	Sharpe	0.041	-1.039	0.977	43	20260625
VIX tercile	Equity-option Greek Markowitz	Low VIX	Sharpe	2.560	1.954	3.218	43	20260625
VIX tercile	Equity-option Greek Markowitz	Mid VIX	Sharpe	0.871	0.304	1.411	43	20260625
VIX tercile	Greek Markowitz + VIX	High VIX	Sharpe	0.418	-0.260	1.202	43	20260625
VIX tercile	Greek Markowitz + VIX	Low VIX	Sharpe	1.065	-0.115	2.165	43	20260625
VIX tercile	Greek Markowitz + VIX	Mid VIX	Sharpe	1.031	0.186	1.855	43	20260625
VIX tercile	VIX hedge sleeve	High VIX	Sharpe	0.215	-6.796	0.734	43	20260625
VIX tercile	VIX hedge sleeve	Low VIX	Sharpe	-4.677	-7.507	-3.120	43	20260625
VIX tercile	VIX hedge sleeve	Mid VIX	Sharpe	-5.496	-8.274	-4.210	43	20260625
Event window	Beta/delta-neutral + VIX	2022 tightening bear	Sharpe	1.444	-0.062	3.644	9	20260625
Event window	Beta/delta-neutral + VIX	COVID crash	Sharpe				3	20260625
Event window	Beta/delta-neutral + VIX	Volmageddon Feb 2018	Sharpe				1	20260625
Event window	Equal premium	2022 tightening bear	Sharpe	-0.156	-3.237	1.003	9	20260625
Event window	Equal premium	COVID crash	Sharpe				3	20260625
Event window	Equal premium	Volmageddon Feb 2018	Sharpe				1	20260625
Event window	Equal risk	2022 tightening bear	Sharpe	-0.899	-3.112	0.310	9	20260625
Event window	Equal risk	COVID crash	Sharpe				3	20260625
Event window	Equal risk	Volmageddon Feb 2018	Sharpe				1	20260625
Event window	Equity-option Greek Markowitz	2022 tightening bear	Sharpe	-2.574	-5.433	-0.978	9	20260625
Event window	Equity-option Greek Markowitz	COVID crash	Sharpe				3	20260625
Event window	Equity-option Greek Markowitz	Volmageddon Feb 2018	Sharpe				1	20260625
Event window	Greek Markowitz + VIX	2022 tightening bear	Sharpe	-0.682	-1.219	-0.140	9	20260625
Event window	Greek Markowitz + VIX	COVID crash	Sharpe				3	20260625
Event window	Greek Markowitz + VIX	Volmageddon Feb 2018	Sharpe				1	20260625
Event window	VIX hedge sleeve	2022 tightening bear	Sharpe	-3.963	-5.637	-2.927	9	20260625
Event window	VIX hedge sleeve	COVID crash	Sharpe				3	20260625
Event window	VIX hedge sleeve	Volmageddon Feb 2018	Sharpe				1	20260625

Table 15: CPCV regime-tagged performance by VIX tercile and event window.

7.5. Resampled universes

The first Monte Carlo tier resamples month indices rather than individual strategy values: one circular block-bootstrap path is applied jointly to every strategy return frame, so cross-sectional dependence is preserved. The default fixed-weight tier uses 1,000 paths and evaluates the train-estimated portfolios. This measures robustness of the fixed portfolio actually used in the headline exercise.

A separate refit tier uses 200 pseudo-training paths for Greek Markowitz + VIX and then scores each refit on the fixed realized test window. This is a procedure and estimation-risk diagnostic, not a replacement for the fixed-weight path distribution. In that refit tier, SPY beta and stress augmentation use slot-calendar dates while the moment inputs are slot-relabeled resampled months. The approximation can affect constraint bounds only; it never enters expected returns or realized P&L.

Universe Family	Basis	Strategy	Realized Value	Path P05 Sharpe	Path P25 Sharpe	Path P50 Sharpe	Path P75 Sharpe	Path P95 Sharpe	Path P50 Max Drawdown	P Sharpe Less Than 0	P Max Drawdown Less Than -0.5	P Default
resampled	full_spread_post_cost	Beta/delta-neutral + VIX	0.678	-0.004	0.401	0.658	0.919	1.305	-0.513	0.051	0.535	0.000
resampled	full_spread_post_cost	Cost-aware Sortino + VIX	0.047	-0.817	-0.289	0.050	0.363	0.828	-1.000	0.459	1.000	0.616
resampled	full_spread_post_cost	Delta-matched equities	1.176	0.418	0.885	1.215	1.613	2.183	-0.774	0.007	0.902	0.000
resampled	full_spread_post_cost	Equal premium	-0.156	-0.796	-0.375	-0.158	0.038	0.291	-0.984	0.703	1.000	0.000
resampled	full_spread_post_cost	Equal risk	-0.107	-0.677	-0.339	-0.107	0.126	0.412	-0.914	0.621	0.996	0.000
resampled	full_spread_post_cost	Equity-option Greek Markowitz	0.913	0.160	0.614	0.894	1.175	1.562	-0.424	0.024	0.269	0.000
resampled	full_spread_post_cost	Greek Markowitz + VIX	0.294	-0.399	0.018	0.269	0.522	0.866	-0.655	0.236	0.858	0.000
resampled	full_spread_post_cost	Underlying Markowitz	0.909	0.133	0.604	0.939	1.276	1.763	-0.232	0.026	0.013	0.000
resampled	full_spread_post_cost	VIX hedge sleeve	-1.777	-2.275	-2.016	-1.830	-1.615	-1.279	-1.000	1.000	1.000	0.623
resampled	gross	Beta/delta-neutral + VIX	1.414	0.737	1.141	1.428	1.728	2.159	-0.547	0.000	0.585	0.000
resampled	gross	Cost-aware Sortino + VIX	1.014	0.431	0.782	1.006	1.248	1.574	-1.000	0.008	1.000	0.905
resampled	gross	Delta-matched equities	1.176	0.418	0.885	1.215	1.613	2.183	-0.774	0.007	0.902	0.000
resampled	gross	Equal premium	-0.030	-0.693	-0.263	-0.033	0.197	0.471	-1.000	0.546	1.000	0.000
resampled	gross	Equal risk	0.082	-0.602	-0.169	0.085	0.334	0.669	-0.982	0.409	1.000	0.000
resampled	gross	Equity-option Greek Markowitz	0.842	0.085	0.529	0.864	1.204	1.657	-0.840	0.038	0.983	0.000
resampled	gross	Greek Markowitz + VIX	1.374	0.643	1.068	1.391	1.721	2.236	-0.602	0.002	0.795	0.000
resampled	gross	Underlying Markowitz	0.909	0.133	0.604	0.939	1.276	1.763	-0.232	0.026	0.013	0.000
resampled	gross	VIX hedge sleeve	-6.429	-9.234	-7.291	-6.400	-5.730	-4.993	-1.000	1.000	1.000	1.000
resampled_stratified	full_spread_post_cost	Beta/delta-neutral + VIX	0.678	0.001	0.408	0.656	0.938	1.317	-0.584	0.049	0.703	0.000
resampled_stratified	full_spread_post_cost	Cost-aware Sortino + VIX	0.047	-0.665	-0.234	0.044	0.326	0.697	-1.000	0.463	1.000	0.691
resampled_stratified	full_spread_post_cost	Delta-matched equities	1.176	0.810	1.023	1.196	1.376	1.617	-0.832	0.000	0.991	0.000
resampled_stratified	full_spread_post_cost	Equal premium	-0.156	-0.848	-0.393	-0.163	0.032	0.297	-0.986	0.700	1.000	0.000
resampled_stratified	full_spread_post_cost	Equal risk	-0.107	-0.773	-0.350	-0.124	0.123	0.426	-0.923	0.639	0.998	0.000
resampled_stratified	full_spread_post_cost	Equity-option Greek Markowitz	0.913	0.315	0.667	0.915	1.136	1.440	-0.380	0.005	0.134	0.000
resampled_stratified	full_spread_post_cost	Greek Markowitz + VIX	0.294	-0.340	0.047	0.273	0.524	0.880	-0.673	0.222	0.881	0.000
resampled_stratified	full_spread_post_cost	Underlying Markowitz	0.909	0.472	0.748	0.929	1.089	1.325	-0.260	0.001	0.000	0.000
resampled_stratified	full_spread_post_cost	VIX hedge sleeve	-1.777	-2.139	-1.932	-1.797	-1.651	-1.427	-1.000	1.000	1.000	0.712
resampled_stratified	gross	Beta/delta-neutral + VIX	1.414	0.658	1.092	1.407	1.709	2.159	-0.546	0.001	0.644	0.000
resampled_stratified	gross	Cost-aware Sortino + VIX	1.014	0.539	0.837	1.022	1.196	1.454	-1.000	0.001	1.000	0.958
resampled_stratified	gross	Delta-matched equities	1.176	0.810	1.023	1.196	1.376	1.617	-0.832	0.000	0.991	0.000
resampled_stratified	gross	Equal premium	-0.030	-0.724	-0.296	-0.041	0.190	0.477	-1.000	0.544	1.000	0.000
resampled_stratified	gross	Equal risk	0.082	-0.567	-0.197	0.072	0.311	0.624	-0.985	0.432	1.000	0.000
resampled_stratified	gross	Equity-option Greek Markowitz	0.842	0.290	0.638	0.853	1.051	1.358	-0.789	0.006	0.983	0.000
resampled_stratified	gross	Greek Markowitz + VIX	1.374	0.696	1.086	1.373	1.657	2.103	-0.571	0.000	0.701	0.000
resampled_stratified	gross	Underlying Markowitz	0.909	0.472	0.748	0.929	1.089	1.325	-0.260	0.001	0.000	0.000
resampled_stratified	gross	VIX hedge sleeve	-6.429	-9.490	-7.568	-6.512	-5.739	-4.887	-1.000	1.000	1.000	1.000

Table 16: Fixed-weight resampled-universe diagnostics.

Strategy	Paths	OK Paths	Realized Sharpe	P05 Sharpe	P50 Sharpe	P95 Sharpe	P50 Max Drawdown	Median Gross NAV	Error Paths
Greek Markowitz + VIX	200	200	1.374	0.999	1.352	2.085	-0.638	1.000	0

Table 17: Refit-path stability diagnostic for Greek Markowitz + VIX.

7.6. Repriced synthetic universes

The second Monte Carlo tier builds a synthetic state world. Each spot-return, implied-vol level, and VIX-level dimension is fit with a GARCH(1,1) marginal when enough train-window observations are available; standardized innovations are then sampled jointly by circular block bootstrap. A Gaussian-copula sensitivity with 250 paths rank-maps innovations through the empirical marginal distributions. The default repriced stress tier uses 1,000 paths and the same fixed train-estimated strategy weights as the headline portfolio.

Synthetic contracts are one-step, one-month options: entry premiums and payoff horizons match. Equity options are priced with Black–Scholes at the simulated equity and IV state. VIX options are priced with Black–76 off the simulated VX-front state at entry and settled against the simulated VIX level at the next step. The generator's initial IV and VIX state levels use within-train forward/backward filling only (`loc[:TRAIN_END]`), so the initial state never crosses into the test window.

This repriced universe is a stress world, not a replay of market option premia. Entry premiums are model-priced from the simulated BS/Black–76 state, so the empirical variance-risk-premium wedge embedded in observed market premia is largely absent by construction. Strategies that harvest that premium should not be expected to retain their realized edge in this world. Rank changes versus the resampled universes are therefore evidence about premium-dependence, not evidence that the headline backtest failed.

The breadth validation in Section 6 repeats the same philosophy for the locked E1 production candidates. Its repriced net rows are even narrower in claim scope: they reprice gross synthetic option paths and then subtract a resampled historical full-cost drag. They are therefore a documented cost-overlay stress, not a synthetic NBBO/CBBO market.

Strategy	Method	Realized Sharpe	P05 Sharpe	P50 Sharpe	P95 Sharpe	P50 Max Drawdown	P Sharpe Less Than 0	P Default
Beta/delta-neutral + VIX	joint_garch_block	1.414	-3.748	-1.073	0.573	-1.000	0.879	0.741
Cost-aware Sortino + VIX	joint_garch_block	1.014	-1.712	0.829	1.949	-1.000	0.256	0.805
Equal premium	joint_garch_block	-0.030	-4.026	0.689	1.794	-1.000	0.357	0.000
Equal risk	joint_garch_block	0.082	-5.142	0.481	1.760	-1.000	0.428	0.000
Equity-option Greek Markowitz	joint_garch_block	0.842	-2.668	-0.541	1.095	-1.000	0.713	0.449
Greek Markowitz + VIX	joint_garch_block	1.374	-3.752	-0.919	0.809	-1.000	0.812	0.681
VIX hedge sleeve	joint_garch_block	-6.429	-4.691	-0.022	1.312	-1.000	0.508	1.000

Table 18: Repriced synthetic-universe diagnostics under the joint GARCH/block-bootstrap state model.

Strategy	Sharpe Realized	Resampled P05	Resampled P50	Resampled P95	Repriced P05	Repriced P50	Repriced P95
Beta/delta-neutral + VIX	1.414	0.737	1.428	2.159	-3.748	-1.073	0.573
Cost-aware Sortino + VIX	1.014	0.431	1.006	1.574	-1.712	0.829	1.949
Delta-matched equities	1.176	0.418	1.215	2.183			
Equal premium	-0.030	-0.693	-0.033	0.471	-4.026	0.689	1.794
Equal risk	0.082	-0.602	0.085	0.669	-5.142	0.481	1.760
Equity-option Greek Markowitz	0.842	0.085	0.864	1.657	-2.668	-0.541	1.095
Greek Markowitz + VIX	1.374	0.643	1.391	2.236	-3.752	-0.919	0.809
Underlying Markowitz	0.909	0.133	0.939	1.763			
VIX hedge sleeve	-6.429	-9.234	-6.400	-4.993	-4.691	-0.022	1.312

Table 19: Comparison of realized, resampled, and repriced-universe Sharpe distributions.

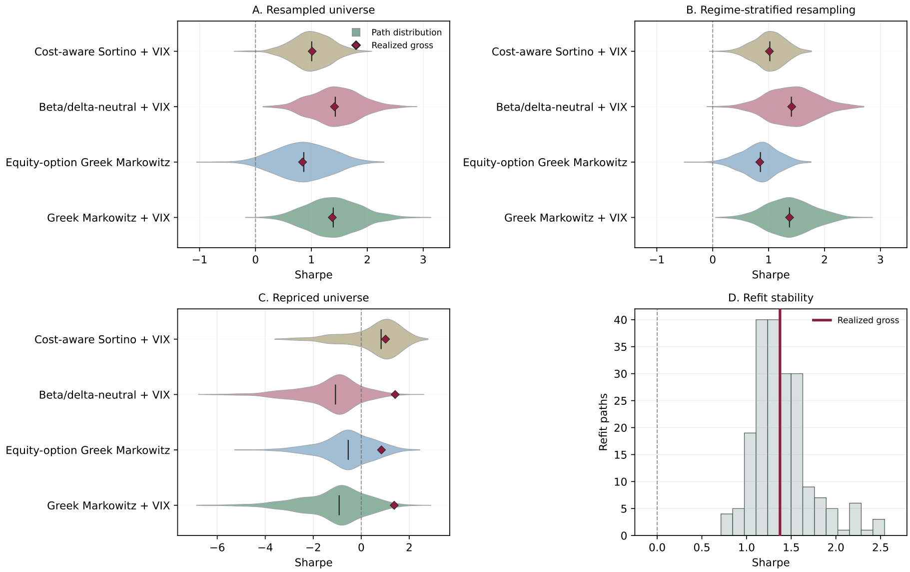

Figure 5: Monte Carlo Sharpe distributions across resampled, regime-stratified, repriced no-VRP stress, and refit-stability universes, with realized gross Sharpes marked where applicable.

7.7. Limitations

These diagnostics broaden the evidence surface but do not remove the central research limitations. CPCV is intentionally non-PIT. The eligible contract universe is fixed at the headline training anchor. Resampled universes preserve observed return rows rather than creating new contracts. Refit resampling uses a slot-calendar approximation for selected constraint inputs. The repriced universe simplifies contract geometry and removes much of the realized option-market premium wedge. For that reason, the section should be read as a robustness and failure-mode map around the headline point-in-time backtest, not as a stronger trading claim.

8. Conclusions

The durable contribution is the option cashflow and risk-accounting framework. A listed option enters the allocation problem only after it has been represented as a premium-funded cashflow, marked under a settlement convention, mapped into delta, gamma, vega, theta, VIX-forward exposure, and residual risk, and forecast under the physical measure. Once those objects exist, the Markowitz algebra is familiar: expected option cashflows and a Greek-induced covariance matrix replace ordinary asset means and covariances.

The validation layer supports a coherent point-in-time research implementation, not production tradability. Gross Sharpe, VIX hedge behavior, and option convexity are not enough. They must survive premium accounting, exact settlement controls, post-cost accounting at both the executable-cost and stress-cost layers, beta and Greek attribution, inference with family-wise corrections, leave-one-out tests, cross-validation and synthetic-universe robustness diagnostics, and no-lookahead checks. VIX options are useful in the framework because they are volatility instruments rather than equity substitutes, but their expiry P&L becomes headline-grade only when exact VRO/SOQ settlement supports the rows being reported.

The trader's use case is a disciplined screening rule rather than an automatic trading signal. A sleeve enters the book only when forecasted option premia survive premium cost, spread, settlement, and Greek-budget checks. Convexity is worth paying for only when state variables and hedge diagnostics justify the premium. Short option exposure is rejected or capped when stress loss, assignment risk, liquidity, or margin screens dominate expected carry. VIX options are treated as volatility instruments. When the validation layer cannot separate option premia from equity drift, volatility carry, or optimistic execution, the portfolio should sit in cash collateral, trade smaller, or use simpler diversifiers.

The breadth diagnostic adds a practical refinement. More names do not automatically save an option-only optimizer; with the baseline historical-mean blend, the 56-name universe mainly gives the inverse problem more noisy directions. Breadth begins to pay only after the mean estimator is made structural-only and the liquidity model caps net contract positions before trading. With VIX included, that combination moves the 57-group \$1 million book from -1.837 to $+1.499$ net Sharpe, with gross Sharpe 1.915, while beating the best capped-naive book by 1.232 net Sharpe. Without VIX, the 56-name optimizer reaches 0.551 net Sharpe, essentially tying capped equal-risk at 0.550, so optimizer edge is not established. Even the positive VIX result is small-account evidence. The locked E1 robustness layer strengthens but does not eliminate the caveats: `larger+VIX` passes the resampled, refit, simulation, reality-check, and rolling OOS screens most cleanly, while `larger` without VIX remains mixed and `orig` without VIX is diagnostic because the hard-cap budget is below one NAV. In the representative month-end contracts studied here, the cap budget supports roughly \$1–\$2 million of full deployment, not institutional scale.

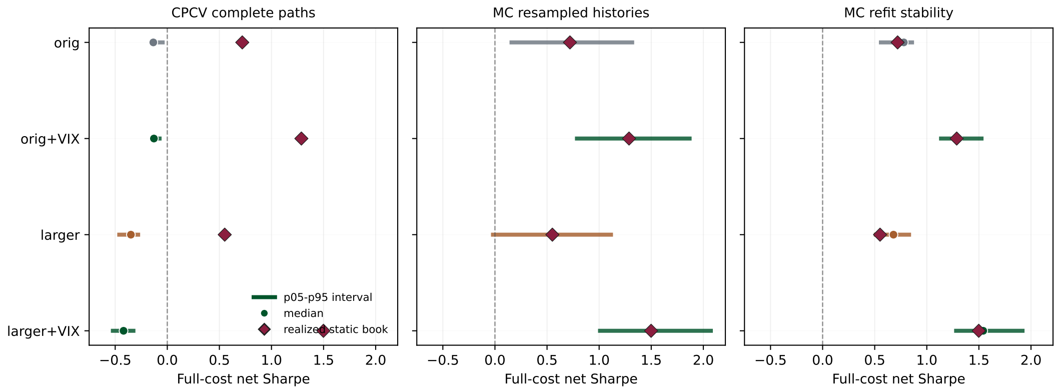

Figure 6: Final distributional validation for the four locked E1 configurations. Each panel reports full-cost net Sharpe distributions for the locked E1 book: the horizontal line is the 5th–95th percentile interval, the circle is the median, and the diamond is the realized static \$1 million book. CPCV complete paths are deliberately harsher because they stitch short cost-timing windows; the Monte Carlo resampled and refit panels show the stronger evidence for the VIX-enabled candidates.

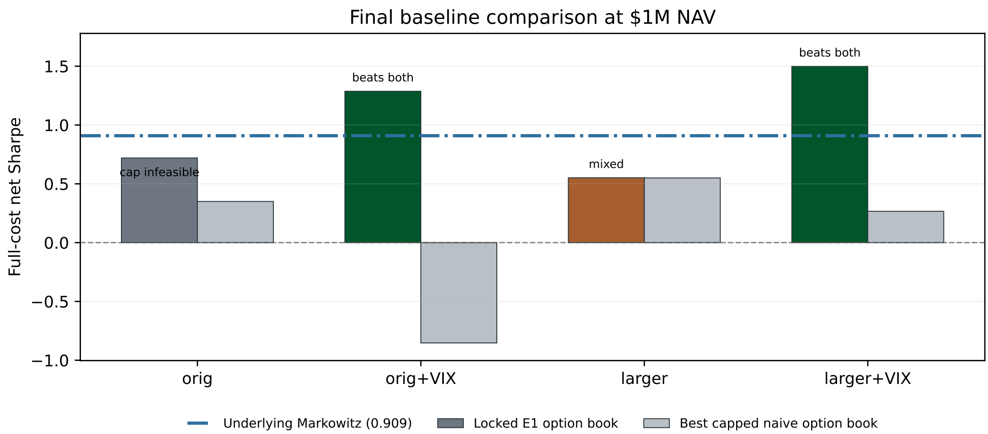

Figure 7: Final baseline comparison. The stock baseline is the ordinary underlying Markowitz portfolio on the same underlying equity universe. The naive option baseline is the better of capped equal-premium and capped equal-risk option allocations for each configuration. The VIX-enabled locked E1 books beat both baselines; the broad no-VIX book essentially ties the naive option book and remains below stock Markowitz; the original no-VIX row is diagnostic because full hard-cap deployment is infeasible.

Figures 6 and 7 are the paper's final empirical scoreboard. The theory earns its strongest empirical support only in the VIX-enabled books, where structural-only means, diagonal residual covariance, N-scaled shrinkage, and net liquidity caps jointly beat both a regular stock Markowitz baseline and naive capped option allocations. The no-VIX breadth result is positive but not strong enough to claim optimizer edge, and the no-VIX eight-name result is a capacity diagnostic rather than a deployable book.

This paper does not claim live-trading performance. The empirical sample is limited to the available OPRA/Databento-derived equity-option panels, VIX option chains, monthly snapshots, and exact VRO/SOQ coverage. Options execution, rates, margin, borrow, assignment, and capacity are modeled through conservative research assumptions rather than broker-level order and fill data. Production deployment would require richer quote histories, executable bid/ask depth, broker margin previews, assignment and exercise modeling, live routing, and reconciliation.

The immediate production-readiness step is forward shadow logging, not capital deployment. Each rebalance should export the locked E1 target contracts, collect market-hours NBBO/CBBO and displayed size, record marketable-limit fill assumptions, broker margin previews, manual or automated rejections, and reconcile positions and cash. Those shadow ledgers can reduce the execution caveat over time, but they remain shadow evidence until real broker orders, fills, margin, exercise/assignment, and statements reconcile.

The conclusion is therefore deliberately restrained. Build the option cashflow. Charge the premium and the implementation frictions. Estimate conditional premia before optimizing, and turn noisy historical means off when T/n becomes thin. Constrain Greeks, beta, VIX exposure, stress loss, and contract-level liquidity before trading, with caps on net positions rather than split solver legs. Use regressions, attribution, regimes, leave-one-out tests, breadth diagnostics, and inference to ask whether the result is more than equity drift, volatility carry, estimator noise, or optimistic execution. Reject any claim that depends on free options, future information, unpriced settlement, optimistic costs, hidden short-option risk, or option depth that the market did not actually offer.

Data availability statement

The reproduction package contains code, generated tables, figures, verification outputs, schemas, and artifact hashes. Raw licensed OPRA/Databento data are not redistributed. Users with the necessary data rights and API keys can rebuild the local feature panels with the documented data-pull commands and environment template. Exact VRO/SOQ files, when provided locally, are versioned by source hash before VIX option rows can enter settlement-gated tables.

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A. Appendix

Reproducibility

Run from the project root after configuring local data paths and credentials where needed:

```
make option-paper
make paper
make verify
make test
```

A.1. Exposures, attribution, and realized-path diagnostics

This first group records what the strategies actually held and where the P&L came from. Figure 8 keeps every generated strategy visible on one growth panel, Figure 9 compares predicted and realized volatility, Tables 20 and 21 record premium and Greek exposures, and Table 22 decomposes mean P&L into Greek and residual components.

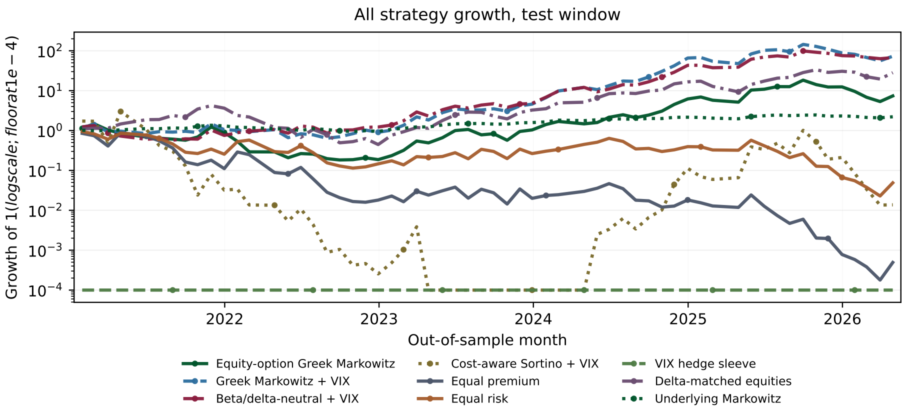

Figure 8: All-strategy out-of-sample growth diagnostic. The main text shows the central strategies; this appendix figure keeps every generated strategy visible for auditability.

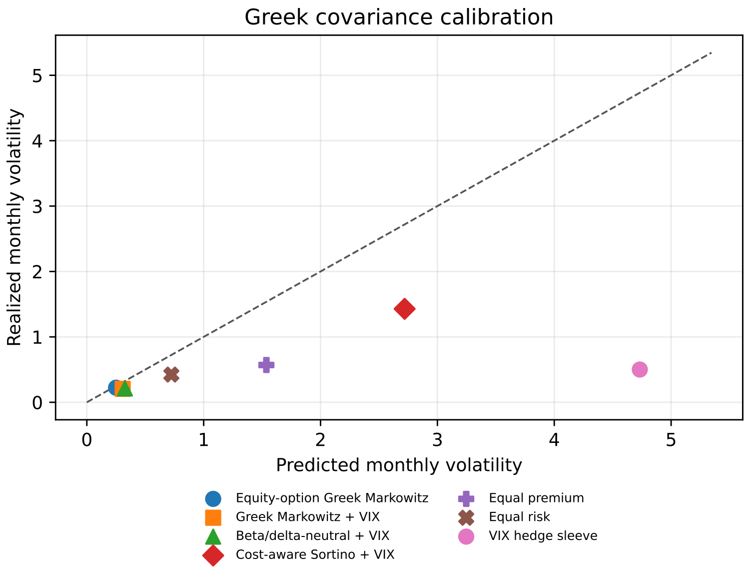

Figure 9: Predicted versus realized monthly volatility by option strategy.

Strategy	Book	Gross NAV	Net NAV	Long premium paid	Short premium sold	Net option premium	Short gross
Equity-option Greek Markowitz	Options	1.000	0.500	0.750	0.250	0.500	0.250
Greek Markowitz + VIX	Options	1.000	0.500	0.750	0.250	0.500	0.250
Beta/delta-neutral + VIX	Options	1.000	0.500	0.750	0.250	0.500	0.250
Cost-aware Sortino + VIX	Options	1.000	0.600	0.800	0.200	0.600	0.200
Equal premium	Options	1.000	1.000	1.000	0.000	1.000	0.000
Equal risk	Options	1.000	1.000	1.000	0.000	1.000	0.000
VIX hedge sleeve	Options	1.000	1.000	1.000	0.000	1.000	0.000
Delta-matched equities	Equities	3.307	3.307				0.000
Underlying Markowitz	Equities	1.000	0.539				0.231

Table 20: Premium and sign exposures. The table records how much option premium is paid, sold, and netted before the strategy return is realized.

Strategy	Book	Net delta	Gross delta	Net gamma	Net vega	Equity vega	VIX vega	Beta SPY proxy	Worst stress return	VIX option gross	Call gross	Put gross
Equity-option Greek Markowitz	Options	2.336	4.055	5.376	0.004	0.004	0.000	3.000	-0.207	0.000	0.682	0.318
Greek Markowitz + VIX	Options	3.307	3.307	5.029	-0.108	0.005	-0.113	2.342	-0.350	0.200	0.633	0.367
Beta/delta-neutral + VIX	Options	-0.086	3.893	2.186	-0.099	0.005	-0.104	-0.145	-0.010	0.200	0.588	0.412
Cost-aware Sortino + VIX	Options	6.288	16.065	129.870	-0.652	0.035	-0.687	3.000	-0.276	0.200	0.679	0.321
Equal premium	Options	1.208	2.202	359.583	0.211	0.041	0.170	0.897	0.258	0.093	0.537	0.463
Equal risk	Options	0.650	1.988	175.662	0.116	0.027	0.089	0.689	0.126	0.065	0.561	0.439
VIX hedge sleeve	Options	0.000	0.000	18.435	1.572	0.000	1.572	-23.137	-0.406	1.000	1.000	0.000
Delta-matched equities	Equities	3.307	3.307	0.000	0.000	0.000	0.000			0.000		
Underlying Markowitz	Equities	0.539	1.000	0.000	0.000	0.000	0.000			0.000		

Table 21: Greek, beta-proxy, VIX-vega, stress, and option-right exposures.

Strategy	Equity delta	Equity gamma	Equity vega	VIX-forward delta	VIX-forward gamma	VIX-option vega	Theta/carry	VX roll	Skew/tail	Residual	Realized ann. mean	Residual vol share
Equity-option Greek Markowitz	0.521	0.898	0.002	0.000	0.000	0.000	-0.003	0.000	0.000	-0.524	0.894	0.570
Greek Markowitz + VIX	0.489	1.475	0.001	0.767	-0.867	-0.227	1.057	0.000	0.000	-1.447	1.246	0.950
Beta/delta-neutral + VIX	0.168	1.486	0.001	0.554	-0.745	-0.190	1.034	0.000	0.000	-1.135	1.172	1.069
Cost-aware Sortino + VIX	2.064	8.577	0.004	0.285	-1.134	-0.112	2.169	0.000	0.000	-6.833	5.020	0.710
Equal premium	0.062	7.928	0.006	-0.016	0.358	0.045	-1.016	0.000	0.000	-7.423	-0.056	0.354
Equal risk	0.027	5.625	0.004	-0.096	0.270	0.049	-0.592	0.000	0.000	-5.184	0.104	0.514
VIX hedge sleeve	0.000	0.000	0.000	2.083	2.615	0.135	-12.254	0.000	0.000	-1.865	-9.285	4.858

Table 22: Annualized mean P&L attribution. Equity-option Greeks and VIX-forward Greeks are separated; components reconcile to realized mean after the residual term.

A.2. Costs, capacity, and settlement diagnostics

The second group makes timing, settlement, and cost conventions visible before any post-cost number is interpreted. Table 23 records return-construction timing, Table 24 audits the point-in-time holding model, and Table 25 documents which required VIX expiries have parsed exact VRO/SOQ settlement. The expiry-level exact-versus-proxy settlement comparison spans 110 expiries and is too long to typeset here; it ships as the machine-readable artifact `artifacts/vix_settlement_audit.csv` (mirrored in `tables/vix_settlement_audit.tex`). Table 26 itemizes the conservative stress-cost layer, Table 27 reports the executable-cost scenarios, and Tables 37 and 38 give the gross and stress-cost NAV diagnostics, including the bootstrap Sharpe confidence intervals quoted in Section 6.

Diagnostic	Value
Return construction	Prior-date option selection, split-adjusted listed-expiry payoff
Minimum option mark filter	0.25
Max train decision date	2020-11-30
Max train realization date	2020-12-31
First test decision date	2020-12-31
First test realization date	2021-01-29
Detected split/unit adjustments	7
Max single option return after repair	48.700
Min single option return after repair	-1.000
Expiry spot source	Raw daily close on listed expiry, or prior trading day if missing
VIX option return construction	Prior-date VIX option selection, VX-forward Greeks, expiry settlement proxy flagged
VIX settlement source	vro_soq_exact:536
VIX first test decision date	2020-12-31

Table 23: Timing and return-construction diagnostics.

Check	Value	Pass
OPRA-derived rows after liquidity filters	160,315	yes
Representative option choices	5,825	yes
Nonmissing option return cells	5,777	yes
Max OPRA spot vs raw close mismatch	2.129e-16	yes
All option choices made before payoff date	True	yes
All payoff dates no later than next rebalance	True	yes
Exact listed-expiry close share	1.000	yes
Max expiry-to-payoff-date lag in days	0	yes
Holding period days min/median/max	15/18.0/21	yes
Rows requiring split conversion during holding	13	yes
Detected raw-close split/unit events	7	yes
Minimum single-option return	-1.000	yes
Maximum single-option return	48.700	yes
Max training decision before first test decision	2020-11-30 before 2020-12-31	yes
Raw VIX option rows after contract parsing	103,650	yes
Representative VIX option choices	630	yes
VIX option holding ledger rows	536	yes
Duplicate VIX date-symbol rows after dedupe	0	yes
Black-76 VX-forward Greek coverage	1.000	yes
VIX settlement source	vro_soq_exact:536	yes
All VIX decisions before payoff dates	True	yes
Minimum VIX long-option return	-1.000	yes

Table 24: Trading-data audit for the point-in-time holding model. Rows include VIX option settlement-source flags; proxy rows, if present, remain diagnostic.

Year	Exact scalar parsed	SOQ component status	RequiredExpiries	FirstExpiration	LastExpiration
2015	False	downloaded	3	2015-12-09	2015-12-30
2015	True	downloaded	10	2015-02-18	2015-11-18
2016	False	unavailable	11	2016-05-04	2016-10-26
2016	True	downloaded	3	2016-01-20	2016-03-16
2016	True	unavailable	4	2016-04-20	2016-12-21
2017	False	unavailable	20	2017-02-15	2017-12-27
2017	True	unavailable	3	2017-01-18	2017-06-21
2018	False	unavailable	6	2018-01-03	2018-10-31
2018	True	unavailable	8	2018-02-14	2018-12-19
2019	True	downloaded	3	2019-10-16	2019-12-18
2019	True	unavailable	8	2019-01-16	2019-09-18
2020	True	downloaded	12	2020-01-22	2020-12-16
2021	True	downloaded	11	2021-01-20	2021-12-22
2022	True	downloaded	10	2022-01-19	2022-12-21
2023	True	downloaded	12	2023-01-18	2023-12-20
2024	True	downloaded	11	2024-01-17	2024-12-18
2025	True	downloaded	11	2025-01-22	2025-12-17
2026	True	downloaded	4	2026-01-21	2026-04-15

Table 25: Public Cboe VRO/SOQ download audit for required VIX expiries. Rows with `Exact scalar parsed = False` are not used in settlement-gated VIX option returns; the excluded expiries cluster in 2015–2018 (Section 2 discusses the resulting coverage bias).

Strategy	Mean monthly cost	Ann. cost drag	Spread share	Fee share	Mean margin/NAV	Mean stress margin/NAV	Mean assignment notional/NAV	Capacity penalized share
Beta/delta-neutral + VIX	0.422	5.068	0.099	0.012	4.029	0.235	0.000	0.273
Cost-aware Sortino + VIX	13.537	162.445	0.003	0.001	2.256	0.241	0.000	0.618
Equal premium	0.259	3.109	0.115	0.025	0.953	0.238	0.000	0.381
Equal risk	0.115	1.381	0.213	0.035	0.920	0.230	0.000	0.347
Equity-option Greek Markowitz	0.733	8.794	0.021	0.003	4.610	0.224	0.039	0.335
Greek Markowitz + VIX	0.610	7.325	0.069	0.008	3.526	0.234	0.000	0.290
VIX hedge sleeve	0.430	5.165	0.347	0.046	0.978	0.244	0.000	0.847

Table 26: Conservative stress-cost layer diagnostics. Costs include round-trip spread crossing, explicit fees, slippage, borrow, capacity penalties, simulated margin funding, and assignment-risk penalties where flagged; this is the layer behind the "Post-cost research" rows of Table 6, not the executable-cost scenarios of Table 27. These are not live broker fills.

Strategy	Scenario	Avg contracts traded	Avg quoted spread	Avg monthly cost	Max capacity used	Rejected trades	Mean margin/NAV	Mean stress capital/NAV
Beta/delta-neutral + VIX	full_spread	56.74896912971811	0.03509309915904966	0.03198047526748731	0.9997072467259026	867	1.6454288980146308	0.11046921803293545
Beta/delta-neutral + VIX	half_spread	56.74896912971811	0.03509309915904966	0.018746121851315658	0.9997072467259026	867	1.6454288980146308	0.11046921803293545
Beta/delta-neutral + VIX	mid	56.74896912971811	0.03509309915904966	0.005511768435143996	0.9997072467259026	867	1.6454288980146308	0.11046921803293545
Cost-aware Sortino + VIX	full_spread	486.94649539475853	0.0764124240818374	0.015719557112203447	0.9984987277951501	265	0.6759167177588942	0.07023236901345059
Cost-aware Sortino + VIX	half_spread	486.94649539475853	0.0764124240818374	0.00972463183676715	0.9984987277951501	265	0.6759167177588942	0.07023236901345059
Cost-aware Sortino + VIX	mid	486.94649539475853	0.0764124240818374	0.003729706561330857	0.9984987277951501	265	0.6759167177588942	0.07023236901345059
Equal premium	full_spread	65.36655726350517	0.03450510899698828	0.027393513458707507	0.9993803841618195	1330	0.5895061728395058	0.14737654320987645
Equal premium	half_spread	65.36655726350517	0.03450510899698828	0.017223026084595215	0.9993803841618195	1330	0.5895061728395058	0.14737654320987645
Equal premium	mid	65.36655726350517	0.03450510899698828	0.007052538710482932	0.9993803841618195	1330	0.5895061728395058	0.14737654320987645
Equal risk	full_spread	43.657154853888706	0.03490292057829765	0.021254428248565684	0.9990134907217686	1222	0.5170241047560368	0.1292560261890092
Equal risk	half_spread	43.657154853888706	0.03490292057829765	0.013262882446192524	0.9990134907217686	1222	0.5170241047560368	0.1292560261890092
Equal risk	mid	43.657154853888706	0.03490292057829765	0.005271336643819363	0.9990134907217686	1222	0.5170241047560368	0.1292560261890092
Equity-option Greek Markowitz	full_spread	18.797109490588205	0.017435798276330978	0.006679109598942776	0.9901002345178692	957	1.3950575743846914	0.07960287865721945
Equity-option Greek Markowitz	half_spread	18.797109490588205	0.017435798276330978	0.0044518190699674685	0.9901002345178692	957	1.3950575743846914	0.07960287865721945
Equity-option Greek Markowitz	mid	18.797109490588205	0.017435798276330978	0.002224528540992161	0.9901002345178692	957	1.3950575743846914	0.07960287865721945
Greek Markowitz + VIX	full_spread	49.294674731990554	0.035957144195152634	0.026579996078377945	0.99884496167564	898	1.5123553671138326	0.1014995147957134
Greek Markowitz + VIX	half_spread	49.294674731990554	0.035957144195152634	0.01563456322750504	0.99884496167564	898	1.5123553671138326	0.1014995147957134
Greek Markowitz + VIX	mid	49.294674731990554	0.035957144195152634	0.0046891303766321375	0.99884496167564	898	1.5123553671138326	0.1014995147957134
VIX hedge sleeve	full_spread	2410.1807707861826	0.15	0.07580327078625228	0.9885078059495911	153	0.14999999999999997	0.03749999999999999
VIX hedge sleeve	half_spread	2410.1807707861826	0.15	0.042053270786252275	0.9885078059495911	153	0.14999999999999997	0.03749999999999999
VIX hedge sleeve	mid	2410.1807707861826	0.15	0.008303270786252284	0.9885078059495911	153	0.14999999999999997	0.03749999999999999

Table 27: Executable-cost, capacity, and market-impact diagnostics by scenario (entry-friction layer). Rows are generated from the cost-scenario ledger and record no-fill/capacity failures rather than assuming midpoint execution.

The CBBO surface is an optional research-cost refinement: panel CBBO spreads take precedence, then exact decision-date surface spreads, then point-in-time inferred CBBO proxy spreads when explicitly enabled, and finally class-default spreads. The coverage table records which source supplied the relative spread by asset class.

The breadth/capacity diagnostic discussed in Section 6 is distributed as a separate artifact bundle under `analysis/artifacts/breadth_solutions/`. The key files are `p1_regularization_results.csv`, `p2_liquidity_results.csv`, `p2_caps_detail.csv`, `p3_combined_results.csv`, `p3_spread_source_coverage.csv`, `robustness/breadth_validation_summary.csv`, `robustness/breadth_cv_pbo_summary.csv`, `robustness/breadth_mc_refit_summary.csv`, and their Markdown/JSON companions. These files are diagnostic outputs, not headline paper tables: large-universe net rows carry the inferred-spread caveat where `inferred_cbbo_proxy` appears, and relaxed rows above the hard-cap capacity threshold report deployed gross exposure explicitly. Repriced synthetic net paths in the robustness folder use a realized-cost overlay, not synthetic NBBO/CBBO.

Strategy	Basis	Paths	P05 Sharpe	P50 Sharpe	P95 Sharpe
larger E1 capped	full_cost_net	11	-0.480	-0.349	-0.261
larger E1 capped	gross	11	1.091	1.147	1.218
larger Equal premium capped	full_cost_net	11	-0.854	-0.792	-0.578
larger Equal premium capped	gross	11	0.879	0.891	0.927
larger Equal risk capped	full_cost_net	11	-0.924	-0.887	-0.517
larger Equal risk capped	gross	11	0.934	0.958	0.998
larger GM paper	full_cost_net	11	-1.565	-1.370	-1.124
larger GM paper	gross	11	0.489	0.581	0.820
larger+VIX E1 capped	full_cost_net	11	-0.541	-0.420	-0.306
larger+VIX E1 capped	gross	11	0.739	0.814	0.864
larger+VIX Equal premium capped	full_cost_net	11	-0.847	-0.786	-0.596
larger+VIX Equal premium capped	gross	11	0.740	0.752	0.765
larger+VIX Equal risk capped	full_cost_net	11	-0.920	-0.871	-0.542
larger+VIX Equal risk capped	gross	11	0.809	0.843	0.878
larger+VIX GM paper	full_cost_net	11	-2.343	-1.866	-1.586
larger+VIX GM paper	gross	11	0.198	0.509	0.758
orig E1 capped	full_cost_net	11	-0.171	-0.134	-0.025
orig E1 capped	gross	11	0.921	0.967	1.043
orig Equal premium capped	full_cost_net	11	-0.192	-0.168	-0.138
orig Equal premium capped	gross	11	0.804	0.847	0.872
orig Equal risk capped	full_cost_net	11	-0.188	-0.168	-0.133
orig Equal risk capped	gross	11	0.805	0.854	0.889
orig GM paper	full_cost_net	11	-1.486	-1.279	-1.071
orig GM paper	gross	11	0.788	0.936	1.009
orig+VIX E1 capped	full_cost_net	11	-0.147	-0.128	-0.055
orig+VIX E1 capped	gross	11	0.778	0.840	0.877
orig+VIX Equal premium capped	full_cost_net	11	-0.320	-0.297	-0.279
orig+VIX Equal premium capped	gross	11	0.325	0.356	0.388
orig+VIX Equal risk capped	full_cost_net	11	-0.303	-0.288	-0.267
orig+VIX Equal risk capped	gross	11	0.349	0.408	0.473
orig+VIX GM paper	full_cost_net	11	-1.543	-1.211	-1.070
orig+VIX GM paper	gross	11	0.602	0.766	0.954

Table 28: Breadth robustness CPCV complete-path Sharpe distribution. The 12 blocked groups generate 66 CPCV splits and 11 complete paths per strategy/basis. These are diagnostics for overfitting and cost-timing fragility, not tradable OOS paths.

scope	config	Basis	N splits	N strategies	PBO	Median lambda	Rank correlation IS OOS
all_configs	all	full_cost_net	78	16	0.154	0.606	0.682
within_config	larger	full_cost_net	78	4	0.103	1.386	0.838
within_config	larger+VIX	full_cost_net	78	4	0.269	1.386	0.690
within_config	orig	full_cost_net	78	4	0.179	1.386	0.715
within_config	orig+VIX	full_cost_net	78	4	0.205	1.386	0.708
all_configs	all	gross	78	16	0.551	-0.118	0.159
within_config	larger	gross	78	4	0.603	-1.386	0.085
within_config	larger+VIX	gross	78	4	0.590	-0.405	0.021
within_config	orig	gross	78	4	0.474	0.405	0.262
within_config	orig+VIX	gross	78	4	0.295	0.405	0.402

Table 29: Breadth robustness probability of backtest overfitting summary. Each config uses 78 total CV/PBO splits: 12 blocked k-fold splits plus 66 CPCV splits.

Universe Family	Basis	Strategy	Realized Value	Path P05 Sharpe	Path P25 Sharpe	Path P50 Sharpe	Path P75 Sharpe	Path P95 Sharpe	Path P50 Max Drawdown	P Sharpe Less Than 0	P Max Drawdown Less Than -0.5	P Default
resampled	full_cost_net	larger E1 capped	0.551	-0.038	0.335	0.553	0.770	1.133	-0.704	0.060	0.935	0.000
resampled	full_cost_net	larger Equal premium capped	0.475	-0.053	0.285	0.470	0.673	0.919	-0.937	0.064	1.000	0.000
resampled	full_cost_net	larger Equal risk capped	0.550	0.024	0.342	0.541	0.752	1.016	-0.908	0.044	0.999	0.000
resampled	full_cost_net	larger GM paper	-1.991	-3.226	-2.246	-2.060	-1.919	-1.763	-1.000	1.000	1.000	1.000
resampled	full_cost_net	larger+VIX E1 capped	1.499	0.989	1.273	1.495	1.735	2.091	-0.428	0.000	0.272	0.000
resampled	full_cost_net	larger+VIX Equal premium capped	0.073	-0.507	-0.146	0.072	0.274	0.527	-0.992	0.410	1.000	0.000
resampled	full_cost_net	larger+VIX Equal risk capped	0.266	-0.315	0.031	0.265	0.478	0.730	-0.969	0.225	1.000	0.000
resampled	full_cost_net	larger+VIX GM paper	-1.837	-3.980	-2.087	-1.910	-1.813	-1.686	-1.000	1.000	1.000	1.000
resampled	full_cost_net	orig E1 capped	0.720	0.140	0.482	0.721	0.965	1.336	-0.679	0.020	0.873	0.000
resampled	full_cost_net	orig Equal premium capped	0.350	-0.143	0.144	0.343	0.557	0.817	-0.917	0.117	1.000	0.000
resampled	full_cost_net	orig Equal risk capped	0.327	-0.169	0.123	0.319	0.533	0.791	-0.918	0.134	1.000	0.000
resampled	full_cost_net	orig GM paper	-1.443	-2.245	-1.755	-1.548	-1.352	-1.151	-1.000	1.000	1.000	0.997
resampled	full_cost_net	orig+VIX E1 capped	1.287	0.768	1.070	1.291	1.520	1.887	-0.535	0.000	0.607	0.000
resampled	full_cost_net	orig+VIX Equal premium capped	-0.853	-1.524	-1.118	-0.886	-0.658	-0.352	-1.000	0.999	1.000	0.000
resampled	full_cost_net	orig+VIX Equal risk capped	-0.977	-1.654	-1.251	-1.004	-0.766	-0.464	-1.000	1.000	1.000	0.000
resampled	full_cost_net	orig+VIX GM paper	-1.196	-1.817	-1.428	-1.266	-1.099	-0.913	-1.000	1.000	1.000	0.997

Table 30: Breadth robustness resampled historical-universe diagnostics. Paths preserve joint monthly return rows and use the corrected full-cost ledger.

Return basis	Strategy	Requested method	Simulation	N paths	Defaulted path share	Ann. return p05	Ann. return p50	Sortino p50	Max DD p50	Terminal wealth p50
Gross before costs	orig E1 capped	circular_block_bootstrap	circular_block_bootstrap	1000	0.000	0.000	0.661	2.313	-0.614	12.653
Gross before costs	orig E1 capped	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1000	0.000	-0.121	0.483	1.847	-0.638	7.170
Gross before costs	orig+VIX E1 capped	circular_block_bootstrap	circular_block_bootstrap	1000	0.000	0.940	2.142	4.632	-0.456	306.240
Gross before costs	orig+VIX E1 capped	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1000	0.000	0.922	2.298	5.800	-0.434	390.332
Gross before costs	larger E1 capped	circular_block_bootstrap	circular_block_bootstrap	1000	0.000	-0.078	0.473	1.946	-0.608	6.929
Gross before costs	larger E1 capped	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1000	0.000	-0.372	0.108	0.883	-0.754	1.667
Gross before costs	larger+VIX E1 capped	circular_block_bootstrap	circular_block_bootstrap	1000	0.000	1.433	2.850	6.042	-0.362	845.857
Gross before costs	larger+VIX E1 capped	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1000	0.000	0.997	2.549	6.365	-0.400	563.366
Full-cost net	orig E1 capped	circular_block_bootstrap	circular_block_bootstrap	1000	0.000	-0.183	0.350	1.618	-0.679	4.481
Full-cost net	orig E1 capped	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1000	0.000	-0.272	0.209	1.170	-0.713	2.583
Full-cost net	orig+VIX E1 capped	circular_block_bootstrap	circular_block_bootstrap	1000	0.000	0.433	1.296	3.242	-0.535	63.753
Full-cost net	orig+VIX E1 capped	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1000	0.000	0.406	1.448	3.866	-0.516	88.000
Full-cost net	larger E1 capped	circular_block_bootstrap	circular_block_bootstrap	1000	0.000	-0.269	0.175	1.200	-0.704	2.244
Full-cost net	larger E1 capped	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1000	0.000	-0.500	-0.115	0.317	-0.852	0.541
Full-cost net	larger+VIX E1 capped	circular_block_bootstrap	circular_block_bootstrap	1000	0.000	0.695	1.659	4.028	-0.428	132.914
Full-cost net	larger+VIX E1 capped	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1000	0.000	0.388	1.447	4.009	-0.498	87.737

Table 31: Breadth robustness E1 tail-path simulations. The EGARCH request falls back to a fixed-parameter GARCH(1,1) residual bootstrap when the monthly sample is too short; the fallback is recorded in the simulation column.

config	strategy	display_strategy	basis	n_months	sharpe	sortino	ann_return	ann_vol	max_drawdown	terminal_wealth	defaulted
larger	larger E1 capped	E1 capped	gross	60	0.666	1.476	0.284	0.976	-0.880	3.489	False
larger	larger E1 capped	E1 capped	full_cost_net	60	0.466	0.958	0.052	0.977	-0.919	1.291	False
larger	larger Equal premium capped	Equal premium capped	gross	60	0.813	2.051	0.457	1.532	-0.812	6.555	False
larger	larger Equal premium capped	Equal premium capped	full_cost_net	60	0.637	1.489	0.098	1.532	-0.890	1.593	False
larger	larger Equal risk capped	Equal risk capped	gross	60	0.834	2.027	0.496	1.443	-0.838	7.505	False
larger	larger Equal risk capped	Equal risk capped	full_cost_net	60	0.651	1.472	0.138	1.443	-0.900	1.907	False
larger	larger GM paper	GM paper	gross	60	0.004	0.004	-1.000	1.198	-1.000	0.000	True
larger	larger GM paper	GM paper	full_cost_net	60	-0.681	-0.674	-1.000	43.477	-1.000	0.000	True
larger+VIX	larger+VIX E1 capped	E1 capped	gross	60	1.560	4.288	1.755	0.889	-0.504	158.719	False
larger+VIX	larger+VIX E1 capped	E1 capped	full_cost_net	60	1.217	2.924	1.060	0.891	-0.696	37.082	False
larger+VIX	larger+VIX Equal premium capped	Equal premium capped	gross	60	0.409	0.893	-0.202	1.485	-0.956	0.324	False
larger+VIX	larger+VIX Equal premium capped	Equal premium capped	full_cost_net	60	0.200	0.401	-0.433	1.485	-0.983	0.059	False
larger+VIX	larger+VIX Equal risk capped	Equal risk capped	gross	60	0.377	0.791	-0.198	1.374	-0.939	0.331	False
larger+VIX	larger+VIX Equal risk capped	Equal risk capped	full_cost_net	60	0.155	0.299	-0.425	1.374	-0.972	0.063	False
larger+VIX	larger+VIX GM paper	GM paper	gross	60	0.057	0.064	-1.000	1.164	-1.000	0.000	True
larger+VIX	larger+VIX GM paper	GM paper	full_cost_net	60	-1.523	-1.404	-1.000	6.269	-1.000	0.000	True
orig	orig E1 capped	E1 capped	gross	60	0.698	1.781	0.350	1.068	-0.827	4.484	False
orig	orig E1 capped	E1 capped	full_cost_net	60	0.584	1.402	0.192	1.067	-0.867	2.407	False
orig	orig Equal premium capped	Equal premium capped	gross	60	0.563	1.192	-0.118	1.579	-0.959	0.534	False
orig	orig Equal premium capped	Equal premium capped	full_cost_net	60	0.453	0.917	-0.278	1.578	-0.969	0.197	False
orig	orig Equal risk capped	Equal risk capped	gross	60	0.581	1.214	-0.086	1.546	-0.955	0.639	False
orig	orig Equal risk capped	Equal risk capped	full_cost_net	60	0.469	0.938	-0.250	1.545	-0.966	0.238	False
orig	orig GM paper	GM paper	gross	60	0.713	1.584	0.285	1.138	-0.864	3.502	False
orig	orig GM paper	GM paper	full_cost_net	60	-0.655	-0.650	-1.000	20.243	-1.000	0.000	True
orig+VIX	orig+VIX E1 capped	E1 capped	gross	60	1.221	3.116	1.333	1.181	-0.678	69.169	False
orig+VIX	orig+VIX E1 capped	E1 capped	full_cost_net	60	1.007	2.362	0.819	1.178	-0.800	19.927	False
orig+VIX	orig+VIX Equal premium capped	Equal premium capped	gross	60	-0.095	-0.164	-0.744	1.643	-1.000	0.001	False
orig+VIX	orig+VIX Equal premium capped	Equal premium capped	full_cost_net	60	-0.265	-0.432	-0.821	1.642	-1.000	0.000	False
orig+VIX	orig+VIX Equal risk capped	Equal risk capped	gross	60	-0.113	-0.187	-0.735	1.565	-0.999	0.001	False
orig+VIX	orig+VIX Equal risk capped	Equal risk capped	full_cost_net	60	-0.293	-0.458	-0.815	1.565	-1.000	0.000	False
orig+VIX	orig+VIX GM paper	GM paper	gross	60	0.979	2.114	0.640	0.811	-0.645	11.849	False
orig+VIX	orig+VIX GM paper	GM paper	full_cost_net	60	-1.327	-1.338	-1.000	1.527	-1.000	0.000	True

Table 32: Breadth robustness rolling 36-month OOS refits. Each monthly row rebuilds representative contracts, moments, and train-volume liquidity caps from its trailing window before scoring the next month.

Spread source	Asset class	Rows	Mean relative spread
default	VIX	536	0.150
panel_cbbo	equity	5777	0.026

Table 33: Cost-input spread-source coverage. Rows distinguish direct panel CBBO spreads, derived CBBO surface spreads, inferred CBBO proxy spreads where enabled, and class-default spread assumptions.

The repaired execution scenarios are execution-sensitivity diagnostics. They use the same decision-date cost-input row that fired the original gate; missing cost input and assignment or dividend hard gates are not repaired.

Scenario	Repaired orders	Partial fills	Avg extra cost per repaired order	Foregone gross from unfilled remainders
full_spread_repaired	4181	4181	0.000	-4.580
half_spread_repaired	4181	4181	0.000	-4.580
mid_repaired	4181	4181	0.000	-4.580

Table 34: Execution-repair diagnostics for `_repaired` scenarios. Counts and costs are diagnostics only and are not headline performance results.

The comparison table reports how repaired execution-sensitivity scenarios differ from the fail-closed cost scenarios. It is a sensitivity check around the execution gates, not a claim that the repaired fills are broker-executed.

Strategy	Scenario	Fail-closed Sharpe	Repaired Sharpe	Fail-closed ann. return	Repaired ann. return	Repair uplift (ann.)
Equity-option Greek Markowitz	mid	1.024	1.195	0.488	0.694	0.206
Greek Markowitz + VIX	mid	0.771	1.355	0.424	0.823	0.400
Beta/delta-neutral + VIX	mid	1.265	1.622	0.691	0.915	0.224
Cost-aware Sortino + VIX	mid	0.142	0.592	0.214	1.424	1.211
Equal premium	mid	0.052	0.017	0.061	0.025	-0.037
Equal risk	mid	0.137	0.143	0.108	0.139	0.031
VIX hedge sleeve	mid	-1.700	-4.256	-1.417	-4.277	-2.860
Equity-option Greek Markowitz	half_spread	0.969	1.122	0.462	0.652	0.191
Greek Markowitz + VIX	half_spread	0.534	1.051	0.293	0.639	0.347
Beta/delta-neutral + VIX	half_spread	0.971	1.275	0.532	0.721	0.189
Cost-aware Sortino + VIX	half_spread	0.094	0.540	0.142	1.298	1.156
Equal premium	half_spread	-0.052	-0.085	-0.061	-0.119	-0.058
Equal risk	half_spread	0.015	0.024	0.012	0.024	0.012
VIX hedge sleeve	half_spread	-1.742	-4.458	-1.552	-4.675	-3.123
Equity-option Greek Markowitz	full_spread	0.913	1.049	0.435	0.610	0.175
Greek Markowitz + VIX	full_spread	0.294	0.748	0.161	0.455	0.294
Beta/delta-neutral + VIX	full_spread	0.678	0.929	0.373	0.527	0.153
Cost-aware Sortino + VIX	full_spread	0.047	0.487	0.070	1.172	1.102
Equal premium	full_spread	-0.156	-0.188	-0.183	-0.263	-0.080
Equal risk	full_spread	-0.107	-0.094	-0.084	-0.091	-0.008
VIX hedge sleeve	full_spread	-1.777	-4.635	-1.687	-5.072	-3.385

Table 35: Fail-closed versus repaired execution-sensitivity performance. Repaired columns remain outside the headline growth exhibits and the reality-check family.

The cost-aware Sortino variant uses train-window entry-cost estimates before solving the Sortino objective. It is included as a research diagnostic and is not part of the headline strategy set or tail-path simulation set.

Status	Method	Objective	Net mean	Gross mean	Entry cost	Downside deviation	Net Sortino	Target	Scenarios	Downside-free	Train scenarios
optimal	linprog_net_mean_downside_free	max_sortino_net_of_costs	0.147	0.184	0.037	0.357	0.412	0.000	69	True	69

Table 36: Cost-aware Sortino objective diagnostics. Entry costs are estimated from the training window and imputed from class defaults when an asset has no train-window cost observation.

Strategy	Downside ann. dev	Max drawdown	Worst month	SR 90% CI lo	SR 90% CI hi	Pred./realized vol	Gross NAV	Net NAV
Equity-option Greek Markowitz	0.505	-0.876	-0.455	0.083	1.679	1.089	1.000	0.500
Greek Markowitz + VIX	0.368	-0.613	-0.371	0.602	2.222	1.385	1.000	0.500
Beta/delta-neutral + VIX	0.344	-0.570	-0.354	0.677	2.220	1.478	1.000	0.500
Cost-aware Sortino + VIX	1.499	-1.005	-1.214	0.466	1.568	1.904	1.000	0.600
Equal premium	1.004	-1.000	-0.662	-0.635	0.437	2.782	1.000	1.000
Equal risk	0.713	-0.976	-0.516	-0.558	0.627	1.814	1.000	1.000
VIX hedge sleeve	3.006		-1.000	-9.004	-4.921	11.150	1.000	1.000
Delta-matched equities	0.473	-0.894	-0.454	0.372	2.182		3.307	3.307
Underlying Markowitz	0.096	-0.303	-0.073	0.103	1.812		1.000	0.539

Table 37: Gross performance diagnostics and NAV accounting. The VIX hedge sleeve's maximum drawdown is undefined ("–") because its compounded wealth path is absorbed at zero (Section 5).

Strategy	Downside ann. dev	Max drawdown	Worst month	SR 90% CI lo	SR 90% CI hi	Pred./realized vol	Gross NAV	Net NAV
Equity-option Greek Markowitz	5.864	-1.243	-10.800	-2.209	-1.164		1.000	0.500
Greek Markowitz + VIX	5.074	-1.183	-9.226	-1.871	-0.994		1.000	0.500
Beta/delta-neutral + VIX	2.997	-1.000	-5.540	-1.961	-1.094		1.000	0.500
Cost-aware Sortino + VIX	154.544	-79.319	-320.564	-2.019	-0.961		1.000	0.600
Equal premium	2.107	-1.037	-2.431	-1.975	-0.816		1.000	1.000
Equal risk	1.065	-1.000	-0.974	-1.576	-0.486		1.000	1.000
VIX hedge sleeve	5.876	-1.159	-9.482	-7.714	-2.676		1.000	1.000
Delta-matched equities	0.473	-0.894	-0.454	0.372	2.182			
Underlying Markowitz	0.096	-0.303	-0.073	0.103	1.812			

Table 38: Stress-cost performance diagnostics and NAV accounting.

A.3. Inference and multiple testing

The third group carries the statistical evidence discussed in Section 6. Table 39 is the full bootstrap interval ledger, Table 40 reports per-variant probabilistic and deflated Sharpe ratios with the centered reality-check family statistics, and Table 41 is the complete factor-regression ledger behind the headline regression table.

Strategy	Metric	Estimate	CI lo	CI hi	N	Method	Block size	Seed	Return basis
Equity-option Greek Markowitz	Sharpe	0.842	0.083	1.679	60	circular block bootstrap	8	20260625	Gross before costs
Equity-option Greek Markowitz	Sortino	1.770	0.141	4.581	60	circular block bootstrap	8	20260625	Gross before costs
Equity-option Greek Markowitz	Calmar	1.020	0.083	2.828	60	circular block bootstrap	8	20260625	Gross before costs
Equity-option Greek Markowitz	Omega	1.888	1.064	3.546	60	circular block bootstrap	8	20260625	Gross before costs
Equity-option Greek Markowitz	Information Ratio	0.598	-0.430	1.571	60	circular block bootstrap	8	20260625	Gross before costs
Greek Markowitz + VIX	Sharpe	1.374	0.602	2.222	60	circular block bootstrap	8	20260625	Gross before costs
Greek Markowitz + VIX	Sortino	3.385	1.188	7.275	60	circular block bootstrap	8	20260625	Gross before costs
Greek Markowitz + VIX	Calmar	2.034	0.700	4.901	60	circular block bootstrap	8	20260625	Gross before costs
Greek Markowitz + VIX	Omega	2.784	1.562	5.251	60	circular block bootstrap	8	20260625	Gross before costs
Greek Markowitz + VIX	Information Ratio	1.200	0.347	2.129	60	circular block bootstrap	8	20260625	Gross before costs
Beta/delta-neutral + VIX	Sharpe	1.414	0.677	2.220	60	circular block bootstrap	8	20260625	Gross before costs
Beta/delta-neutral + VIX	Sortino	3.410	1.335	6.541	60	circular block bootstrap	8	20260625	Gross before costs
Beta/delta-neutral + VIX	Calmar	2.057	0.701	5.611	60	circular block bootstrap	8	20260625	Gross before costs
Beta/delta-neutral + VIX	Omega	2.948	1.693	5.333	60	circular block bootstrap	8	20260625	Gross before costs
Beta/delta-neutral + VIX	Information Ratio	1.147	0.332	1.962	60	circular block bootstrap	8	20260625	Gross before costs
Cost-aware Sortino + VIX	Sharpe	1.014	0.466	1.568	60	circular block bootstrap	8	20260625	Gross before costs
Cost-aware Sortino + VIX	Sortino	3.349	1.250	6.124	60	circular block bootstrap	8	20260625	Gross before costs
Cost-aware Sortino + VIX	Calmar	4.993	0.053	5.602	60	circular block bootstrap	8	20260625	Gross before costs
Cost-aware Sortino + VIX	Omega	2.471	1.510	4.021	60	circular block bootstrap	8	20260625	Gross before costs
Cost-aware Sortino + VIX	Information Ratio	0.971	0.373	1.518	60	circular block bootstrap	8	20260625	Gross before costs
Equal premium	Sharpe	-0.030	-0.635	0.437	60	circular block bootstrap	8	20260625	Gross before costs
Equal premium	Sortino	-0.055	-0.946	0.920	60	circular block bootstrap	8	20260625	Gross before costs
Equal premium	Calmar	-0.056	-1.051	0.849	60	circular block bootstrap	8	20260625	Gross before costs
Equal premium	Omega	0.977	0.624	1.416	60	circular block bootstrap	8	20260625	Gross before costs
Equal premium	Information Ratio	-0.120	-0.779	0.351	60	circular block bootstrap	8	20260625	Gross before costs
Equal risk	Sharpe	0.082	-0.558	0.627	60	circular block bootstrap	8	20260625	Gross before costs
Equal risk	Sortino	0.146	-0.821	1.305	60	circular block bootstrap	8	20260625	Gross before costs
Equal risk	Calmar	0.107	-0.667	0.954	60	circular block bootstrap	8	20260625	Gross before costs
Equal risk	Omega	1.064	0.662	1.621	60	circular block bootstrap	8	20260625	Gross before costs
Equal risk	Information Ratio	-0.058	-0.732	0.485	60	circular block bootstrap	8	20260625	Gross before costs
VIX hedge sleeve	Sharpe	-6.429	-9.004	-4.921	60	circular block bootstrap	8	20260625	Gross before costs
VIX hedge sleeve	Sortino	-3.089	-3.253	-2.904	60	circular block bootstrap	8	20260625	Gross before costs
VIX hedge sleeve	Calmar		-10.058	-8.262	60	circular block bootstrap	8	20260625	Gross before costs
VIX hedge sleeve	Omega	0.040	0.016	0.071	60	circular block bootstrap	8	20260625	Gross before costs
VIX hedge sleeve	Information Ratio	-6.132	-8.858	-4.691	60	circular block bootstrap	8	20260625	Gross before costs
Delta-matched equities	Sharpe	1.176	0.372	2.182	60	circular block bootstrap	8	20260625	Gross before costs
Delta-matched equities	Sortino	2.317	0.615	5.886	60	circular block bootstrap	8	20260625	Gross before costs
Delta-matched equities	Calmar	1.225	0.388	3.716	60	circular block bootstrap	8	20260625	Gross before costs
Delta-matched equities	Omega	2.367	1.301	5.182	60	circular block bootstrap	8	20260625	Gross before costs
Delta-matched equities	Information Ratio	1.041	0.189	2.072	60	circular block bootstrap	8	20260625	Gross before costs
Underlying Markowitz	Sharpe	0.909	0.103	1.812	60	circular block bootstrap	8	20260625	Gross before costs
Underlying Markowitz	Sortino	1.856	0.172	4.958	60	circular block bootstrap	8	20260625	Gross before costs
Underlying Markowitz	Calmar	0.590	0.045	2.544	60	circular block bootstrap	8	20260625	Gross before costs
Underlying Markowitz	Omega	1.929	1.074	3.877	60	circular block bootstrap	8	20260625	Gross before costs
Underlying Markowitz	Information Ratio	0.072	-0.796	0.866	60	circular block bootstrap	8	20260625	Gross before costs
Equity-option Greek Markowitz	Sharpe	-1.443	-2.209	-1.164	60	circular block bootstrap	8	20260625	Post-cost research
Equity-option Greek Markowitz	Sortino	-1.347	-1.895	-1.123	60	circular block bootstrap	8	20260625	Post-cost research
Equity-option Greek Markowitz	Calmar	-6.357	-11.009	-1.856	60	circular block bootstrap	8	20260625	Post-cost research
Equity-option Greek Markowitz	Omega	0.098	0.045	0.218	60	circular block bootstrap	8	20260625	Post-cost research
Equity-option Greek Markowitz	Information Ratio	-1.545	-2.532	-1.238	60	circular block bootstrap	8	20260625	Post-cost research
Greek Markowitz + VIX	Sharpe	-1.259	-1.871	-0.994	60	circular block bootstrap	8	20260625	Post-cost research
Greek Markowitz + VIX	Sortino	-1.198	-1.683	-0.971	60	circular block bootstrap	8	20260625	Post-cost research
Greek Markowitz + VIX	Calmar	-5.136	-8.117	-1.087	60	circular block bootstrap	8	20260625	Post-cost research
Greek Markowitz + VIX	Omega	0.116	0.054	0.269	60	circular block bootstrap	8	20260625	Post-cost research
Greek Markowitz + VIX	Information Ratio	-1.345	-2.133	-1.069	60	circular block bootstrap	8	20260625	Post-cost research
Beta/delta-neutral + VIX	Sharpe	-1.371	-1.961	-1.094	60	circular block bootstrap	8	20260625	Post-cost research
Beta/delta-neutral + VIX	Sortino	-1.300	-1.790	-1.073	60	circular block bootstrap	8	20260625	Post-cost research
Beta/delta-neutral + VIX	Calmar	-3.895	-6.033	-1.289	60	circular block bootstrap	8	20260625	Post-cost research
Beta/delta-neutral + VIX	Omega	0.151	0.075	0.321	60	circular block bootstrap	8	20260625	Post-cost research
Beta/delta-neutral + VIX	Information Ratio	-1.508	-2.254	-1.234	60	circular block bootstrap	8	20260625	Post-cost research
Cost-aware Sortino + VIX	Sharpe	-1.057	-2.019	-0.961	60	circular block bootstrap	8	20260625	Post-cost research
Cost-aware Sortino + VIX	Sortino	-1.019	-1.756	-0.933	60	circular block bootstrap	8	20260625	Post-cost research
Cost-aware Sortino + VIX	Calmar	-1.985	-4.376	-0.441	60	circular block bootstrap	8	20260625	Post-cost research
Cost-aware Sortino + VIX	Omega	0.016	0.007	0.032	60	circular block bootstrap	8	20260625	Post-cost research
Cost-aware Sortino + VIX	Information Ratio	-1.003	-2.017	-0.902	60	circular block bootstrap	8	20260625	Post-cost research
Equal premium	Sharpe	-1.331	-1.975	-0.816	60	circular block bootstrap	8	20260625	Post-cost research
Equal premium	Sortino	-1.502	-1.923	-1.066	60	circular block bootstrap	8	20260625	Post-cost research
Equal premium	Calmar	-3.051	-4.189	-1.543	60	circular block bootstrap	8	20260625	Post-cost research
Equal premium	Omega	0.359	0.206	0.547	60	circular block bootstrap	8	20260625	Post-cost research
Equal premium	Information Ratio	-1.372	-2.081	-0.833	60	circular block bootstrap	8	20260625	Post-cost research
Equal risk	Sharpe	-0.934	-1.576	-0.486	60	circular block bootstrap	8	20260625	Post-cost research
Equal risk	Sortino	-1.199	-1.714	-0.697	60	circular block bootstrap	8	20260625	Post-cost research
Equal risk	Calmar	-1.276	-1.962	-0.685	60	circular block bootstrap	8	20260625	Post-cost research
Equal risk	Omega	0.505	0.326	0.702	60	circular block bootstrap	8	20260625	Post-cost research
Equal risk	Information Ratio	-1.082	-1.764	-0.626	60	circular block bootstrap	8	20260625	Post-cost research
VIX hedge sleeve	Sharpe	-3.413	-7.714	-2.676	60	circular block bootstrap	8	20260625	Post-cost research
VIX hedge sleeve	Sortino	-2.445	-3.171	-2.131	60	circular block bootstrap	8	20260625	Post-cost research
VIX hedge sleeve	Calmar	-12.391	-15.461	-7.145	60	circular block bootstrap	8	20260625	Post-cost research
VIX hedge sleeve	Omega	0.016	0.004	0.032	60	circular block bootstrap	8	20260625	Post-cost research
VIX hedge sleeve	Information Ratio	-3.357	-7.275	-2.673	60	circular block bootstrap	8	20260625	Post-cost research
Delta-matched equities	Sharpe	1.176	0.372	2.182	60	circular block bootstrap	8	20260625	Post-cost research
Delta-matched equities	Sortino	2.317	0.615	5.886	60	circular block bootstrap	8	20260625	Post-cost research
Delta-matched equities	Calmar	1.225	0.388	3.716	60	circular block bootstrap	8	20260625	Post-cost research
Delta-matched equities	Omega	2.367	1.301	5.182	60	circular block bootstrap	8	20260625	Post-cost research
Delta-matched equities	Information Ratio	1.041	0.189	2.072	60	circular block bootstrap	8	20260625	Post-cost research
Underlying Markowitz	Sharpe	0.909	0.103	1.812	60	circular block bootstrap	8	20260625	Post-cost research
Underlying Markowitz	Sortino	1.856	0.172	4.958	60	circular block bootstrap	8	20260625	Post-cost research
Underlying Markowitz	Calmar	0.590	0.045	2.544	60	circular block bootstrap	8	20260625	Post-cost research
Underlying Markowitz	Omega	1.929	1.074	3.877	60	circular block bootstrap	8	20260625	Post-cost research
Underlying Markowitz	Information Ratio	0.072	-0.796	0.866	60	circular block bootstrap	8	20260625	Post-cost research

Table 39: Full bootstrap inference ledger for gross and stress-cost strategy metrics.

Variant	Sharpe	Monthly Sharpe	Probabilistic Sharpe	Deflated Sharpe	DSR trials	Reality check p	Centered max stat	Centered max p95	N	Block size	Seed
Equity-option Greek Markowitz::gross	0.8421194565895299	0.24309894747589325	0.9786072533439618	3.408658092459508e-11	9	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Greek Markowitz + VIX::gross	1.3743885399954887	0.3967517967687661	0.9994784200061717	9.942036443637217e-08	9	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Beta/delta-neutral + VIX::gross	1.4135317239912606	0.40805146067721504	0.9996222330231731	1.6814438922155963e-07	9	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Cost-aware Sortino + VIX::gross	1.0144053186224946	0.29283359188704267	0.9978781622127668	4.038713604286699e-13	9	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Equal premium::gross	-0.029556763365263826	-0.00853230264265457	0.4740471815096092	1.4656856535628173e-15	9	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Equal risk::gross	0.08183086586465663	0.02362253621748983	0.5729898956153834	2.909213396306826e-15	9	0.078	3.160753023196929	3.424388436686476	60	8	20260625
VIX hedge sleeve::gross	-6.429184989935394	-1.8559458423045507	4.582392667763317e-06	2.8202688354747757e-12	9	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Delta-matched equities::gross	1.1763702732937484	0.3395888469764097	0.9953982143625639	7.018551285374242e-08	9	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Underlying Markowitz::gross	0.908883330100838	0.2623720176478412	0.982650547013001	3.947580211258329e-10	9	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Equity-option Greek Markowitz::mid	1.0240615920008287	0.2956211179042176	0.9958848001741835	0.27266130728744786	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Greek Markowitz + VIX::mid	0.7711086511912625	0.22259989366986233	0.9600370155790526	0.13421561083076827	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Beta/delta-neutral + VIX::mid	1.2645573353367627	0.3650462589811979	0.9977069013011275	0.5054711870627524	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Cost-aware Sortino + VIX::mid	0.14216345544131048	0.04103905463398399	0.6252900904885451	0.006071127745085553	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Equal premium::mid	0.05215826661430436	0.015056794635116447	0.5468293019458382	0.0032539003952041755	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Equal risk::mid	0.1368325564521835	0.039500156650786405	0.6215589535720385	0.005580984991423783	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
VIX hedge sleeve::mid	-1.6996724333246915	-0.4906531684570985	1.410586661600098e-07	1.9995616984377406e-19	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Equity-option Greek Markowitz::half_spread	0.9688501246318463	0.2796829401302995	0.9934794075177362	0.2887758280750587	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Greek Markowitz + VIX::half_spread	0.5335726940127596	0.15402916926025031	0.8852349965736346	0.07082348692549895	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Beta/delta-neutral + VIX::half_spread	0.9706071354884369	0.28019014547581034	0.9852643350252832	0.3143355313904279	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Cost-aware Sortino + VIX::half_spread	0.09439727908111553	0.027250147244125143	0.5836065916661326	0.007304383187074222	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Equal premium::half_spread	-0.05205224625655659	-0.01502618919407382	0.4548272428986697	0.0034731006492324528	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Equal risk::half_spread	0.01517449190340474	0.004380498492623262	0.5134507484415902	0.004628787852991816	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
VIX hedge sleeve::half_spread	-1.741614934592496	-0.5027609256558251	1.7627571606747332e-08	9.478226752602493e-21	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Equity-option Greek Markowitz::full_spread	0.9134753475315865	0.26369761889772425	0.9899408092589378	0.2987736265465256	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Greek Markowitz + VIX::full_spread	0.2943184002523529	0.08496240380657799	0.7446656012700859	0.032333086899338205	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Beta/delta-neutral + VIX::full_spread	0.677919250357419	0.19569843084134259	0.9355215240422323	0.16054099931371685	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Cost-aware Sortino + VIX::full_spread	0.04656344262125476	0.013441708065888566	0.5412897255880229	0.008369236383558824	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Equal premium::full_spread	-0.15639381474734904	-0.04514700552198722	0.37082218454392535	0.003544055760555844	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
Equal risk::full_spread	-0.10651899845248258	-0.030749386215175076	0.4080645087724906	0.003685898263592443	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625
VIX hedge sleeve::full_spread	-1.7767478772756298	-0.5129029326135905	2.5459797323930304e-09	7.7677544773028065e-22	7	0.078	3.160753023196929	3.424388436686476	60	8	20260625

Table 40: Reality-check inference across gross and executable-cost variants. Probabilistic and deflated Sharpe ratios are computed in per-month units; the centered max statistic and its bootstrap 95th percentile define the family-wise reality check ($p = 0.078$ for this variant family).

Strategy	Ann. alpha	Alpha HAC t	Alpha HAC se	HAC lags	R^2	Residual ann. vol	N	Beta SPY	Beta AAPL	Beta AMZN	Beta GOOGL	Beta JPM	Beta META	Beta MSFT	Beta NVDA	Beta TSLA	Beta VX front	Beta dVIX	Beta dVVIX
Equity-option Greek Markowitz	0.405	1.005	0.403	3	0.618	0.634	58	0.005	-1.331	-0.875	0.631	0.590	0.126	2.585	-0.089	1.136	-1.368	0.046	0.003
Greek Markowitz + VIX	0.675	1.814	0.372	3	0.513	0.616	58	1.908	-1.462	-0.747	0.625	-0.127	-0.026	1.877	0.222	0.871	-0.767	0.056	-0.001
Beta/delta-neutral + VIX	0.845	2.425	0.348	3	0.429	0.601	58	0.715	-1.475	-0.612	0.472	-0.578	-0.170	1.336	0.370	0.815	-0.522	0.038	-0.001
Cost-aware Sortino + VIX	4.091	2.422	1.689	3	0.451	3.650	58	13.774	-9.541	-2.823	0.167	-3.209	-0.676	11.207	0.013	4.966	-4.302	0.195	0.008
Equal premium	-0.416	-0.499	0.834	3	0.253	1.647	58	2.522	-1.714	-0.069	1.943	-1.879	-0.344	3.001	-0.374	0.404	-0.749	0.032	-0.001
Equal risk	-0.183	-0.294	0.621	3	0.253	1.115	58	2.041	-1.795	-0.113	1.331	-1.484	-0.316	1.968	-0.082	0.321	-0.379	0.020	-0.003
VIX hedge sleeve	-8.807	-12.880	0.684	3	0.252	1.264	58	3.562	-0.597	-0.589	0.670	-3.047	0.423	-0.958	-0.030	-0.131	-1.180	0.093	-0.013
Delta-matched equities	0.000	1.990	0.000	3	1.000	0.000	58	-0.000	0.088	0.539	0.230	0.590	0.011	0.442	0.779	0.628	-0.000	-0.000	0.000
Underlying Markowitz	0.000	2.378	0.000	3	1.000	0.000	58	-0.000	-0.085	0.159	-0.096	-0.050	0.029	0.280	0.191	0.111	-0.000	-0.000	0.000

Table 41: Full factor regression ledger, including SPY, all eight underlying equities, VX front returns, VIX changes, and VVIX changes. The delta-matched-equities and underlying-Markowitz benchmark rows are exact linear combinations of the regressors ($R^2 = 1.000$, zero residual variance); their alpha t-statistics are numerical artifacts of near-zero standard errors and should be ignored.

A.4. Tail-path simulation diagnostics

The next tables simulate alternative out-of-sample paths from the realized monthly strategy returns. They are path-risk diagnostics only: simulated returns do not enter contract selection, optimizer weights, expected-return forecasts, or any trading decision. The block bootstrap preserves short-run dependence in the realized option return path. The volatility-clustered simulation fits an EGARCH(1,1)-t model (Nelson, 1991) only when the OOS sample is long enough; otherwise it reports a fixed-parameter GARCH(1,1) residual-bootstrap fallback rather than pretending that the EGARCH parameters are precisely estimated. Simulated wealth paths are floored at zero (absorbing default), and the summary reports the defaulted-path share directly. Table 42 gives drawdown-breach probabilities, Table 43 summarizes path centers and left tails, and Table 44 records the fitted model status per strategy.

Return basis	Strategy	Requested method	Simulation	Breach 10%	Breach 25%	Breach 50%	Breach 75%	Breach 90%
Gross before costs	Equity-option Greek Markowitz	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	0.983	0.770	0.298
Gross before costs	Equity-option Greek Markowitz	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1.000	1.000	0.976	0.742	0.424
Gross before costs	Greek Markowitz + VIX	circular_block_bootstrap	circular_block_bootstrap	1.000	0.996	0.795	0.135	0.005
Gross before costs	Greek Markowitz + VIX	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1.000	0.996	0.814	0.373	0.146
Gross before costs	Beta/delta-neutral + VIX	circular_block_bootstrap	circular_block_bootstrap	1.000	0.996	0.585	0.063	0.001
Gross before costs	Beta/delta-neutral + VIX	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1.000	0.969	0.592	0.176	0.053
Gross before costs	Delta-matched equities	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	0.902	0.548	0.175
Gross before costs	Delta-matched equities	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1.000	0.998	0.778	0.282	0.068
Gross before costs	Underlying Markowitz	circular_block_bootstrap	circular_block_bootstrap	0.981	0.432	0.013	0.000	0.000
Gross before costs	Underlying Markowitz	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	0.962	0.300	0.011	0.000	0.000
Gross before costs	Equal premium	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	1.000	0.999	0.995
Gross before costs	Equal premium	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1.000	1.000	1.000	1.000	0.999
Gross before costs	Equal risk	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	1.000	0.973	0.841
Gross before costs	Equal risk	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1.000	1.000	1.000	0.993	0.938
Gross before costs	VIX hedge sleeve	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	1.000	1.000	1.000
Gross before costs	VIX hedge sleeve	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1.000	1.000	1.000	1.000	1.000
Full-spread post-cost	Equity-option Greek Markowitz	circular_block_bootstrap	circular_block_bootstrap	1.000	0.857	0.269	0.006	0.001
Full-spread post-cost	Equity-option Greek Markowitz	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1.000	0.973	0.533	0.091	0.006
Full-spread post-cost	Greek Markowitz + VIX	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	0.858	0.232	0.009
Full-spread post-cost	Greek Markowitz + VIX	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1.000	0.999	0.933	0.645	0.337
Full-spread post-cost	Beta/delta-neutral + VIX	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	0.535	0.034	0.000
Full-spread post-cost	Beta/delta-neutral + VIX	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1.000	0.991	0.768	0.315	0.083
Full-spread post-cost	Delta-matched equities	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	0.902	0.548	0.175
Full-spread post-cost	Delta-matched equities	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1.000	0.998	0.778	0.282	0.068
Full-spread post-cost	Underlying Markowitz	circular_block_bootstrap	circular_block_bootstrap	0.981	0.432	0.013	0.000	0.000
Full-spread post-cost	Underlying Markowitz	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	0.962	0.300	0.011	0.000	0.000
Full-spread post-cost	Equal premium	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	1.000	0.990	0.908
Full-spread post-cost	Equal premium	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1.000	1.000	1.000	0.999	0.979
Full-spread post-cost	Equal risk	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	0.996	0.869	0.560
Full-spread post-cost	Equal risk	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1.000	1.000	0.997	0.857	0.508
Full-spread post-cost	VIX hedge sleeve	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	1.000	1.000	1.000
Full-spread post-cost	VIX hedge sleeve	egarch_or_ewma	garch11_residual_fallback_insufficient_egarch_obs	1.000	1.000	1.000	0.999	0.993

Table 42: Drawdown-breach probabilities from block-bootstrap and volatility-clustered path simulations. The thresholds are intentionally wide because option-only post-cost paths can experience large drawdowns.

Return basis	Strategy	Simulation	N paths	Ann. return p50	Sortino p50	Max DD p05	Max DD p50	Terminal wealth p05	Defaulted path share
Gross before costs	Equity-option Greek Markowitz	circular_block_bootstrap	1000	0.525	1.826	-0.976	-0.840	0.128	0.000
Gross before costs	Equity-option Greek Markowitz	garch11_residual_fallback_insufficient_egarch_obs	1000	0.181	1.238	-0.996	-0.871	0.023	0.002
Gross before costs	Greek Markowitz + VIX	circular_block_bootstrap	1000	1.382	3.451	-0.816	-0.602	3.033	0.000
Gross before costs	Greek Markowitz + VIX	garch11_residual_fallback_insufficient_egarch_obs	1000	1.086	2.713	-0.983	-0.684	0.327	0.007
Gross before costs	Beta/delta-neutral + VIX	circular_block_bootstrap	1000	1.354	3.505	-0.777	-0.547	4.583	0.000
Gross before costs	Beta/delta-neutral + VIX	garch11_residual_fallback_insufficient_egarch_obs	1000	1.130	3.130	-0.910	-0.553	1.634	0.002
Gross before costs	Delta-matched equities	circular_block_bootstrap	1000	1.006	2.410	-0.964	-0.774	0.717	0.000
Gross before costs	Delta-matched equities	garch11_residual_fallback_insufficient_egarch_obs	1000	1.029	2.629	-0.921	-0.638	0.867	0.000
Gross before costs	Underlying Markowitz	circular_block_bootstrap	1000	0.176	1.943	-0.422	-0.232	1.035	0.000
Gross before costs	Underlying Markowitz	garch11_residual_fallback_insufficient_egarch_obs	1000	0.143	1.656	-0.395	-0.199	0.974	0.000
Gross before costs	Equal premium	circular_block_bootstrap	1000	-0.784	-0.064	-1.000	-1.000	0.000	0.000
Gross before costs	Equal premium	garch11_residual_fallback_insufficient_egarch_obs	1000	-0.854	-0.613	-1.000	-1.000	0.000	0.138
Gross before costs	Equal risk	circular_block_bootstrap	1000	-0.450	0.153	-0.999	-0.982	0.001	0.000
Gross before costs	Equal risk	garch11_residual_fallback_insufficient_egarch_obs	1000	-0.601	-0.349	-1.000	-0.995	0.000	0.002
Gross before costs	VIX hedge sleeve	circular_block_bootstrap	1000	-1.000	-3.084	-1.000	-1.000	0.000	1.000
Gross before costs	VIX hedge sleeve	garch11_residual_fallback_insufficient_egarch_obs	1000	-1.000	-3.272	-1.000	-1.000	0.000	1.000
Full-spread post-cost	Equity-option Greek Markowitz	circular_block_bootstrap	1000	0.367	2.139	-0.637	-0.424	0.922	0.000
Full-spread post-cost	Equity-option Greek Markowitz	garch11_residual_fallback_insufficient_egarch_obs	1000	0.140	0.922	-0.790	-0.511	0.424	0.000
Full-spread post-cost	Greek Markowitz + VIX	circular_block_bootstrap	1000	0.001	0.435	-0.859	-0.655	0.234	0.000
Full-spread post-cost	Greek Markowitz + VIX	garch11_residual_fallback_insufficient_egarch_obs	1000	-0.182	-0.093	-0.988	-0.835	0.020	0.000
Full-spread post-cost	Beta/delta-neutral + VIX	circular_block_bootstrap	1000	0.232	1.124	-0.731	-0.513	0.556	0.000
Full-spread post-cost	Beta/delta-neutral + VIX	garch11_residual_fallback_insufficient_egarch_obs	1000	0.080	0.632	-0.929	-0.642	0.171	0.000
Full-spread post-cost	Delta-matched equities	circular_block_bootstrap	1000	1.006	2.410	-0.964	-0.774	0.717	0.000
Full-spread post-cost	Delta-matched equities	garch11_residual_fallback_insufficient_egarch_obs	1000	1.029	2.629	-0.921	-0.638	0.867	0.000
Full-spread post-cost	Underlying Markowitz	circular_block_bootstrap	1000	0.176	1.943	-0.422	-0.232	1.035	0.000
Full-spread post-cost	Underlying Markowitz	garch11_residual_fallback_insufficient_egarch_obs	1000	0.143	1.656	-0.395	-0.199	0.974	0.000
Full-spread post-cost	Equal premium	circular_block_bootstrap	1000	-0.520	-0.290	-0.999	-0.984	0.002	0.000
Full-spread post-cost	Equal premium	garch11_residual_fallback_insufficient_egarch_obs	1000	-0.682	-0.793	-1.000	-0.998	0.000	0.000
Full-spread post-cost	Equal risk	circular_block_bootstrap	1000	-0.312	-0.172	-0.988	-0.914	0.016	0.000
Full-spread post-cost	Equal risk	garch11_residual_fallback_insufficient_egarch_obs	1000	-0.298	-0.407	-0.984	-0.902	0.019	0.000
Full-spread post-cost	VIX hedge sleeve	circular_block_bootstrap	1000	-1.000	-1.650	-1.000	-1.000	0.000	0.623
Full-spread post-cost	VIX hedge sleeve	garch11_residual_fallback_insufficient_egarch_obs	1000	-0.776	-0.870	-1.000	-1.000	0.000	0.012

Table 43: Tail-path simulation summary. Median return and Sortino describe the simulated path center; the fifth-percentile maximum drawdown and terminal wealth columns describe the left tail of realized-path resampling. Paths that lose all NAV are absorbed at zero wealth and counted in the defaulted-path share column.

Return basis	Strategy	Method	Status	Reason	N obs	Source start	Source end	Block length	Path count	Lag1 autocorr
Gross before costs	Equity-option Greek Markowitz	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.096
Gross before costs	Equity-option Greek Markowitz	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	0.096
Gross before costs	Greek Markowitz + VIX	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.003
Gross before costs	Greek Markowitz + VIX	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	0.003
Gross before costs	Beta/delta-neutral + VIX	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.083
Gross before costs	Beta/delta-neutral + VIX	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	-0.083
Gross before costs	Delta-matched equities	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.014
Gross before costs	Delta-matched equities	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	0.014
Gross before costs	Underlying Markowitz	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.041
Gross before costs	Underlying Markowitz	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	0.041
Gross before costs	Equal premium	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.256
Gross before costs	Equal premium	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	-0.256
Gross before costs	Equal risk	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.188
Gross before costs	Equal risk	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	-0.188
Gross before costs	VIX hedge sleeve	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.125
Gross before costs	VIX hedge sleeve	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	-0.125
Full-spread post-cost	Equity-option Greek Markowitz	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.075
Full-spread post-cost	Equity-option Greek Markowitz	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	0.075
Full-spread post-cost	Greek Markowitz + VIX	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.090
Full-spread post-cost	Greek Markowitz + VIX	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	0.090
Full-spread post-cost	Beta/delta-neutral + VIX	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.110
Full-spread post-cost	Beta/delta-neutral + VIX	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	0.110
Full-spread post-cost	Delta-matched equities	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.014
Full-spread post-cost	Delta-matched equities	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	0.014
Full-spread post-cost	Underlying Markowitz	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.041
Full-spread post-cost	Underlying Markowitz	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	0.041
Full-spread post-cost	Equal premium	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.242
Full-spread post-cost	Equal premium	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	-0.242
Full-spread post-cost	Equal risk	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.155
Full-spread post-cost	Equal risk	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	-0.155
Full-spread post-cost	VIX hedge sleeve	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.113
Full-spread post-cost	VIX hedge sleeve	garch11_residual_fallback_insufficient_egarch_obs	fallback	insufficient_egarch_obs	60	2021-01-29	2026-04-30		1000	0.113

Table 44: Simulation assumptions and model status. Fallback rows are deliberate diagnostics: when the monthly OOS sample is too short for EGARCH estimation, the appendix uses a fixed-parameter GARCH(1,1) residual bootstrap and records the reason.

A.5. Additional robustness results

The final group holds the robustness cuts: liquidity tiers (Table 45), SPY-return regimes (Table 46, Figure 10), VIX-level regimes (Table 47, Figure 11), forecast ablations (Table 49), leave-one-underlying-out tests (Table 50, Figure 12), and the walk-forward rolling diagnostic (Table 51). These are uncertainty screens, not proof that a premium is stable.

Liquidity tier	Strategy	Ann. return	Ann. vol	Sharpe	Calmar	Omega
all_eligible	Greek Markowitz	1.2463192201296207	0.9068172382561567	1.374388539995489	2.033918581869829	2.784363176427202
all_eligible	Equal premium	-0.05552159864526898	1.878473564887012	-0.02955676336526383	-0.05553271091792252	0.9771765473237329
all_eligible	Equal risk	0.10434553413680825	1.275136625211684	0.08183086586465663	0.10693488072635121	1.0641387065829775
all_eligible	Delta neutral	1.1720642765798035	0.8291743699040248	1.4135317239912606	2.0573541792299705	2.947610573038661
all_eligible	VIX hedge sleeve	-9.28491238522107	1.444181867492721	-6.429184989935395	nan	0.04042569095025844
tight_spread_quartile	Greek Markowitz	0.7303532638080936	1.0047849887103797	0.726875174305188	0.8741163613036393	1.8086255408376313
tight_spread_quartile	Equal premium	0.8328849438491032	1.954819126128125	0.42606752344334964	0.8353858747460982	1.3816341770771015
tight_spread_quartile	Equal risk	0.8890568650908819	1.24447750676842	0.7144017149811958	1.0024475264243362	1.7054047852220204
tight_spread_quartile	Delta neutral	0.616948156897824	1.839928133485132	0.3353110078974774	0.6186798622336975	1.2799856377055105
top_volume_quartile	Greek Markowitz	2.526134501171973	2.0209005780985363	1.250004343879604	3.05152660125299	3.1037344122269954
top_volume_quartile	Equal premium	-1.1699626268279582	2.3381506856559016	-0.5003794811025015	-1.1699626271251244	0.6936318778579352
top_volume_quartile	Equal risk	-0.3863656502558095	2.1839471975979072	-0.1769116261971753	-0.3863658542494987	0.8758611931290181
top_volume_quartile	Delta neutral	2.1254579520308203	1.5251047754203533	1.3936471685658425	2.5319548593205963	3.1972700102186558
top_volume_quartile	VIX hedge sleeve	-9.28491238522107	1.444181867492721	-6.429184989935395	nan	0.04042569095025844

Table 45: Liquidity-tier performance. The tiers test whether the option allocation survives when the universe is restricted to high-volume, tight-spread, high-open-interest, or jointly liquid contracts. Tier membership is classified point-in-time from liquidity observed through the training-window end (2020-12-31), so an allocator at the decision date could have known the tier labels; an earlier draft classified tiers on end-of-sample liquidity.

Strategy	Regime	Months	Ann. return	Ann. vol	Sharpe	Sharpe CI lo	Sharpe CI hi	Sortino	Calmar	Omega	Info. ratio
Equity-option Greek Markowitz	Down	20	-0.821	0.844	-0.973	-2.216	-0.064	-1.188	-0.926	0.514	-0.409
Equity-option Greek Markowitz	Flat	20	0.716	0.913	0.785	-0.385	1.734	1.744	1.493	1.830	-0.100
Equity-option Greek Markowitz	Up	20	2.787	1.175	2.373	0.685	4.553	8.069	6.615	6.928	1.808
Greek Markowitz + VIX	Down	20	0.665	0.748	0.888	-0.223	1.909	2.009	1.283	1.922	1.525
Greek Markowitz + VIX	Flat	20	0.691	0.924	0.747	-0.526	2.010	1.692	1.388	1.809	0.206
Greek Markowitz + VIX	Up	20	2.384	0.985	2.421	0.831	4.232	6.598	7.182	5.575	1.737
Beta/delta-neutral + VIX	Down	20	1.368	0.718	1.905	1.135	3.046	4.579	4.716	3.996	2.505
Beta/delta-neutral + VIX	Flat	20	0.725	0.830	0.874	-0.354	1.983	2.391	1.964	2.125	0.219
Beta/delta-neutral + VIX	Up	20	1.423	0.951	1.496	-0.182	3.218	3.419	3.357	3.019	0.777
Cost-aware Sortino + VIX	Down	20	0.441	2.357	0.187	-0.836	1.086	0.333	0.448	1.143	0.387
Cost-aware Sortino + VIX	Flat	20	1.150	3.794	0.303	-1.495	1.332	0.672	1.044	1.268	0.019
Cost-aware Sortino + VIX	Up	20	13.468	6.833	1.971	1.144	2.919	9.405	0.761	5.711	1.872
Equal premium	Down	20	-0.696	1.854	-0.375	-1.677	0.397	-0.636	-0.714	0.752	-0.121
Equal premium	Flat	20	-2.279	0.894	-2.550	-4.886	-1.126	-2.202	-2.293	0.167	-3.328
Equal premium	Up	20	2.809	2.352	1.195	0.314	2.010	3.236	3.712	2.597	0.917
Equal risk	Down	20	-0.092	1.420	-0.065	-1.141	0.782	-0.114	-0.109	0.953	0.265
Equal risk	Flat	20	-1.548	0.690	-2.245	-4.086	-0.950	-2.026	-1.616	0.171	-3.267
Equal risk	Up	20	1.954	1.419	1.377	0.302	2.376	3.665	3.495	2.843	0.915
VIX hedge sleeve	Down	20	-7.880	1.343	-5.868	-10.036	-4.146	-3.008		0.020	-5.401
VIX hedge sleeve	Flat	20	-8.261	1.953	-4.230	-7.825	-2.729	-2.796		0.109	-3.965
VIX hedge sleeve	Up	20	-11.714	0.276	-42.492	-107.213	-26.949	-3.453		0.000	-40.570
Delta-matched equities	Down	20	-1.817	0.660	-2.753	-4.506	-1.423	-2.266	-1.856	0.149	-2.234
Delta-matched equities	Flat	20	1.474	0.671	2.195	1.478	3.035	11.412	10.359	8.822	1.568
Delta-matched equities	Up	20	3.630	0.698	5.203	3.511	8.613	34.716	26.891	45.819	4.428
Underlying Markowitz	Down	20	-0.340	0.120	-2.841	-5.016	-1.410	-2.337	-0.751	0.134	1.316
Underlying Markowitz	Flat	20	0.217	0.179	1.213	-0.052	2.292	3.100	2.201	2.579	-0.562
Underlying Markowitz	Up	20	0.658	0.172	3.829	2.463	6.121	15.942	17.455	15.149	-0.128

Table 46: Performance by SPY-return tercile. Down, flat, and up regimes are descriptive cuts of the held-out sample, not fitted optimizer inputs.

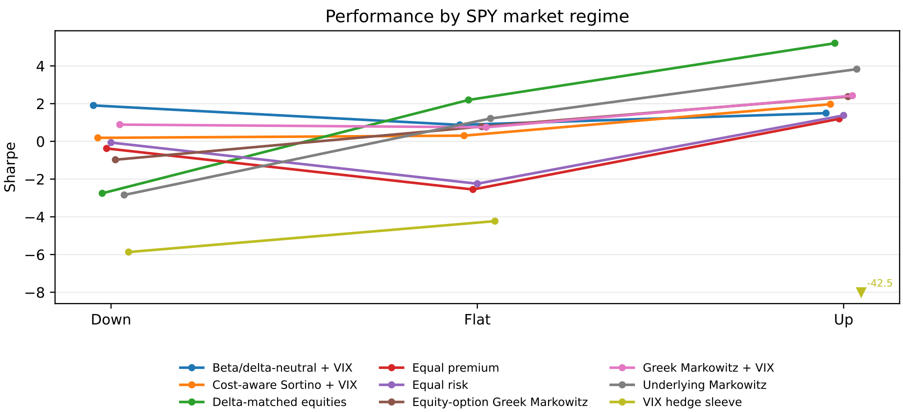

Figure 10: Sharpe ratios by SPY-return tercile. The figure visualizes Table 46. Values below −8 are clipped and annotated with the true Sharpe so that a single failed sleeve does not compress the axis.

Strategy	Regime	Months	Ann. return	Ann. vol	Sharpe	Sharpe CI lo	Sharpe CI hi	Sortino	Calmar	Omega	Info. ratio
Equity-option Greek Markowitz	Low VIX	20	1.585	0.919	1.724	1.033	2.559	4.862	5.911	3.697	1.954
Equity-option Greek Markowitz	Mid VIX	20	1.104	1.216	0.907	-0.388	1.845	2.157	2.351	2.017	0.841
Equity-option Greek Markowitz	High VIX	20	-0.006	1.029	-0.006	-1.030	1.097	-0.009	-0.007	0.996	-0.808
Greek Markowitz + VIX	Low VIX	20	1.386	0.846	1.638	0.265	3.472	4.774	4.419	3.257	2.330
Greek Markowitz + VIX	Mid VIX	20	1.533	1.054	1.455	0.412	2.331	3.812	4.985	3.198	1.408
Greek Markowitz + VIX	High VIX	20	0.821	0.838	0.979	-0.096	2.111	2.047	2.057	2.047	-0.223
Beta/delta-neutral + VIX	Low VIX	20	1.176	0.761	1.546	0.252	3.145	3.811	5.194	2.948	2.396
Beta/delta-neutral + VIX	Mid VIX	20	0.974	0.965	1.009	-0.446	2.152	2.326	2.350	2.250	1.028
Beta/delta-neutral + VIX	High VIX	20	1.366	0.786	1.737	0.804	2.874	4.716	4.063	4.231	0.383
Cost-aware Sortino + VIX	Low VIX	20	6.858	5.503	1.246	0.027	2.324	3.829	3.003	2.699	1.358
Cost-aware Sortino + VIX	Mid VIX	20	7.748	5.957	1.301	0.559	1.872	6.246	8.754	3.637	1.286
Cost-aware Sortino + VIX	High VIX	20	0.455	2.740	0.166	-0.906	1.000	0.322	0.457	1.139	-0.362
Equal premium	Low VIX	20	0.249	1.592	0.157	-1.511	1.123	0.283	0.286	1.132	0.551
Equal premium	Mid VIX	20	0.944	2.287	0.413	-0.680	1.136	0.923	1.004	1.371	0.398
Equal premium	High VIX	20	-1.360	1.717	-0.792	-3.832	0.263	-1.240	-1.372	0.524	-1.345
Equal risk	Low VIX	20	0.598	1.122	0.533	-0.926	1.641	0.983	0.824	1.488	1.038
Equal risk	Mid VIX	20	0.348	1.456	0.239	-0.779	0.996	0.488	0.436	1.204	0.231
Equal risk	High VIX	20	-0.633	1.264	-0.501	-2.543	0.500	-0.787	-0.679	0.676	-1.227
VIX hedge sleeve	Low VIX	20	-9.290	1.621	-5.730	-20.577	-3.438	-3.036	-9.290	0.063	-5.189
VIX hedge sleeve	Mid VIX	20	-10.031	1.331	-7.534	-41.311	-4.785	-3.203		0.038	-7.879
VIX hedge sleeve	High VIX	20	-8.533	1.406	-6.069	-9.076	-4.610	-3.031		0.018	-7.261
Delta-matched equities	Low VIX	20	2.273	0.692	3.283	2.654	4.097	19.209	15.977	17.371	2.903
Delta-matched equities	Mid VIX	20	1.918	0.917	2.091	1.374	2.983	6.416	6.081	5.040	1.689
Delta-matched equities	High VIX	20	-0.904	0.917	-0.986	-2.049	-0.174	-1.201	-0.980	0.495	-1.403
Underlying Markowitz	Low VIX	20	0.350	0.174	2.008	0.968	3.220	5.384	4.524	4.537	1.879
Underlying Markowitz	Mid VIX	20	0.386	0.194	1.994	1.369	2.695	7.002	7.409	5.008	0.746
Underlying Markowitz	High VIX	20	-0.201	0.181	-1.108	-2.171	-0.324	-1.403	-0.622	0.473	-1.495

Table 47: Performance by VIX-level tercile.

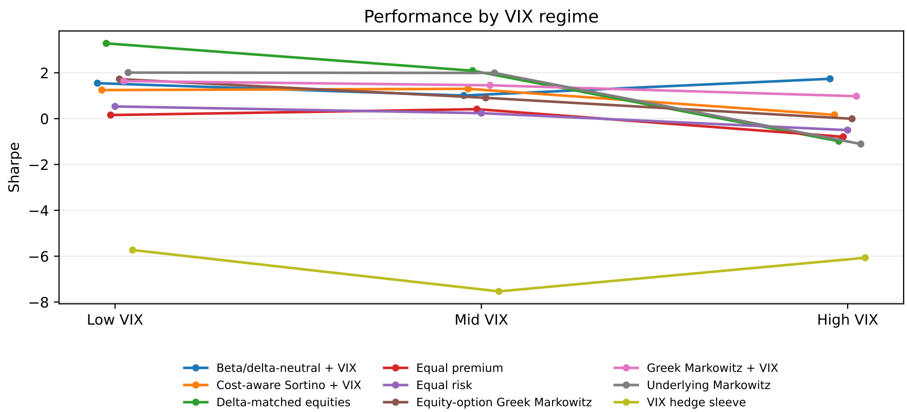

Figure 11: Sharpe ratios by VIX-level tercile. The figure visualizes Table 47; values below −8 would be clipped and annotated as in Figure 10.

The vol-of-vol regime diagnostic conditions on the prior decision date's VIX chain feature. These features are not optimizer inputs; the table is a descriptive robustness screen for realized returns after the feature was observable.

Strategy	Bucket	Months	Mean monthly return	Annualized Sharpe
Beta/delta-neutral + VIX	1	7	0.101	1.215
Beta/delta-neutral + VIX	2	7	0.026	0.672
Beta/delta-neutral + VIX	3	7	0.099	1.238
Cost-aware Sortino + VIX	1	7	0.051	0.136
Cost-aware Sortino + VIX	2	7	0.642	1.122
Cost-aware Sortino + VIX	3	7	0.748	1.096
Delta-matched equities	1	7	0.097	1.911
Delta-matched equities	2	7	0.084	0.830
Delta-matched equities	3	7	0.198	2.224
Equal premium	1	7	-0.039	-0.273
Equal premium	2	7	0.022	0.133
Equal premium	3	7	-0.021	-0.147
Equal risk	1	7	0.008	0.069
Equal risk	2	7	-0.002	-0.023
Equal risk	3	7	0.025	0.236
Equity-option Greek Markowitz	1	7	0.052	0.558
Equity-option Greek Markowitz	2	7	0.014	0.205
Equity-option Greek Markowitz	3	7	0.127	0.890
Greek Markowitz + VIX	1	7	0.102	1.352
Greek Markowitz + VIX	2	7	0.030	0.895
Greek Markowitz + VIX	3	7	0.101	0.986
Underlying Markowitz	1	7	0.010	0.794
Underlying Markowitz	2	7	0.015	0.753
Underlying Markowitz	3	7	0.033	1.530
VIX hedge sleeve	1	7	-0.876	-12.999
VIX hedge sleeve	2	7	-0.996	-318.638
VIX hedge sleeve	3	7	-0.968	-51.079

Table 48: VIX chain vol-of-vol regime performance. Regime labels use the prior decision date's diagnostic feature and do not feed expected-return forecasts.

Ablation	Ann. return	Ann. vol	Sharpe	Calmar	Omega	Active assets
carry_only	6.701470998996984	1.9164807271628133	3.496758878925927	2.809625893980088e-05	8.037348345976534	54
variance_risk_premium_only	9.631391759962566	0.6700827342459892	14.373436693306676	nan	nan	54
skew_tail_only	4.4890452067116735	3.2048741838250296	1.4006931159319274	0.0030793538397452446	2.691677332065947	18
vix_regime_only	4.590815059335688	2.544245648201727	1.8043914362516356	0.0042982130732032746	3.3862924961758023	54
relative_value_only	2.5789118807598532	1.4076551542781675	1.8320622582327668	7.262838528939611	6.695662484712621	54
full_conditional_model	4.113956564258067	2.9681846972242787	1.3860177124776858	2.187721423486171	3.1853840590190656	54

Table 49: Forecast ablation performance. Each row isolates one economically identified component of the conditional option-premium forecast.

Exclusion	Remaining underlyings	Option assets	Ann. return	Ann. vol	Sharpe	Sharpe CI lo	Sharpe CI hi	Sortino	Calmar	Omega	Info. ratio	Net delta	Beta SPY
All underlyings	8	54	1.246	0.907	1.374	0.601	2.209	3.385	2.034	2.784	1.200	3.307	1.297
No AAPL	7	48	1.253	0.949	1.321	0.571	2.091	3.233	2.016	2.673	1.147	3.216	1.240
No AMZN	7	48	1.286	0.920	1.397	0.578	2.242	3.710	2.097	2.899	1.223	3.344	1.178
No GOOGL	7	49	1.214	0.895	1.357	0.547	2.213	3.325	1.936	2.740	1.180	3.246	1.265
No JPM	7	49	1.202	0.917	1.311	0.650	2.015	3.032	2.540	2.741	1.106	3.186	1.105
No META	7	47	1.250	0.944	1.323	0.566	2.149	3.188	2.011	2.658	1.163	3.381	1.375
No MSFT	7	49	1.410	0.955	1.476	0.750	2.228	3.807	2.386	3.053	1.312	3.406	1.627
No NVDA	7	46	1.085	0.821	1.322	0.547	2.151	3.256	1.893	2.750	1.157	3.571	0.775
No TSLA	7	47	1.198	0.842	1.424	0.670	2.251	3.612	1.988	3.118	1.265	3.330	1.287
No META/NVDA/TSLA	5	32	0.934	0.837	1.115	0.336	1.940	2.279	1.407	2.352	0.968	4.019	1.217

Table 50: Leave-one-underlying-out robustness for the VIX-enabled Greek Markowitz option portfolio.

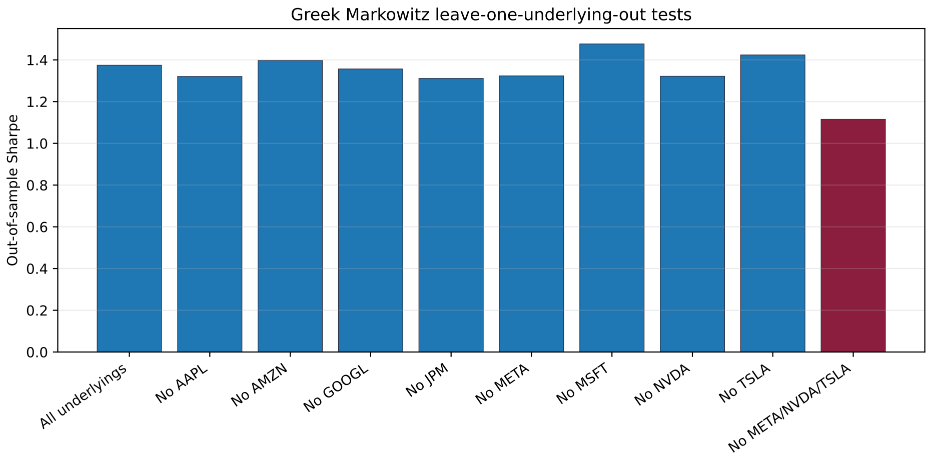

Figure 12: Out-of-sample Sharpe ratios for leave-one-underlying-out variants (Table 50).

Diagnostic	Value
Rolling 36M OOS months	20.000
Rolling 36M OOS ann. return	1.813
Rolling 36M OOS ann. vol	0.847
Rolling 36M OOS Sharpe	2.141
Rolling 36M OOS Sortino	16.252
Rolling 36M OOS Calmar	14.824
Rolling 36M OOS Omega	13.319

Table 51: Walk-forward rolling 36-month out-of-sample diagnostic. At each evaluation date the representative contracts, conditional premia, and covariances are re-estimated on the trailing 36-month window and weights come from the same constrained maximum-Sharpe solver as the headline strategies (Section 5).

A.6. Instrument and settlement convention reference

Every candidate option is first translated into a cashflow convention. Equity options record premium, underlying, strike, expiry, right, exercise style, mark, Greek model, payoff source, and split adjustment. VIX options record OSI symbol, VX-forward anchor, VIX settlement source, Black–76 Greek model, and whether expiry P&L used exact VRO/SOQ or a settlement proxy. Short options additionally record assignment-risk flags, simulated margin funding, and short-premium budget use.

A weak implementation-quality score does not automatically exclude a contract. It widens the haircut, raises the stress or capacity penalty, or downgrades the claim. This is why the paper can show VIX option diagnostics without treating proxy settlement as exact listed settlement.

A.7. Production-readiness and forward shadow checklist

The locked E1 evidence is not promoted to live-trading evidence until the following operational gates are populated with broker-neutral or broker-native ledgers. A forward shadow run can begin with delayed or exported platform data, but it remains shadow-only: IBKR's free trial describes delayed market data for most products and simulated account values based on market conditions, commissions, and fees;[3] that is useful for workflow testing, not a substitute for paid live data or real fills.

Gate	Required production evidence	Current status
Exact quote source	Market-hours OPRA/NBBO or broker CBBO with bid, ask, displayed size, timestamps, and symbol map	shadow ledger required
Fill model	Marketable-limit order logs, partial fills, fees, unfilled remainders, and no midpoint-only fills	not live-filled
Margin model	Broker preview before order plus internal stress margin and collateral use	shadow/input only
Assignment/exercise	Assignment ledger, early-exercise flags, and exercise/expiry processing	research flags only
Ex-dividend closeout	Rule and audit for short calls near ex-dividend or hard-to-borrow windows	required before live
VIX settlement gate	Exact VRO/SOQ for every VIX expiry used in claims	required and audited
Order sizing cap	Displayed-size participation, train-volume cap, and account-level notional cap	modeled, not broker proven
Kill switch	Manual and automated stop conditions for quote failures, drawdown, margin, and stale data	required before live
Reconciliation	Vendor quote, broker order/fill, broker position, cash, and statement reconciliation	not present
Broker statement audit	End-of-day positions, commissions, fees, interest, exercise/assignment, and cash movements	required before live

Table 52: Production-readiness checklist. A forward shadow-trading layer may write target, quote, order, shadow-fill, margin, rejection, and reconciliation ledgers, but only real broker orders, fills, margin previews, and statement reconciliation can support a production-tradability claim.

A.8. Proofs, solver reductions, and implementation notes

Proposition A.1 (No free option exposure)**.** *Let $\mathcal{C}$ include at least one binding scale control among gross premium, short-premium, margin/collateral, stress loss, or contract-level liquidity caps. Then no sequence of option positions can take unbounded premium, payoff, or Greek exposure while remaining feasible and using zero capital.*

Proof. Each listed control is positively homogeneous in the premium-weight vector. If a sequence w_k has $\|w_k\| \to \infty$, at least one controlled exposure also diverges unless the sequence lies in the null space of every cashflow, Greek, stress, margin, and liquidity functional. Such a null direction has neither priced premium exposure nor admissible tradable contract size in this model. Therefore any economically nonzero option direction must consume one of the controlled budgets before scale can diverge. □

Proposition A.2 (Cost and constraint monotonicity)**.** *For*

$$V(\mathcal{C}, c) = \max_{w \in \mathcal{C}} \widehat{\mu}^\top w - \frac{\gamma}{2} w^\top \Sigma w - c^\top |w|,$$

with $\Sigma \succeq 0$, if $\mathcal{C}_1 \subseteq \mathcal{C}_0$ and $c_1 \geq c_0$ elementwise, then $V(\mathcal{C}_1, c_1) \leq V(\mathcal{C}_0, c_0)$.

[3] Interactive Brokers Free Trial: `https://www.interactivebrokers.com/en/trading/free-trial.php`.

Proof. Every $w \in \mathcal{C}_1$ is feasible for $\mathcal{C}_0$, and $-c_1^\top |w| \le -c_0^\top |w|$. Taking maxima over the smaller feasible set with the larger cost vector cannot increase the objective. $\square$

Proposition A.3 (Net liquidity caps are representation-invariant)**.** *Let a signed option weight be represented as $w = w^+ - w^-$, with $w^+, w^- \ge 0$. The constraint $|w_i| \le \bar{q}_i$ depends only on the net position and is invariant to the chosen split representation. Separate split-leg caps $w_i^+ \le \bar{q}_i$, $w_i^- \le \bar{q}_i$ are not equivalent and can consume liquidity budget with offsetting pseudo-cash pairs.*

Proof. The net cap maps every pair (w_i^+, w_i^-) with the same difference to the same constraint value $|w_i^+ - w_i^-|$. Split-leg caps depend on the individual decomposition: for any feasible net $w_i = 0$, the pair $w_i^+ = w_i^- = \bar{q}_i$ satisfies split caps but uses two gross legs and no net contract exposure. In a homogenized conic solve that offsetting pair can help satisfy gross-budget equalities while consuming scarce contract-level cap budget. Binding $|w_i|$ removes that representation artifact. $\square$

Proposition A.4 (Historical means become fragile when n/T rises)**.** *Suppose contract-level historical mean estimates have independent estimation errors with variance bounded below by σ_η^2/T. Across n candidate directions, the expected maximum absolute noise component grows on the order of $\sigma_\eta\sqrt{(\log n)/T}$. Consequently, holding T fixed while expanding n raises the optimizer's opportunity to select noise as if it were alpha.*

Proof. For sub-Gaussian errors with scale $\sigma_\eta/\sqrt{T}$, standard maximal inequalities give $\mathbb{E}\max_{1\le i\le n}|\eta_i| = O(\sigma_\eta\sqrt{(\log n)/T})$, and the same rate is attained up to constants for independent Gaussian errors. A mean–variance optimizer loads directions with favorable estimated means after risk scaling, so as n/T rises the best-looking historical-mean direction is increasingly likely to be an estimation outlier. This matches the empirical diagnostic: the historical mean helps in the eight-name case where T/n is healthier, but hurts in the broad universe where the inverse problem is thinner. $\square$

Proposition A.5 (Greek-factor covariance identity and positive definiteness)**.** *Under Assumption 1, with $\Omega_t = \mathrm{Var}_t(f_{t+1})$ and $\Sigma_{\varepsilon,t} = \mathrm{Var}_t(\varepsilon_{t+1})$,*

$$\mathrm{Var}_t(r_{O,t+1}) = B_t\Omega_t B_t^\top + \Sigma_{\varepsilon,t},$$

and the sum is positive semidefinite. If in addition $\Sigma_{\varepsilon,t} \succ 0$, then $\Sigma_{O,t} \succ 0$.

Proof. $\mathrm{Var}_t(B_t f_{t+1} + \varepsilon_{t+1}) = B_t\Omega_t B_t^\top + \Sigma_{\varepsilon,t} + B_t\Sigma_{f\varepsilon,t} + \Sigma_{f\varepsilon,t}^\top B_t^\top$, and $\Sigma_{f\varepsilon,t} = \mathrm{Cov}_t(f_{t+1}, \varepsilon_{t+1}) = 0$ by the projection construction. Both remaining terms are covariance forms, hence positive semidefinite. For any $x \ne 0$, $x^\top\Sigma_{O,t}x \ge x^\top\Sigma_{\varepsilon,t}x \ge \lambda_{\min}(\Sigma_{\varepsilon,t})\|x\|^2 > 0$. $\square$

This is the population statement, the option-risk-model analogue of an equity factor covariance model. As recorded in Section 4, the estimated $\widehat{\Sigma}_{\varepsilon,t}$, built from pairwise-complete residual moments on an unbalanced contract panel, need not be positive semidefinite in finite samples and is projected onto the PSD cone before entering $\widehat{\Sigma}_{O,t}$.

Proposition A.6 (Unconstrained maximum Sharpe)**.** *Let $\Sigma \succ 0$ and $\mu \neq 0$, and define* $\mathrm{SR}(q) = q^\top \mu / \sqrt{q^\top \Sigma q}$ *on the domain $q \neq 0$. Then (i)* $\mathrm{SR}(cq) = \mathrm{SR}(q)$ *for all $c > 0$; (ii) the stationary points of* SR *restricted to $\{q : q^\top \Sigma q = 1\}$ are $q_\pm = \pm \Sigma^{-1}\mu / \sqrt{\mu^\top \Sigma^{-1} \mu}$; (iii) the maximizer is the positive root q_+, unique up to positive scaling, with optimal value* $\mathrm{SR}^* = \sqrt{\mu^\top \Sigma^{-1} \mu}$*.*

Proof. (i) is immediate. By (i) it suffices to maximize $q^\top \mu$ on the compact ellipsoid $\{q : q^\top \Sigma q = 1\}$, where a maximizer exists. The Lagrange condition $\mu = 2\lambda \Sigma q$ with $\Sigma \succ 0$ gives $q = \Sigma^{-1}\mu/(2\lambda)$, and $q^\top \Sigma q = 1$ forces $2\lambda = \pm\sqrt{\mu^\top \Sigma^{-1} \mu} \neq 0$ because $\mu \neq 0$. The objective at $q_\pm$ equals $\pm\sqrt{\mu^\top \Sigma^{-1} \mu}$, so the maximum is attained at q_+ and (iii) follows. □

Proposition A.7 (Conic reduction of the constrained problem)**.** *Let $K \subseteq \mathbb{R}^n$ be a closed convex cone containing some q with $q^\top \mu > 0$, and let $\Sigma \succ 0$. Then*

$$\sup_{q \in K \setminus \{0\}} \mathrm{SR}(q) = \max\left\{ y^\top \mu : \ y \in K, \ y^\top \Sigma y \leq 1 \right\},$$

and the right-hand side is a convex quadratically constrained program (the Charnes–Cooper transformation).

Proof. For $q \in K \setminus \{0\}$, the point $y = q/\sqrt{q^\top \Sigma q}$ lies in K (cone), satisfies $y^\top \Sigma y = 1$, and has $y^\top \mu = \mathrm{SR}(q)$, so the right side is at least the left. Conversely, the right side is positive by assumption, and any feasible y with $y^\top \mu > 0$ has $\mathrm{SR}(y) = y^\top \mu / \sqrt{y^\top \Sigma y} \geq y^\top \mu$, so the left side is at least the right. The feasible set is closed, convex, and bounded because $\Sigma \succ 0$, and the objective is linear, so the maximum is attained. □

The budget functionals defining $\mathcal{K}_t$ in Section 4 are positively homogeneous of degree one with strictly positive limits (gross NAV, short premium, per-contract, per-underlying, Greek, beta, and stress budgets), so they fix the admissible scale along the optimal ray of Proposition 4.2; genuinely directional restrictions, such as a long-only sleeve, form the conic part to which Proposition A.7 applies. The implemented solver accordingly either maximizes the ratio directly over $\mathcal{K}_t$ under a gross-NAV normalization or solves the reduced convex program $\max\{\mu^\top y : y^\top \Sigma y \leq 1, \ y \in \mathcal{K}_t\}$ and rescales the solution to the gross budget.

The reproducibility package exposes the model code, empirical pipeline, publication tables, figures, strategy-return ledgers, factor regressions, attribution, regime splits, VIX settlement records, claim-audit records, and `tables/empirical_summary.json`. The audit layer independently checks data lineage, point-in-time timing, optimizer constraints, output reproducibility, figure visibility, bibliography scope, caveat text, and LaTeX rendering.

Moment	Value
Median residual mean	0.076
Median residual vol	2.076
Median absolute residual autocorr(1)	0.098
Systematic covariance rank	27.000

Table 53: Training-window residual diagnostics for the local Greek-factor approximation.

The production gap remains visible. Exact VRO/SOQ settlement, executable option quotes, calibrated costs, margin, assignment, borrow, capacity, order routing, and broker reconciliation would be required for production claims. Those claims are not made in this article. The current contribution is the option-only theory object, a point-in-time research implementation, and an audit layer that prevents stronger wording when the data do not support it.

A.9. Distributional-robustness assumptions and ledgers

Table 54 records the state-model and repricing assumptions behind Section 7. In particular, it documents the one-step pricing-tenor convention, the VIX forward convention, the path count, the block length, and the seed used by the synthetic repricing tier.

Section	Item	Value	Notes
State Dimension	spot:AAPL	mu=0.019431; omega=0.00113634; alpha=0.0777105; beta=0.748476	n_obs=70; fallback=False; reason=
State Dimension	spot:AMZN	mu=0.0316915; omega=0.00203097; alpha=0; beta=0.704828	n_obs=70; fallback=False; reason=
State Dimension	spot:GOOGL	mu=0.0182634; omega=0.00348159; alpha=0.143728; beta=2.13318e-14	n_obs=70; fallback=False; reason=
State Dimension	spot:JPM	mu=0.0162303; omega=0.00362136; alpha=0.211243; beta=0	n_obs=70; fallback=False; reason=
State Dimension	spot:META	mu=0.0213947; omega=0.00167887; alpha=0.315841; beta=0.392632	n_obs=70; fallback=False; reason=
State Dimension	spot:MSFT	mu=0.0262454; omega=0.000436407; alpha=0.140304; beta=0.728673	n_obs=70; fallback=False; reason=
State Dimension	spot:NVDA	mu=0.049128; omega=0.00330517; alpha=0.0538818; beta=0.712314	n_obs=70; fallback=False; reason=
State Dimension	spot:TSLA	mu=0.0189024; omega=0.000354872; alpha=0.0604221; beta=0.939578	n_obs=70; fallback=False; reason=
State Dimension	spot:VX_FRONT	mu=-0.00580358; omega=0.0268687; alpha=0.446917; beta=0.0984006	n_obs=70; fallback=False; reason=
State Dimension	iv:AAPL	mu=0.0081573; omega=0.0225031; alpha=0.213579; beta=0.606376	n_obs=69; fallback=False; reason=
State Dimension	iv:AMZN	mu=0.000709312; omega=0.0592597; alpha=0.042638; beta=0.453598	n_obs=69; fallback=False; reason=
State Dimension	iv:GOOGL	mu=-0.00220899; omega=0.0485703; alpha=0.00707529; beta=0.485269	n_obs=69; fallback=False; reason=
State Dimension	iv:JPM	mu=-0.00210938; omega=0.0415749; alpha=0.0733827; beta=0.415499	n_obs=69; fallback=False; reason=
State Dimension	iv:META	mu=-0.00157152; omega=0.0244162; alpha=0.205676; beta=0.547104	n_obs=69; fallback=False; reason=
State Dimension	iv:MSFT	mu=0.00613006; omega=0.0939383; alpha=0.221406; beta=1.84521e-17	n_obs=69; fallback=False; reason=
State Dimension	iv:NVDA	mu=0.000877734; omega=0.0172009; alpha=0.162236; beta=0.739677	n_obs=69; fallback=False; reason=
State Dimension	iv:TSLA	mu=-0.00981235; omega=0.0226323; alpha=0.187789; beta=0.505187	n_obs=69; fallback=False; reason=
State Dimension	iv:VX_FRONT	mu=-0.00212937; omega=0.0459306; alpha=0.360287; beta=2.13779e-17	n_obs=69; fallback=False; reason=
State Dimension	vix:VIX	mu=0.00114242; omega=0.0338075; alpha=0.49259; beta=0.0963354	n_obs=68; fallback=False; reason=
Kernel	Simulation Method	joint_garch_block	joint block rows preserve empirical cross-dependence; gaussian_copula is rank-mapped sensitivity
Kernel	Strike Rule	K_t = entry state_t * exp(train median log moneyness)	Static moneyness, repriced each month
Kernel	Original Tenor Reference	train median tenor_days / 365	Stored as tenor_years for audit only
Kernel	Pricing Tenor Rule	0.08333333333333333	Synthetic contracts are one-step (1-month) options so premium and payoff horizons match; realized-kernel tenors average 0.6 months
Kernel	Skew Freeze	contract median IV / underlying ATM median IV	54 contracts
Kernel	Rate	0.02	Continuously-compounded annual rate used in BS and Black-76
Kernel	IV Floors Caps	0.05 to 2.0	Applied to ATM and skew-adjusted contract IV
Kernel	VIX Level Caps	5.0 to 150.0	Applied during state reconstruction
Kernel	Min Mark Rule	0.25	Entry denominator uses max(model premium, min_mark) in dollars
Kernel	Spot Anchor	contract train-end spot	State spot paths are relative to 1.0 and scaled by anchor_spot at repricing
Kernel	VIX Forward Convention	simulated spot:VX_FRONT entry forward; VIX level settlement proxy at t+1	Premium uses the VX-front state, payoff uses simulated VIX level
Kernel	Seed	20260625	
Kernel	Path Count	1000	
Kernel	Horizon Months	60	
Kernel	Block Length	6	
Kernel	Min GARCH Obs	60	

Table 54: Repriced-universe state-model and pricing assumptions.

The full fold schedule is intentionally not typeset because the blocked and CPCV ledgers are large. The machine-readable schedule and fold ledger ship as `artifacts/cv_fold_schedule.csv` and `artifacts/cv_fold_ledger.csv`; the split membership, CPCV path metrics, path-month returns, PBO summary, regime tags, runtime log, and context-consistency checks are in the adjacent `artifacts/cv_*.csv` files.